Anurag ENGINEERING COLLEGE
(An Autonomous Institution)

Accredited By A+ Grade

FIRST YEAR & SECOND YEAR COURSE STRUCTURE AND SYLLABUS

B.Tech.

ELECTRONICS AND COMMUNICATION ENGINEERING

R25 Regulation Under CBCS



A Four-Year Regular Degree Program
(Applicable for the batches admitted from the Academic Year 2025 – 26)
&
A Three-Year Regular Degree Program under Lateral Entry Scheme (LES)
(Applicable for the batches admitted from the Academic Year 2026 – 27)

ANURAG ENGINEERING COLLEGE

(An Autonomous Institution)

Ananthagiri (V&M), Kodad, Suryapet(Dt), Telangana (India)

ACADEMIC REGULATIONS FOR B. TECH. REGULAR STUDENTS

WITH EFFECT FROM THE ACADEMIC YEAR 2025-26 (R-25)

1.0 Under-Graduate Degree Programme in Engineering & Technology (UGP in E&T)

Anurag Engineering College offers new regulations termed as R-25 regulations for four-year (eight semesters) **Bachelor of Technology** (B.Tech.) degree programme, under Choice Based Credit System (CBCS), with effect from the academic year **2025-26.**

2.0 Eligibility for Admission

2.1 Admissions to the undergraduate (UG) programme shall be made either on the basis of the merit rank obtained by the qualified students at the entrance test conducted by Telangana Government (EAPCET) or the University or on the basis of any other order of merit approved by the University, subject to reservations as prescribed by the government from time to time.

2.2 The medium of instruction for the entire undergraduate programme in Engineering & Technology will be **English** only.

3.0 B.Tech. Programme Structure

3.1 A student after securing admission shall complete the B.Tech. programme in a minimum period of **four** academic years and a maximum period of **eight** academic years starting from the date of commencement of first year first semester, failing which student shall forfeit seat in B.Tech. course. Each student has to secure a minimum of 160 credits out of 164 credits for successful completion of the undergraduate programme and award of the B.Tech. degree.

3.2 **UGC/ AICTE** specified definitions/ descriptions are adopted appropriately for various terms and abbreviations used in these academic regulations/ norms.

3.2.1 Semester Scheme

The undergraduate programme is of four academic years and there shall be two semesters in each academic year. There shall be a minimum of 15 weeks of instruction, excluding the mid- term and semester-end exams. Around 15 instruction hours, 30 instruction hours and 45 hours of learning need to be followed per one credit of theory course, practical course and project/field-based learning respectively. In each semester, there shall be 'Continuous Internal Evaluation (CIE)' and 'Semester End Examination (SEE)' under Choice Based Credit System (CBCS) and credit-based semester system (CBSS) indicated by UGC. The curriculum/course structure suggested by AICTE are followed.

3.2.2 Credit Courses

All courses offered in each semester are to be registered by the student. Against each course in the course structure, the L: T: P: C (lecture periods: tutorial periods: practical periods: credits) pattern has been defined.

- One credit is allocated for one hour per week in a semester for lecture (L) or Tutorial (T) session.
- One credit is allocated for two hours per week in a semester for Laboratory/ Practical (P) session.
- One credit is allocated for three hours per week in a semester for Project/Mini-Project session.

For example, a theory course with three credit weightage requires three hours of classroom instruction per week, totaling approximately 45 hours of instruction over the entire semester.

3.2.3 Subject Course Classification

All subjects/courses offered for the undergraduate programme in E&T (B.Tech. degree programmes) are broadly classified as follows.

S. No.	Broad Course Classification	Course Group/ Category	Course Description
1	Foundation Courses (FnC)	BS – Basic Sciences	Includes Mathematics, Physics and Chemistry courses
2		ES - Engineering Sciences	Includes Fundamental Engineering Courses
3		HS – Humanities and Social Sciences	Includes courses related to Humanities, Social Sciences and Management
4	Core Courses (CoC)	PC – Professional Core	Includes core courses related to the parent branch of Engineering.
5	Elective Courses (E&C)	PE – Professional Electives	Includes elective courses related to the parent branch of Engineering.
6		OE – Open Electives	Elective courses which include inter-disciplinary courses or courses in an area outside the parent branch of Engineering.
7	Project Core	Project Work	B.Tech. Project Work
7	Other Core Courses (OCC)	Industry Training/ Internship/ Industry Oriented Mini-project/Skill Development Courses	Industry Training/ Internship/ Industry Oriented Mini-Project/Skill Development Courses
8			
9		Seminar	Seminar based on core contents related to parent branch of Engineering.
10	Skill Development Courses (SDC)	-	Courses designed to help individuals gain, improve, or refine specific skills
11	Value Added Courses (VAC)	-	Courses to build professional values, traditional knowledge and sensitization of societal issues

4.0 Mandatory Induction Programme

An induction program of one week duration for the UG students entering the institution, right at the start shall be implemented. Normal classes commence only after the induction programme is conducted. Following activities could be part of the induction programme: i) Physical Activity,

ii) Creative Arts, iii) Imparting Universal Human Values, iv) Literary Activities, v) Lectures by Eminent People, vi) Visits to Local Areas and vii) Familiarization to department as well as entire institute and viii) Making students understand Innovative practices at the college premises etc.

5.0 Course Registration

5.1 A faculty advisor / mentor shall be assigned to a group of around 20 students, who will advise the students about the undergraduate programme, its course structure and curriculum, choices/options of the courses, based on their competence, progress, pre-requisites and interest.

5.2 The academic section of the college invites 'registration forms' from students before the beginning of the semester through 'on-line registration', ensuring 'date and time stamping'. The online registration requests for semester courses shall be completed two weeks before the commencement of SEEs (Semester End Examinations) of the preceding semester.

5.3 A student can apply for **on-line** registration, **only after** obtaining the '**written approval**' from faculty advisor/mentor, which should be submitted to the college academic section through the Head of the Department. A copy of it shall be retained with the Head of the Department, faculty advisor/ mentor and the student.

5.4 A student shall register for all the courses offered in a semester as specified in the course structure.

5.5 Course options exercised through **on-line** registration are final and **cannot** be changed; further, alternative choices also will not be considered. However, if the course that has already been listed for registration by the Head of the Department in a semester could not be offered due to any inevitable or unexpected reasons, then the student shall be allowed to have alternative choice either for a new course (subject to offering of such a course), or for another existing course. Such alternative arrangements will be made by the Head of the Department, with due notification and time-framed schedule, within **a week**, but before the commencement of class- work of the semester.

5.6 The Head of the Department / Course Coordinator should review vacant slots in the timetable of each section once in every week or fortnight. The vacant slots in the time-table may be allocated to the subject teachers who could not take classes in proportion to the number of weeks completed from the commencement of the semester.

5.7 Two faculty members may be allocated for the tutorial session of Mathematics-1 course for better interaction/practice and to minimize the failures in the subject.

5.8 **Professional Electives**: The students have to choose six Professional Electives (PE-I to PE- VI) from the six baskets of professional electives given. Students have the flexibility to choose from the list of professional electives offered by the Institute or opt to register for the equivalent Massive Open Online Courses (MOOCs) as listed from time to time by the Institution.

5.9 **Open Electives**: Students have to choose three Open Electives (OE-I, II & III) from three baskets of Open Electives given by other than the parent department. However, the student can opt for an Open Elective course offered by his/her parent (Own) department, if the student has not studied that course so far. Similarly, Open Elective courses being studied should not match any courses of the forthcoming semesters.

5.10 Provision for Early Registration of MOOCs:

For a professional elective in a semester, students are allowed to register for an equivalent NPTEL/MOOCs course listed from time to time by the Institution one semester in advance. For example, a Professional Elective of III Year II Sem shall be allowed to register under NPTEL/MOOCs platform in III year I Sem.

The credits earned in one semester in advance can be submitted in the subsequent semester for the assessment. The students who have registered in advance in an equivalent NPTEL/MOOCs course and fail to secure any pass grade in the NPTEL/MOOCs course, can register for the regular course offered in the following semester of their course structure.

5.11 **Conversion of Marks Secured in** NPTEL/MOOCs **into Grades**: Marks secured in the internal and external evaluations of a NPTEL/MOOCs course shall be scaled to 40 and 60 marks respectively. The sum of these two components shall be considered as the total marks out of 100. The corresponding grade shall then be determined as per the marks-to-grades conversion rules specified in Clause 10.3.

5.12 NPTEL/MOOCs are allowed only for professional elective courses and for a few Minors & Honors courses

5.13 Additional learning resources:

Students are encouraged to acquire additional course-related knowledge by auditing learning resources from NPTEL/MOOCs platforms for each course offered in their course structure. These additional courses are not meant for earning credits but are intended to enhance knowledge. The Institution shall notify such courses from time to time through their portals for the benefit of students. They are categorized into three types: prerequisite, reinforcement, and aspirational. Prerequisite courses help students gain familiarity and provide sufficient background. Reinforcement courses aim to offer different perspectives on learning, while aspirational courses focus on next-level or advanced learning.

6.0 Rules to offer Elective courses

6.1 An elective course may be offered to the students, only if a minimum of 25% of class strength opts for it.

6.2 Same elective course for different sections may be offered by different faculty members. The selection of elective course by students will be based on first come first serve and / or CGPA criterion.

6.3 If the number of students registrations are more than the strength of one section, then it is choice of the concerned Department to offer the same course for more than one section based on the resources available in the department.

7.0 Attendance requirements:

7.1 A student shall be eligible to appear for the semester-end examinations, if the student acquires a minimum of 75% of aggregate attendance of all the courses for that semester.

7.2 Shortage of attendance in aggregate upto 10% (securing 65% and above but below 75%) in each semester may be condoned by the college academic committee on genuine and valid grounds, based on the student's representation with supporting evidence.

7.3 A stipulated fee shall be payable for condoning of shortage of attendance as notified in the respective college websites.

7.4 **Two hours** of attendance for each theory course shall be considered, if the student appears for the mid-term examination of that course.

7.5 Shortage of attendance below 65% in aggregate shall in **no** case be condoned.

7.6 Students whose shortage of attendance is not condoned in any semester, are not eligible to take their semester-end examinations of that semester. They get detained and will not be promoted to the next

semester. Their registration for that semester shall stand cancelled, including internal marks. They may seek re-registration(re-admission) for that semester in the next academic year.

8.0 A student fulfilling the attendance requirement in the present semester shall not be eligible for readmission into the same semester Criteria for Earning of Credits in a Course

8.1 A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each course, if the student secures not less than 35% (21 marks out of 60 marks) in the semester end examinations (SEE), and a minimum of 40% (40 marks out of 100 marks) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together; in terms of letter grades, this implies securing **'C'** grade or above in that course.

8.2 A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to Field Based Research Project / Industry Oriented Mini Project / Internship, if the student secures not less than 40% marks (i.e. 40 out of 100 allotted marks) in each of them. The student is deemed to have failed, if he/she (i) does not submit a report on Field-Based Research Project/Industry Oriented Mini Project/ Internship, or (ii) not make a presentation of the same before the evaluation committee as per schedule, or (iii) secures less than 40% marks in Field-Based Research Project / Industry Oriented Mini Project / Internship evaluations.

8.3 A student eligible to appear in the semester-end examination for any course, is absent from it or failed (thereby failing to secure **'C'** grade or above) may re-appear for that course in the supplementary examination as and when it is conducted. In such cases, internal marks assessed in continuous internal evaluation (CIE) earlier for that course will be carried over, and added to the marks obtained in the SEE supplementary/make-up examination. If the student secures sufficient marks for passing, **'C'** grade or above shall be awarded as specified in clause 10.3.

9.0 Distribution of Marks and Evaluation

9.1 The performance of a student in every course (including Value Added Courses and Skill Development Courses, Laboratory/Practical and Project Work) will be evaluated for 100 marks each, with 40 marks allotted for CIE (Continuous Internal Evaluation) and 60 marks for SEE (Semester End-Examination), irrespective of the credits allocated.

9.2 Continuous Internal Evaluation (CIE)

9.2.1 Theory Courses:

For theory courses, during a semester, there shall be two mid-term examinations. Each Mid-Term examination consists of two parts i) **Part – A** for 10 marks, ii) **Part – B** for 20 marks, totaling to 30 marks. 30 marks. Total duration of mid-term examination is two hours.

1. Mid Term Examination for 30 marks:
   a. Part - A : Objective/quiz paper for 10 marks.
   b. Part - B : Descriptive paper for 20 marks.

The objective/quiz paper is set with multiple choice, fill-in the blanks and match the following type of questions for a total of 10 marks.

The descriptive paper shall contain 6 questions out of which, the student has to answer 4 questions, each carrying 5 marks. The **average of the two Mid Term Examinations** shall be taken as the final marks for Mid Term Examination (for 30 marks).

While the first mid-term examination shall be conducted on 50% of the syllabus, the second mid-term examination shall be conducted on the remaining 50% of the syllabus. Questions will be drawn from the mid-term exam syllabus, ensuring uniform coverage of all topics.

The remaining 10 marks of Continuous Internal Evaluation are distributed as follows:

2. Five marks for the assignment for 5 marks. Student shall submit two assignments and the **average of 2 Assignments** each for 5 marks shall be taken. The first assignment should be submitted before the conduct of the first mid-term examination, and the second assignment should be submitted before the conduct of the second mid-term examination.
3. Five marks for the Viva-Voce/PPT/Poster Presentation/ Case Study on a topic in the concerned subject. This assessment shall be completed before II Mid-Term Examination. The Principals shall schedule these sessions in their semester plan.

9.2.2 Engineering Drawing and Computer Aided Drafting Course:

For this course, 20 marks will be allocated for day-to-day assessments conducted during drawing practice sessions**,** and another 20 marks will be allocated for the mid-term examination**.** In the mid-term examination, students shall attempt any four out of six given questions**.** The first mid-term exam will be conducted in the conventional mode using a drawing board**,** while the second mid-term exam will be conducted using a CAD package**.**

9.3 A **Computer-Based Test (CBT**) in each course is available for students who either:

1. missed one of the two mid-term examinations due to unavoidable circumstances, or
2. attended both mid-term examinations but wish to improve their internal marks

The CBT will be conducted at the end of the semester and will carry a total of 30 marks. The marks obtained in the CBT will be considered equivalent to those obtained in one mid-term examination. Zero marks will be awarded to students who are absent from the mid-term examination. The average of the best two scores from the three exams (the two mid-term exams and the CBT), combined with other internal assessment components, will constitute the Continuous Internal Improvement (CII) marks for that specific course. CBT exams shall be conducted by the University.

9.4 Semester End Examination for theory courses

9.4.1 Theory Courses:

The semester end examinations (SEE), for theory courses, will be conducted for 60 marks consisting of two parts viz. i) **Part- A** for 10 marks and ii) **Part - B** for 50 marks.

- **Part-A** is compulsory, consists of five short answer questions covering all units of syllabus; each question carries two marks.
- **Part-B** consists of five questions carrying 10 marks each. There shall be two questions asked in the question paper from each unit with either-or choice and the student should answer either of the two questions. The student shall answer one question from each of five units.

9.4.2 Engineering Drawing and Computer Aided Drafting Course:

Question paper consists of five questions carrying 12 marks each. There shall be two questions asked in the question paper from each unit with either-or choice and the student should answer either of the two questions. The student shall answer one question from each of five units. There shall be no section with short answer questions.

9.4.3 Duration of SEE:

The duration of Semester End Examination of theory and drawing courses is 3 hours.

9.5 Semester End Examination for Practical Courses

For practical courses there shall be a **Continuous Internal Evaluation (CIE)** during the semester for 40 marks and semester-end examination for 60 marks.

The breakup of the continuous internal evaluation for 40 marks is as follows:

1. 10 marks for a write-up on day-to-day experiments in the laboratory (in terms of aim, components/procedure, expected outcome).
2. 10 marks for viva-voce (or) tutorial (or) case study (or) application (or) poster presentation of the course concerned.
3. 10 marks for the internal practical examination conducted by the laboratory teacher concerned.
4. The remaining 10 marks are for Laboratory Record Report/Project and Presentation, which consists of the Design (or) Software / Hardware Model Presentation (or) App Development (or) Prototype submission which shall be evaluated after completion of laboratory course and before semester end practical examination.

The **Semester End Examination** for practical courses shall be conducted with an external examiner and the laboratory course teacher. The external examiner shall be appointed from the inside or outside college and not from a group colleges.

In the Semester End Examination for practical courses held for 3 hours, rubrics of evaluation for 60 marks is as given below:

1. 10 marks for write-up/ programe
2. 15 for experiment/program
3. 15 for evaluation of results or excution
4. 10 marks for presentation on another experiment/program in the same laboratory course and
5. 10 marks for viva-voce on concerned laboratory course.

For any change of experiment, 5 marks will be deducted from the total of 60 marks. If second time change of experiment is requested, another five marks will be deducted from the 60 marks. No third change will be permitted.

9.6 Field-based Research Project:

There shall be a Field-based Research Project in the intervening summer between II-II and III- I Semesters. Students will register for this project immediately after II Year II Semester examinations and pursue it during summer vacation. The Field-based Research Project shall be submitted in a report form and presented before the committee in III year I semester. It shall be evaluated for 100 external marks. The evaluation committee shall consist of an External Examiner, Head of the Department, Supervisor of the Project and a Senior Faculty Member of the department. There shall be no internal marks for Field-based Research Project. Student shall have to earn 40% marks, i.e 40 marks out of 100 marks. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the committee as per schedule, or (iii) secures less than 40% marks in this course.

9.7 Internship/Industry Oriented Mini Project:

There shall be an Internship/Industry Oriented Mini Project in collaboration with an industry from their specialization. Students shall register for this project immediately after III Year II Semester Examinations and pursue it during summer vacation. Internship should be carried out at an organization (or) Industry. The Industry Oriented Mini Project shall be submitted in a report form and presented before the committee in IV Year I Semester before the semester end examination. It shall be evaluated for 100 external marks. The committee consists of an External Examiner, Head of the Department, Supervisor of the Industry Oriented Mini Project/Internship, and a Senior Faculty Member of the Department.

9.7.1 For evaluating industry-oriented mini-projects, it is preferable to appoint an external examiner from the industry, ideally from one of the organizations/ industries with which the institute has established

/ proposing to establish collaborations.

9.8 UG Project Work:

9.8.1 The UG project work shall be initiated at the beginning of the IV Year II Semester and the duration of the project work is one semester. The student must present in consultation with his/her supervisor, the title, objective and plan of action of his/her Project work to the departmental committee for approval within two weeks from the commencement of IV Year II Semester. Only after obtaining the approval of the departmental committee, the student can start his/her project work.

9.8.2 Student has to submit project work report at the end of IV Year II Semester. The project work shall be evaluated for 100 marks. Out of which 40 marks and 60 marks are allocated for CIE and External Evaluation respectively.

9.8.3 For internal evaluation, the departmental committee consisting of Head of the Department, Project Supervisor and a Senior Faculty Member shall evaluate the project work for 40 marks. The distribution of marks is as follows:

- Objective(s) of the work done - 05 Marks
- Methodology adopted - 15 Marks
- Results and Discussions - 15 Marks
- Conclusions and Outcomes - 05 Marks

Total - 40 Marks

9.8.4 The External Evaluation shall be conducted by the external examiner for a total of 60 marks.

- the presentation of the work - 20 Marks
- communication skills - 15 Marks
- viva-voce - 25 Marks

Total - 60 Marks

The topics for main Project shall be different from the topic of Industry Oriented Mini Project/ Internship/SDC. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the External Examiner as per schedule, or (iii) secures less than 40% marks in the sum total of the CIE and SEE taken together.

9.8.5 For conducting viva-voce exam of project work, the institute appoints an external examiner. The external examiner may be selected from the list of experts submitted by the HoD.

9.8.6 A student who has failed, may re-appear once for the above evaluation, when it is scheduled again; if student fails in such 'one re-appearance' evaluation also, he/she has to appear for the same in the next subsequent year, as and when it is scheduled.

9.9 Skill Development Courses:

Four Skill Development Courses are included in the Curriculum in II-I, II-II, III-I and III-II semesters. Each Skill Development Course carries one credit. The evaluation pattern will be same as that of a laboratory course including the internal and external assessments.

The objective of Skill Courses is to develop the cognitive skills as well as the psycho-motor skills.

9.10 Value-Added Courses:

The evaluation of Value-Added Courses shall be similar to that of theory courses. However, the scheduling of these mid-term exams and semester-end examinations may not be combined with main-stream examinations. One hour /45 mins proctored mid-term examination shall be conducted in the regular class by the same subject teacher. It should not impact the conduct of other classes on that day.

The scheduling of the semester-end examinations shall also be intimated by the University time to time.

10.0 Grading Procedure

10.1 Absolute grading system is followed for awarding the grades to each course.

10.2 Grades will be awarded to indicate the performance of students in each Theory, Laboratory, Industry-Oriented Mini Project/ Internship/ Skill development course and Project Work. Based on the percentage of marks obtained (Continuous Internal Evaluation plus Semester End Examination, both taken together) as specified in clause 8 above, a letter grade shall be given as explained in the following clause.

10.3 To measure the performance of a student, a 10-point grading system is followed. The mapping between the percentage of marks secured and the corresponding letter grade is as follows:

Range of % of Marks Secured in a Course	Letter Grade	Grade Points (GP)
Greater than or equal to 90	O (Outstanding)	10
80 and less than 90	A^+ (Excellent)	9
70 and less than 80	A (Very Good)	8
60 and less than 70	B^+ (Good)	7
50 and less than 60	B (Average)	6
40 and less than 50	C (Pass)	5
Below 40	F (FAIL)	0
Absent	Ab	0

10.4 A student shall be declared successful or 'passed' in a semester, if he/she secures **'C'** grade or above in every course (ie GP ≥ 5)

10.5 A student who has obtained an '**F**' grade in any course shall be deemed to have '**failed**' and is required to re-appear for a supplementary exam as and when conducted. In such cases, internal marks in those courses will remain the same as those obtained earlier.

10.6 To a student who has not appeared for an examination in any course, '**Ab**' grade will be allocated in that course, and he/she is deemed to have '**Failed**'. Such student will be required to re-appear for supplementary/make-up exam as and when conducted. The internal marks in those courses will remain the same as those obtained earlier.

10.7 The students earn a Grade Point (G) in each course, on the basis of letter grade secured in that course. Every student who passes a course will receive grade point **GP >= 5** ('C' grade or above).

10.8 The 'Credit Points' (C) are computed by multiplying the grade point with credits for a given course.

Credit Points (C) = Grade Point (G) x Credits

10.9 The Semester Grade Point Average (SGPA) is calculated only when all the courses offered in a semester are cleared by a student. It is calculated by dividing the sum of credit points (ΣCG) secured from all courses registered in a semester, by the total number of credits registered during that semester. SGPA is rounded off to **two** decimal places. SGPA for each semester is thus computed as

$$\text{SGPA} = \{ \sum_{i=1}^{N} C_i G_i \} / \{ \sum_{i=1}^{N} C_i \}$$

where 'i' is the course indicator index (considering all courses in a semester), 'N' is the no. of courses registered for the semester (as listed under the course structure of the branch), C_i is the no. of credits allotted to the ith course, and G_i represents the grade points corresponding to the letter grade awarded for that i^{th} course.

10.10 If a student earns more than 160 credits, only the courses corresponding to the best 160 credits shall be considered for the computation of CGPA of B.Tech. degree.

10.11 The Cumulative Grade Point Average (CGPA) is a measure of the overall cumulative performance of a student in all semesters considered for registration. The CGPA is the ratio of the total credit points secured by a student for the courses correspond to best 160 credits out of **all** registered courses in **all** semesters, and the total number of credits correspond to those selected courses. CGPA is rounded off to **two** decimal places. CGPA is thus computed at the end of each semester, from the I year II semester onwards, as per the formula where '**M'** is the total no. of courses corresponding to the best 160 credits from the courses registered in all eight semesters, 'j' is the course indicator index (takes into account all courses from 1 to 8 semesters), C_j is the no. of credits allotted to the jth course, and G_j represents the grade points (GP) corresponding to the letter grade awarded for that j^{th} course.

Illustration of the Calculation of SGPA:

Course	Credits	Letter Grade	Grade Points	Credit Points
Course 1	4	A	8	4 x 8 = 32
Course 2	3	O	10	3 x 10 = 30
Course 3	3	C	5	3 x 5 = 15
Course 4	3	B	6	3 x 6 = 18
Course 5	3	A	8	3 x 8 = 24
Course 6	2	A+	9	2 x 9 = 18
Course 7	1	C	5	1 x 5 = 5
Course 8	1	O	10	1 x 10 = 10
	20			152

SGPA = 152/20 = 7.6

The CGPA of the entire B.Tech. programme shall be calculated considering the best 160 credits earned by the student.

10.12 For merit ranking or comparison purposes or for any other listing, **only** the '**rounded off'** values of the CGPAs will be used.

10.13 SGPA of a semester will be mentioned in the semester Memorandum of Grades if all courses of that semester are cleared in first attempt. Otherwise, the SGPA shall be mentioned only on the Memorandum of Grades in which sitting he passed his last exam in that semester.

11.0 Declaration of Results and issue of Grade Memo

11.1 While declaring the results, the web-version should display the marks earned by the students with the internal and external marks break-up. However, in the memorandum of grades, the marks need not be shown.

11.2 After the completion of each semester, a certificate of memorandum of grades shall be issued to all the registered students, indicating the letter grades and credits earned. It will show the details of the courses registered (course code, course title, no. of credits), letter grade and credits earned.

12.0 Withholding of Results

12.1 If the student has not paid the fees to the University at any stage, or has dues pending due to any reason whatsoever, or if any case of indiscipline is pending, the result of the student may be withheld, and the student will not be allowed to go into the next higher semester. The award or issue of the degree may also be withheld in such cases.

13.0 Supplementary Examinations:

13.1 At the end of each semester, along with regular semester examinations, supplementary examinations shall be conducted for the students who have back-log subjects. Advanced supplementary examinations in IV Year II Semester courses may be conducted for those who failed in any course offered in IV Year II Semester. It may enable the students to receive their B.Tech. provisional certificate at an early date. Advanced supply examinations may be scheduled within one month period after the declaration of the final semester results. There shall be no supplementary examination in the successive semester. The students who could not secure any pass grade in advance supplementary examinations have to wait for regular series examination of next batch to write their back-log examination.

14.0 Promotion Rules

S.No.	Promotion	Conditions to be Fulfilled
1	First year first semester to first year second semester	Regular course of study of first year first semester and fulfilment of attendance requirement.
2	First year second semester to Second year first semester	(i) Regular course of study of first year second semester and fulfilment of attendance requirement (ii) Must have secured at least 25% of the total credits up to first year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
3.	Second year first semester to Second year second semester	Regular course of study of second year first semester and fulfilment of attendance requirement.
4	Second year second semester to Third year first semester	(i) Regular course of study of second year second semester and fulfilment of attendance requirement. (ii) Must have secured at least 25% of the total credits up to second year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
5	Third year first semester to Third year second semester	Regular course of study of third year first semester and fulfilment of attendance requirement.
6	Third year second semester to Fourth year first semester	Regular course of study of third year second semester and fulfilment of attendance requirement.
7	Fourth year first semester to Fourth year second semester	Regular course of study of fourth year first semester and fulfilment of attendance requirement.

15.0 Re-admission after Detention

i) A student detained due to lack of credits, shall be promoted to the next academic year only after acquiring the required number of credits.

ii) A student detained due to shortage of attendance shall be admitted in the same semester in the successive academic years.

iii) When a student is readmitted in the following academic years, the academic regulations under which the student seeks re-admission shall only be applicable to this student, not the academic regulations in which he got admitted in his/her first year of study.

16.0 Credit Exemption

A student (i) shall register for all courses covering 164 credits as specified and listed in the course structure and (ii) earn 160 or more credits to successfully complete the undergraduate programme.

- Best 160 credits shall be considered for CGPA computation. The student can avail exemption of courses ***totaling up to 4 credits*** other than Professional core courses, Laboratory Courses, Seminars, Project Work and Field Based Research Project / Industry Oriented Mini Project / Internship, for optional drop out from these 164 credits registered;
- The semester grade point average (SGPA) of each semester shall be mentioned at the bottom of the grade card, when all the subjects in that semester have been passed by the student.
- Credits earned by the student in either a Minor or Honors program cannot be counted towards the required 160 credits for the award of the B.Tech. degree

17.0 Award of Degree

17.1 A student who registers for all the courses specified in the course structure and secures the required number of 160 credits within 8 academic years from the date of commencement of the first academic year, shall be declared to have qualified for the award of B.Tech. degree in the branch of Engineering selected at the time of admission.

17.2 A student who qualifies for award of the degree as listed in item 17.1 shall be placed in the following classes.

17.3 A student with final CGPA (at the end of the undergraduate programme) ≥ 7.5, and fulfilling the following conditions - shall be placed in '**First Class with Distinction**':

(i) Should have passed all the courses in '**First Appearance**'

(ii) Should not have been detained or prevented from writing the semester end examinations in any semester due to shortage of attendance or any other reason.

A student not fulfilling any of the above conditions with final CGPA **≥** 7.5 shall be placed in **'First Class'.**

17.4 Students with final CGPA (at the end of the undergraduate programme) >= 6.5 but < 7.5 shall be placed in **'First Class'.**

17.5 Students with final CGPA (at the end of the undergraduate programme) >= 5.5 but < 6.5, shall be placed in '**Second Class**'.

17.6 All other students who qualify for the award of the degree (as per item 17.1), with final CGPA (at the end of the undergraduate programme) >= 5.00 but < 5.5, shall be placed in '**pass class**'.

17.7 Grace Marks

Grace marks shall be given to those students who complete the course work of four-year B. Tech. degree, not secured pass grade in not more than three subjects and adding a specified grace marks enables the student to pass the subject(s) as well as gets eligibility to receive the provisional degree certificate.

Grace marks for students admitted under the R-25 Academic Regulations should not exceed **0.15%** of

the total maximum marks in all eight semesters (excluding the marks allocated for value added courses and skill development courses).

18.0 Award of Gold Medals

18.1 Students fulfilling the conditions listed under item 17.3 alone will be eligible for award of '**Gold Medal**'.

18.2 If more than one student secures the same highest CGPA, then the following tie resolution criteria, in the same order of preference shall be followed for selecting the Gold Medal winner, until the tie is resolved: 1) more number of times secured highest SGPAs, ii) more number of O and A+ grades in that order and iii) highest SGPA in the order of first semester to eight semester.

19.0 Conversion of CGPA into equivalent Percentage of Marks

19.1 The following formula shall be used for the conversion of CGPA into equivalent marks, whenever it is necessary

Percentage (%) of Marks = (Final CGPA – 0.5) x 10

20.0 Transitory Regulations for the students re-admitted in R-25 Regulations:

20.1 Transitory regulations are applicable to the students detained due to shortage of attendance as well as detained due to the shortage of credits and seek permission to re-join the B.Tech. programme, where R-25 regulations are in force.

20.2 A student detained due to shortage of attendance and re-admitted in R-25 regulations: Such students shall be permitted to join the same semester, but in R-25 Regulations

20.3 A student detained due to shortage of credits and re-admitted in R-25 regulations: Such students shall be promoted to the next semester in R-25 regulations, only after acquiring the required number of credits as per the corresponding regulations of his/her previous semester.

20.4 A student who has failed in any course in a specific regulation has to pass those courses in the same regulations.

20.5 If a student is readmitted to R-25 Regulations and has any course with 80% of syllabus common with his/her previous regulations, that particular course in R-25 Regulations will be substituted by an equivalent course of R-22 regulations by the Institute. All these details are summarized in a set of look-up Table; one set for each B. Tech. branch.

20.6 Look Up Table of equivalence courses

20.6.1 A lookup table will be provided for the benefit of students and Principals. This lookup table will include all the courses to be registered by students who have been re-admitted under the R-25 Academic Regulations from the R-22 Academic Regulations. Separate lookup tables will be provided for the following categories of students:

1. Students re-admitted into the I Year II Semester of the R-25 Regulations
2. Students re-admitted into the II Year I Semester of the R-25 Regulations
3. Students re-admitted into the II Year II Semester of the R-25 Regulations
4. Students re-admitted into the III Year I Semester of the R-25 Regulations
5. Students re-admitted into the III Year II Semester of the R-25 Regulations
6. Students re-admitted into the IV Year I Semester of the R-25 Regulations
7. Students re-admitted into the IV Year II Semester of the R-25 Regulations

For every B.Tech. branch there shall be separate set of seven lookup tables.

20.6.2 Applicability of Look-up Table: The above look-up table shall be applicable for i) students who seek

re admission from R-22 regulations to R-25 regulation and are going to be re-admitted in the same JNTUH affiliated autonomous college and ii) detained students of one JNTUH affiliated autonomous college who seek admission into another JNTUH affiliated autonomous college.

For the above category of students, the principals of the affiliated colleges need not consult the University for the equivalence courses. However the Principals need to inform in the specified format, the list of such students and equivalences derived from the transitory regulations.

20.6.3 These look-Up Tables are not applicable for the students who seek transfer from i) other Universities to JNTUH affiliated autonomous colleges, ii) autonomous to non-autonomous colleges, III) one autonomous to another autonomous colleges and iv) non-autonomous to autonomous colleges under JNTUH. Such students should consult the University regarding equivalent courses, as was in previous practice.

20.7 The R-25 Academic Regulations are applicable to a student from the year of re-admission.

However, the student is required to complete the study of B.Tech. degree within the stipulated period of eight academic years from the year of first admission.

21.0 Student Transfers

21.1 There shall be no branch transfers after the completion of admission process.

21.2 There shall be no transfers from one college to another within the constituent colleges and units of Jawaharlal Nehru Technological University Hyderabad.

21.3 The students seeking transfer to colleges affiliated to JNTUH from various other Universities/institutions is having back-logs at the previous University/institute, have to pass the courses offered at JNTUH which are equivalent to the failed courses at the previous University/institute.

21.4 The transferred students from other Universities/Institutions to JNTUH affiliated colleges, shall be given a chance to write CBTs for getting CIE component in the **equivalent course(s)** as per the clearance letter issued by the University.

22.0 Value Added Courses

22.1 Faculty members who have received a certificate in Innovation and Entrepreneurship / Entrepreneurship from a reputed foundation/organization may be given preference to teach the "Innovation and Entrepreneurship" course. This certificate course should include an assessment. Total training duration (online or physical), excluding assessment, should be at least 30 hours. Faculty members from all disciplines with innovative mindset and aptitude to co-create an entrepreneurial ecosystem are eligible to teach this subject.

22.2 Faculty members who have credited a course on Intellectual Property Rights in their UG or PG programme or credited an equivalent course in MOOCs platform/ reputed foundation/ organization in which assessment is a part, may be given preference to teach the elective course on Intellectual Property Rights.

22.3 To ensure quality delivery and standardization in teaching the **Indian Knowledge System (IKS)** and other value-added courses, the following guidelines must be adhered to: i) faculty members must undergo a Faculty Development Program (FDP) organized by UGC-MMTTC (Malaviya Mission Teacher Training Centre), **or** Any other recognized and competent institution/organization offering similar certified programs, ii) the total instructional duration of the FDP should be a around 32 hours or more, III) all sessions in the FDP must be conducted by certified and qualified resource persons with recognized expertise in the respective domains,
iv) A formal assessment component must be included as part of the FDP.

23.0 Mapping with the Sustainable Development Goals

All the courses specified in the course structure of every programme are mapped with the one or

more sustainable development goals.

24.0 Scope

24.1 The academic regulations should be read as a whole, for the purpose of any interpretation.

24.2 In case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Principal is final.

24.3 The institution may change or amend the academic regulations, course structure or syllabi at any time, and the changes or amendments made shall be applicable to all students with effect from the dates notified by the principal.

24.4 Where the words “he”, “him”, “his”, occur in the regulations, they include “she”, “her”, “hers”.

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ANURAG
Engineering Engineers

ANURAG ENGINEERING COLLEGE

(An Autonomous Institution)

Ananthagiri (V&M), Kodad, Suryapet(Dt), Telangana (India)

ACADEMIC REGULATIONS FOR B.TECH. (LATERAL ENTRY SCHEME) FROM THE AY 2026-27

Eligibility for the award of B.Tech. Degree (LES)

1. The LES students after securing admission shall pursue a course of study for not less than three academic years and not more than six academic years.
2. The student shall register for 123/124 credits and secure 120 credits with CGPA ≥ 5 from II year to IV-year B.Tech. programme (LES) for the award of B.Tech. degree.
3. The student can avail exemption of courses ***totaling up to 3/4 credits*** other than Professional core courses, Laboratory Courses, Seminars, Project Work and Field Based Research Project / Industry Oriented Mini Project / Internship, for optional drop out.
4. The students, who fail to fulfil the requirement for the award of the degree in six academic years from the year of admission, shall forfeit their seat in B.Tech.
5. The attendance requirements of B.Tech. (Regular) shall be applicable to B.Tech. (LES).
6. **Promotion rule**

S. No	Promotion	Conditions to be fulfilled
1	Second year first semester to Second year second semester	Regular course of study of second year first semester and fulfilment of attendance requirement.
2	Second year second semester to Third year first semester	(i) Regular course of study of second year second semester and fulfilment of attendance requirement. (ii) Must have secured at least 25% of the total credits up to second year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
3	Third year first semester to Third year second semester	Regular course of study of third year first semester and fulfilment of attendance requirement.
4	Third year second semester to Fourth year first semester	Regular course of study of third year second semester and fulfilment of attendance requirement.
5	Fourth year first semester to Fourth year second semester	Regular course of study of fourth year first semester and fulfilment of attendance requirement.

7. All the other regulations as applicable to B.Tech. 4-year degree course (Regular) will hold good for B. Tech. (Lateral Entry Scheme).
8. LES students are not permitted to exit the B.Tech. program after completion of second year (B.Tech. II Year II Semester).

ANURAG ENGINEERING COLLEGE

(An Autonomous Institution)

Department of Electronics and Communication Engineering

COURSE STRUCTURE and SYLLABUS (R25).

I YEAR I SEMESTER COURSE STRUCTURE

S.No.	Course Code	Course Title	L	T	P	Credits
1	2599101	Matrices and Calculus	3	1	0	4
2	2599106	Engineering Physics	3	0	0	3
3	2505103	C Programming & Data Structures	3	0	0	3
4	2503101	Computer Aided Engineering Graphics	2	0	2	3
5	2502103	Basic Electrical Engineering	3	0	0	3
6	2599107	Engineering Physics Lab	0	0	2	1
7	2505104	C Programming & Data Structures Lab	0	0	2	1
8	2502104	Basic Electrical Engineering Laboratory	0	0	2	1
9	2505105	IT Workshop	0	0	2	1
10		Induction Program				
Total			**14**	**1**	**10**	**20**

I YEAR II SEMESTER COURSE STRUCTURE

S.No.	Course Code	Course Title	L	T	P	Credits
1	2599201	Ordinary Differential Equations and Vector Calculus	3	0	0	3
2	2599204	Engineering Chemistry	3	0	0	3
3	2599205	English for Skill Enhancement	3	0	0	3
4	2504201	Electronic Devices and Circuits	3	0	0	3
5	2504202	Network Analysis and Synthesis	3	1	0	4
6	2599206	Engineering Chemistry Lab	0	0	2	1
7	2503202	Engineering Workshop	0	0	2	1
8	2599207	English Language and Communication Skills Lab	0	0	2	1
9	2504203	Electronic Devices and Circuits Lab	0	0	2	1
Total			**15**	**1**	**8**	**20**

T-Tutorial P-Practical D -Drawing L- Lectures C-

II YEAR I SEMESTER COURSE STRUCTURE

S.No.	Course Code	Course Title	L	T	P	Credits
1	2504303	Probability Theory and Stochastic Processes	3	0	0	3
2	2504304	Signals and Systems	3	0	0	3
3	2504305	Digital Logic Design	3	0	0	3
4	2504306	Electronic Circuit Analysis	3	0	0	3
5	2505301	Python Programming	3	0	0	3
6	2504307	Modelling and Simulation Lab	0	0	2	1
7	2504308	Digital Logic Design Lab	0	0	2	1
8	2504309	Electronic Circuit Analysis Laboratory	0	0	2	1
9	2505302	Python Programming Lab	0	0	2	1
10	2505303	Skill Development Course-1 Linux and Shell Scripting	0	0	2	1
11	2599304	Environmental Science / Indian Knowledge System	1	0	0	1
Total			**16**	**0**	**10**	**21**

T-Tutorial P-Practical D -Drawing L- Lectures C-Credits

II YEAR II SEMESTER COURSE STRUCTURE

S.No.	Course Code	Course Title	L	T	P	Credits
1	2599402	Numerical Methods and Complex Variables	3	0	0	3
2	2504403	Electromagnetic Fields and Transmission Lines	3	0	0	3
3	2504404	Analog and Digital Communications	3	0	0	3
4	2504405	Linear and Digital IC Applications	3	0	0	3
5	2504406	Control Systems	3	0	0	3
6	2588401	Innovation and Entrepreneurship	2	0	0	2
7	2599403	Computational Mathematics Lab	0	0	2	1
8	2504407	Analog and Digital Communications Laboratory	0	0	2	1
9	2504408	Linear and Digital IC Applications Laboratory	0	0	2	1
10	2504409	Skill Development Course – 2 Web and Mobile Applications	0	0	2	1
Total			**17**	**0**	**8**	**21**

T-Tutorial P-Practical D -Drawing L- Lectures C-Credits

III YEAR I SEMESTER COURSE STRUCTURE

S.No.	Course Code	Course Title	L	T	P	Credits
1		Digital Signal Processing	3	0	0	3
2		RISC and Microcontroller architectures	3	0	0	3
3		CMOS VLSI Design	3	0	0	3
4		Professional Elective-I	3	0	0	3
5		Open Elective-I	2	0	0	2
6		RISC and Microcontroller Interfacing Laboratory	0	0	2	1
7		CMOS VLSI Design Laboratory	0	0	2	1
8		Digital Signal Processing Laboratory	0	0	2	1
9		Field-based Research Project	0	0	4	2
10		FPGA based System Design Lab	0	0	2	1
11		Intellectual Property Rights	1	0	0	1
Total Credits			**15**	**0**	**12**	**21**

T-Tutorial **P-Practical** **D -Drawing** **L- Lectures** **C-Credits**

III YEAR II SEMESTER COURSE STRUCTURE

S.No.	Course Code	Course Title	L	T	P	Credits
1		Antenna Design and Wave Propagation	3	0	0	3
2		IoT Architectures and Protocols	3	0	0	3
3		Business Economics and Financial Analysis	3	0	0	3
4		Professional Elective-II	3	0	0	3
5		Open Elective – II	2	0	0	2
6		Advanced Communications Lab	0	0	2	1
7		IoT Architectures and Protocols Laboratory	0	0	2	1
8		VLSI Design Verification Laboratory	0	0	2	1
9		Advanced English Communication Skills Laboratory	0	0	2	1
10		45G Practical Lab/Robotic Lab/Drone Lab	0	0	2	1
11		Gender Sensitization Lab* / Human Values and Professional Ethics Lab*	1	0	0	1
Total Credits			**15**	**0**	**10**	**20**

T-Tutorial **P-Practical** **D -Drawing** **L- Lectures** **C-Credits**

IV YEAR I SEMESTER COURSE STRUCTURE

S.No.	Course Code	Course Title	L	T	P	Credits
1		Microwave and Optical Communications	3	0	0	3
2		Embedded System Design	3	0	0	3
3		Fundamentals of Management for Engineers	3	0	0	3
4		Professional Elective-III	3	0	0	3
5		Professional Elective – IV	3	0	0	3
6		Open Elective – III	2	0	0	2
7		Microwave and Optical Communications Laboratory	0	0	2	1
8		Embedded System Design Lab	0	0	2	1
9		Industry Oriented Mini Project/ Internship	0	0	4	2
Total Credits			**17**	**0**	**08**	**21**

T-Tutorial P-Practical D -Drawing L- Lectures C-Credits

IV YEAR II SEMESTER COURSE STRUCTURE

S.No.	Course Code	Course Title	L	T	P	Credits
1		Professional Elective – V	3	0	0	3
2		Professional Elective – VI	3	0	0	3
3		Project Work	0	0	28	14
Total Credits			**06**	**0**	**28**	**20**

T-Tutorial P-Practical D -Drawing L- Lectures C-Credits

LIST OF PROFESSIONAL ELECTIVES

III B. TECH I SEM

Professional Elective-I:		
S.NO	**COURSE CODE**	**COURSE NAME**
1		Sustainability for Electronics
2		CMOS Fabrication and Technology
3		Data Communications and Computer Networks
4		Computer Organization and Operating Systems
5		Introduction to Semiconductor Devices

III B. TECH II SEM

Professional Elective-II:		
S.NO	**COURSE CODE**	**COURSE NAME**
1		5G Communications
2		Electronic Measurements and Instrumentation
3		Low Power VLSI Design
4		Image and Video Processing
5		5G Wireless standard Design

IV B. TECH I SEM

Professional Elective-III:		
S.NO	**COURSE CODE**	**COURSE NAME**
1		Biomedical Signal and Image Processing
2		Wireless Communication Networks
3		Design for Testability
4		Unmanned Aerial Vehicles and Satellite Imaging

IV B. TECH I SEM

Professional Elective-IV:		
S.NO	**COURSE CODE**	**COURSE NAME**
1		Artificial Neural Networks and Deep Learning
2		Satellite Communications
3		Analog and Mixed Signal IC Design
4		Biomedical Instrumentation
5		Microsensors & Nano sensors

IV B. TECH II SEM

Professional Elective-V:		
S.NO	**COURSE CODE**	**COURSE NAME**
1		AI for Signal and Image Processing
2		Radar Systems
3		Intelligent e - Computer Aided Design
4		Network Security and Cryptography
5		Introduction to wireless and Cellular Communications

IV B. TECH II SEM

Professional Elective-VI:		
S.NO	**COURSE CODE**	**COURSE NAME**
1		DSP Processors and Architectures
2		Quantum Technologies
3		RF Circuit Design
4		Model Based System Engineering

LIST OF OPEN ELECTIVES

III B. TECH ECE I SEM OPEN ELECTIVE-I			
S. No	**Course Code**	**Subject**	**Offering Department**
1		Principles of Communication	Electronics & Communication Engineering
2		Fundamentals of Cyber Physical Systems	
III B. TECH ECE II SEM OPEN ELECTIVE- II			
S.No.	**Course Code**	**Subject**	**Offering Department**
1		Fundamentals of Image Processing	Electronics & Communication Engineering
2		Automotive Electronics	
IV B. TECH ECE I SEM OPEN ELECTIVE- III			
S.No.	**Course Code**	**Subject**	**Offering Department**
1		Introduction to wireless Communications	Electronics & Communication Engineering
2		Electronics for Health Care	

VISION AND MISSION OF THE INSTITUTE

VISION OF THE INSTITUTE

To be a leading institution in engineering education, fostering innovation, research, and ethical values to serve society.

MISSION OF THE INSTITUTE

- To provide quality education through effective teaching-learning processes.
- To promote research, innovation, and entrepreneurship.
- To collaborate with industry and society for sustainable development.
- To develop competent and socially responsible professionals.

QUALITY POLICY

To ensure high standards in imparting professional education by providing world-class infrastructure, top-quality-faculty and decent work culture to sculpt the students into Socially Responsible Professionals through creative team-work, innovation and research.

SHORT TERM GOALS

- To be a Nationally acclaimed centre for Research and Development.
- To be an Incubation center for Techno – Managerial Enterprises.

LONG TERM GOALS

- To evolve into a Deemed University offering viable programmes of relevance for upliftment of rural students and populace.
- To be recognized Internationally as a Research and Development Centre by forging alliances with research organizations, Government entities, industries and alumni.

VISION AND MISSION OF THE DEPARTMENT

VISION OF THE DEPARTMENT

Our vision is to develop the department into a full-fledged center of learning in various fields of Electronics & Communication Engineering keeping in view the latest development.

MISSION OF THE DEPARTMENT

M1: To turn out full-fledged Engineers in the field of Electronics Communication Engineering with an overall background suitable for making a successful career either in industry/research or higher education in India and abroad.

M2: To inculcate professional behavior, strong ethical values, innovative research capabilities and leadership abilities in the young minds so as to work with a commitment to the progress of the nation.

ANURAG
Engineering Engineers

PROGRAMME EDUCATIONAL OBJECTIVES (PEO'S)

PEO1: Graduates will have successful technical or professional careers, including supportive and leadership roles on multidisciplinary teams.

PEO2: Graduates will be able to acquire, use and develop skills as required for effective professional practices.

PEO3: Graduates will be able to attain holistic education that is an essential prerequisite for being a responsible member of society.

PROGRAM SPECIFIC OUTCOMES (PSO'S)

PSO I. Analyze, Design, and develop solutions for complex real-time problems in Embedded Systems, Communication Systems, and Electronic Circuits using various simulation tools.

PSO II. Integrate, Innovate, and develop reliable Electronic Systems, IoT Solutions, and VLSI applications to support industry needs, societal advancement, sustainable development, and technology-driven startups

PROGRAM OUTCOMES (PO'S)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8

I B.Tech. I Semester

		L	T	P	C
Course Name	**: MATRICES AND CALCULUS**				
Course Code	**: 2599101**	3	1	0	4
Course Category	: **Theory**				

CIE: 40 SEE: 60 Total: 100

Course Objectives: To learn

- Applying basic operations on matrices and their properties. Concept of a rank of the matrix and applying this concept to know solving the system of linear equations.
- Concept of eigen values and eigen vectors and to reduce the quadratic form to canonical form
- Concept of the mean value theorems and their application to the mathematical problems
- Finding maxima and minima of functions of three variables.
- Evaluation of multiple integrals and their applications.

UNIT-I: Matrices **8 Hrs**

Rank of a matrix by Echelon form and Normal form – Inverse of Non-singular matrices by Gauss-Jordan method.

System of linear equations: Solving system of non-homogeneous equations by Gauss elimination method and Gauss Seidel Iteration Method.

UNIT-II: Eigen values and Eigen vectors **10 Hrs**

Linear Transformation and Orthogonal Transformation: Cayley-Hamilton Theorem (without proof), Eigen values – Eigen vectors and their properties – Diagonalization of a matrix.

Quadratic forms and Nature of the Quadratic Forms – Reduction of Quadratic form to canonical form by Orthogonal Transformation.

UNIT-III: Single Variable Calculus **10 Hrs**

Limit and Continuous of functions and its properties. Mean value theorems: Rolle's theorem – Lagrange's Mean value theorem with applications – Cauchy's Mean value Theorem-Taylor's Series – Maclaurin's Series (All the theorems without proof).

Curve Tracing: Curve tracing in cartesian coordinates.

UNIT-IV: Multivariable Calculus (Partial Differentiation and applications) **10 Hrs**

Definitions of Limit and continuity – Partial Differentiation: Total derivative – Jacobian – Functional dependence & independence.

Applications: Maxima and minima of functions of three variables using method of Lagrange multipliers.

UNIT-V: Multivariable Calculus (Integration) **10 Hrs**

Evaluation of Double Integrals (Cartesian and polar coordinates) – change of order of integration (only Cartesian form) – Change of variables for double integrals (Cartesian to polar). Evaluation of Triple Integrals

Applications: Areas by double integrals and volumes by triple integrals.

Self-study content

UNIT-I	***UNIT- II***	***UNIT- III***	***UNIT-IV***	***UNIT-V***
Vector Spaces & Linear Transformations	*Diagonalization & Matrix Similarity:*	*Improper Integrals and Beta-Gamma Functions*	*Taylor's Series Expansion for Functions of Two Variables*	*Applications of Multiple Integrals in Physics*

TEXT BOOKS:

1. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.
2. R.K. Jain and S.R.K. Iyengar, Advanced Engineering Mathematics, Narosa Publications, 5th Editon, 2016.

REFERENCES:

1. Erwin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, 9thEdition, Pearson, Reprint,

2002.
3. N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2008.
4. H. K. Dass and Er. Rajnish Verma, Higher Engineering Mathematics, S Chand and Company Limited, New Delhi.

Course Outcomes: After learning the contents of this paper, the student must be able to

CO-1. Write the matrix representation of a set of linear equations and to analyse the solution of the system of equations

CO-2. Find the Eigen values and Eigen vectors. Reduce the quadratic form to canonical form using orthogonal transformations.

CO-3. Solve the applications of the mean value theorems.

CO-4. Find the extreme values of functions of Three variables with/ without constraints.

CO-5. Evaluate the multiple integrals and apply the concept to find areas, volumes.

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	3	1	1	1	-	-	-	-	-	1	2	1
CO-2	3	3	1	1	2	-	-	-	-	-	1	2	1
CO-3	3	2	1	1	1	-	-	-	-	-	1	1	1
CO-4	3	3	1	1	1	-	-	-	-	-	1	2	1
CO-5	3	3	2	1	1	-	-	-	-	-	1	2	1

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



I B.Tech. I Semester

		L	T	P	C
Course Name	**: ENGINEERING PHYSICS**				
Course Code	**: 2599106**	3	0	0	3
Course Category	: **Theory**				

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To understand fundamental concepts of quantum mechanics and their applications in solids and nanomaterials.
- To give students a clear understanding of how semiconductors and nanomaterials work, how they are made, and how they are used in modern technology
- To introduce quantum computing principles, quantum gates, and basic quantum algorithms.
- To learn the properties and applications of magnetic and dielectric materials.
- To explore the working and applications of lasers and fiber optics in modern technology.

UNIT-I: 12 Hrs

Quantum Mechanics: Introduction, de-Broglie hypothesis, Davisson-Germer Experiment Heisenberg uncertainty principle, physical significance of wave function, postulates of quantum mechanics (Elementary treatment): operators in quantum mechanics, eigen values and eigen functions, expectation value; Schrödinger's time independent wave equation, particle in a 1D box.

Bloch's theorem (Elementary treatment), Kronig-Penney model (Elementary treatment): E-K diagram, effective mass of electron (Elementary treatment), formation of energy bands, classification of solids.

UNIT-II: 12 Hrs

Semiconductor Physics: Intrinsic and extrinsic semiconductors, variation of Fermi energy level with temperature and doping concentration (Elementary treatment), direct and indirect band gap semiconductors, Hall Effect**,** construction, principle, V-I characteristics and applications of P-N Junction diode, LED and solar cell.

Nanotechnology: Introduction to nanoscience and nanotechnology, Classification of nanomaterials: 0D,1D,2D and 3D, surface to volume ratio, quantum confinement, synthesis methods: Ball milling, sol-gel method, Characterization Techniques: X-ray diffraction (XRD) and Scanning Electron Microscopy (SEM), applications of nanomaterials**.**

UNIT-III: 12Hrs

Quantum Computing: Introduction, linear algebra for quantum computation(Elementary treatment), Dirac's Bra and Ket notation and their properties, superposition, entanglement, Hilbert space, Bloch's sphere, concept of quantum computer: classical bits, Qubits, multiple Qubit system, quantum computing system for information processing (Elementary treatment), evolution of quantum systems, quantum measurements, quantum gates (Elementary treatment): Pauli, Hadamard, and CNOT gates, challenges and advantages of quantum computing over classical computation.

UNIT-IV: 12 Hrs

Dielectric materials: Introduction to dielectric materials, types of polarization: electronic, ionic (Quantitative) & orientation (Elementary treatment); ferroelectric, piezoelectric, pyroelectric materials and their properties and applications

Magnetic materials: Introduction to magnetic materials, origin of magnetic moment-classification of magnetic materials: dia, para, ferro, anti-ferro and ferri magnetic materials. Weiss domain theory of ferromagnetism, Hysteresis and B-H curve, soft and hard magnetic materials, applications.

UNIT-V: **12 Hrs**

Laser: Introduction to laser, characteristics of laser, Einstein coefficients and their relations, metastable state, population inversion, pumping, lasing action, He-Ne laser, Nd-YAG Laser, semiconductor diode laser, applications of Lasers.

Fiber Optics: Introduction to fiber optics, total internal reflection, Physical components of optical fiber, acceptance angle, numerical aperture, classification of optical fibers (based on refractive index profile), losses in optical fiber, applications: optical fiber for communication system, sensor for structural health monitoring.

Self-study content

UNIT-I	***UNIT- II***	***UNIT- III***	***UNIT-IV***	***UNIT-V***
Introduction to Density of States (DOS) in solids, Fermi-Dirac distribution function	*Perovskite solar cells, Graphene and its properties*	*Quantum supremacy experiments (e.g., Google's Sycamore processor) Introduction to IBM Qi skit platform*	*Spintronics and magnetic memory devices*	*Optical sensors in smart cities*

Textbooks:

1. **M.N Avadhanulu, P.G Kshirsagar, T VS Arun Murthy, A Textbook of Engineering Physics, S Chand,2019.**
2. Shatendra Sharma and Jyotsna Sharma, Engineering Physics, Pearson India Education Services Pvt. Ltd,2018.
3. Michael A. Nielsen & Isaac L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press.

References:

1. Charles Kittel, *Introduction to Solid State Physics*, John Wiley & Sons, Inc.
2. Md. N. Khan, S. Panigrahi, Principles of Engineering Physics 1 & 2, Cambridge University Press, 2016
3. Prof. A. K. Katiyar, Dr. C. K. Pandey, Engineering Physics: Theory and Practical, Wiley India Pvt. Ltd., India,2017.
4. Halliday, Resnick and Walker, Fundamentals of Physics, John Wiley &Sons,11th Edition, 2018.

Course Outcomes:

1. **CO1**: Apply quantum mechanical principles to explain particle behavior and energy band formation in solids
2. **CO2**:. Able to explain the principles, types, fabrication methods, and applications of semiconductors and nanomaterials in simple terms.
3. **CO3**: Understand quantum computing concepts, use quantum gates, and explain basic quantum algorithms.
4. **CO4**: Classify magnetic and dielectric materials and explain their properties, synthesis, and applications.
5. **CO5**: Explain the principles of lasers and fiber optics and their applications in communication and sensing.

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	1	3	1	1								2	
CO-2	2	2	2	2								1	
CO-3	2	3	2	2								1	
CO-4	2	2		1								2	
CO-5	2	2		1	2							2	

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



I B.Tech. I Semester

		L	T	P	C
Course Name	**: C PROGRAMMING AND DATA STRUCTURES**				
Course Code	**: 2505103**	3	0	0	3
Course Category	**: Theory**				

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To introduce computer systems, environments, programming basics, and the fundamentals of C language.
- To understand structured programming concepts using statements, functions, and recursion.
- To apply arrays, pointers, and searching/sorting techniques for computational problem solving.
- To develop applications using strings, structures, and derived data types.
- To implement fundamental data structures (linked lists, stacks, and queues) for problem-solving.

UNIT-I: **12Hrs**

Introduction to Computers – Computer Systems, Computing Environments, Computer Languages, Creating and running programs

Introduction to C Language – Background, Simple C programs, Identifiers, Basic data types, Variables, Constants, Input / Output, Structure of a C Program – Operators, Bit-wise operators, Expressions Expression Evaluation, Type conversions.

UNIT-II: **12Hrs**

Statements – if and switch statements, Repetition statements – while, for, do-while statements, other statements related to looping – break, continue, go to, Designing Structured Programs- Functions, basics, user defined functions, inter function communication, standard functions, Recursion.

UNIT-III: **12Hrs**

Arrays – Concepts, using arrays in C, two –dimensional arrays, multi-dimensional arrays. Searching and sorting

Pointers – Introduction, Pointers for inter function communication, pointers to pointers, compatibility, Pointer Applications – Passing an array to a function.

UNIT-IV: **12Hrs**

Strings – Concepts, C Strings, String Input / Output functions, arrays of strings, string manipulation functions, string / data conversion.

Derived types – Structures – Declaration, definition and initialization of structures, accessing structures, operations on structures, nested structures, The Type def, enumerated types

UNIT-V: **12Hrs**

Data Structures – Introduction to Data Structures, abstract data types, Linear list – singly linked list implementation, insertion, deletion and searching operations on linear list, Stacks-Operations, array and linked representations of stacks, stack applications, Queues-operations, array and linked representations.

Self-study content

UNIT-I	***UNIT- II***	***UNIT- III***	***UNIT-IV***	***UNIT-V***
Compare arrays and linked lists	*dynamic memory allocation functions in C*	*C program to create and traverse a singly linked list*	*an algorithm and C program to implement singly linked list operations.*	*insertion and deletion operations in linked list.*

Textbooks:

1. C Programming & Data Structures, B.A.Forouzan and R.F. Gilberg, Third Edition, Cengage Learning.
2. Problem Solving and Program Design in C, J.R. Hanly and E.B. Koffman, Fifth Edition, Pearson Education.
3. The C Programming Language, B.W. Kernighan and Dennis M.Ritchie, PHI/Pearson Education

References:

1. **C** & Data structures – P. Padmanabham, 3rd Edition, B.S. Publications.
2. C Programming with problem solving, J.A. Jones & K. Harrow, Dreamtech Press
3. Programming in C – Stephen G. Kochan, III Edition, Pearson Education.
4. C for Engineers and Scientists, H. Cheng, McGraw-Hill International Edition
5. Data Structures using C – A. M. Tanenbaum, Y. Langsam, and M.J. Augenstein, Pearson Education / PHI
6. C Programming & Data Structures, E. Balagurusamy, TMH.
7. C Programming & Data Structures, P. Dey, M Ghosh R Thereja, Oxford University Press
8. C & Data structures – E V Prasad and N B Venkateswarlu, S. Chand & Co.

Course Outcomes:

After completion of the Course, the student should be able to

CO-1. Understand computer systems, environments, C syntax, operators, and simple programs.
CO-2. Apply structured programming constructs, functions, and recursion to solve problems.
CO-3. Use arrays, pointers, searching and sorting techniques for solving computational tasks.
CO-4. Develop programs using strings, structures, and derived data types.
CO-5. Implement linear data structures (linked lists, stacks, queues) to solve engineering problems.

Course Articulation Matrix:

CO	Program Outcomes (PO)											Program Specific Outcomes (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	2	1		1					1		3	1
CO-2	3	3	2		2					1		3	2
CO-3	2	3	3		3	1				2	1	3	2
CO-4	2	2	2		2	1				2	2	2	2
CO-5	2	3	3		3	2		1		2	2	3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



I B.Tech. I Semester

		L	T	P	C
Course Name	**: COMPUTER AIDED ENGINEERING GRAPHICS**				
Course Code	**: 2503101**	2	0	2	3
Course Category	: **Theory**				

CIE: 40 SEE: 60 Total: 100

Course Objectives

- To introduce the fundamentals of engineering drawing and projection systems.
- To develop skills in constructing orthographic, isometric, and sectional views.
- To train students in interpreting and creating technical drawings using CAD tools.
- To familiarize students with dimensioning standards and drafting conventions.
- To bridge manual drafting techniques with computer-aided drafting practices.

UNIT-I: **13Hrs**

Introduction to Engineering Graphics (Conventional):

Principles of Engineering Graphics and their Significance, Geometrical Constructions, Scales, Plain and Diagonal, Conic Sections including the Rectangular Hyperbola, General method only. Cycloid, Epicycloids and Hypocycloid.

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UNIT-II: **12Hrs**

Orthographic Projections (Conventional and Computer Aided)

Principles of Orthographic Projections, Conventions, Projections of Points and Lines, Projections of Plane regular geometric figures. Computer aided orthographic projections, points, lines and planes. Introduction to Computer aided drafting, views, commands and conics.

UNIT-III: **12Hrs**

Projections of Regular Solids (Conventional and Computer Aided)

Sections or Sectional views of Right Regular Solids, Prism, Cylinder, Pyramid, Cone, and Computer aided projections of solids, sectional views

UNIT-IV: **10Hrs**

Development of Surfaces (Conventional)

Prism, Cylinder, Pyramid and Cone.

UNIT-V: **13Hrs**

Isometric Projections (Conventional and Computer Aided)

Principles of Isometric Projection, Isometric Scale, Isometric Views, Conventions, Isometric Views of Lines, Plane Figures, Simple and Compound Solids, Isometric Projection of objects having non, isometric lines. Conversion of Isometric Views to Orthographic Views and Vice- versa, Conventions. Conversion of orthographic projection in to isometric view.

Self-study content

UNIT-I	*UNIT- II*	*UNIT- III*	*UNIT-IV*	*UNIT-V*
*Draw involute for gear tooth profile, * Create simple 3D objects.*	*True length determination.*	*Projection of Frustums, Projection of Combined Solids*	*Development of Oblique Solids. Assembly Drawings.*	*3D to 2D Drawing Extraction*

Note:

1. The End Semester Examination will be in conventional mode.
2. CIE–I will be in conventional mode.
3. CIE–II will be using Computer.

Textbooks:

1. Engineering Drawing, N.D.Bhatt, Charotar,54th Edition, 2023.
2. Engineering Drawing and graphics Using AutoCAD, T. Jeyapoovan and Vikas, S. Chand and company Ltd., 3rdEdition,2010.

References:

1. Engineering Drawing, Basant Agrawal and C.M.Agrawal, McGraw Hill,3rdEdition,2019.
2. Engineering Graphics and Design, WILEY, John Wiley and Sons Inc,3rdEdition, 2020.
3. Engineering Drawing, M.B.Shah and B.C.Rane,Pearson,2ndEdition, 2009.
4. Engineering Drawing, N.S.Parthasarathy and Vela Murali, Oxford,1st Edition,2015.
5. Computer Aided Engineering Drawing, K.Balaveera Reddy, CBS Publishers, 2nd Edition,2015.

Course Outcomes: After completion of the course, the student should be able to

CO-1. Understand and apply the principles of orthographic and isometric projections.
CO-2. Create sectional views and dimensioned drawings using BIS standards.
CO-3. Use CAD software to generate 2D engineering drawings.
CO-4. Visualize and construct solid models from 2D views.
CO-5. Demonstrate drafting skills for practical and industrial applications.

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	1	-	-	2	-	-	3	-	2	2	2	1	2
CO-2	2	2	3	3	3	2	3	-	-	-	2	3	1
CO-3	1	2	3	-	-	2	3	3	3	2	2	1	3
CO-4	-	-	3	1	3	2	-	-	-	-	2	3	2
CO-5	3	2	1	3	2	3	2	2	2	-	2	3	2

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



I B.Tech. I Semester

Course Name**: BASIC ELECTRICAL ENGINEERING**
Course Code**: 2502103**
Course Category**: Theory**

L	T	P	C
3	0	0	3

CIE: 40 SEE: 60 Total: 100

Course Objectives: The objective of this course is:

- To understand and apply basic laws in D.C. electrical circuits.
- To impart the knowledge of AC circuits, Phasor algebra related to alternating quantities.
- To understand the magnetic circuits and principle of operation of transformer.
- To understand the construction and working of DC machines and Induction motors.
- To acquire knowledge of various electrical installations and Energy calculations.

UNIT-I: **12 Hrs**

D.C. Circuits: Electrical circuit elements (R, L and C), voltage and current sources, Ohm's Law, KVL & KCL, Resistive Networks-Series and parallel circuits, Voltage and Current division rules, Mesh analysis, Nodal Analysis, Simple problems.

UNIT-II: **13 Hrs**

A.C. Circuits: Representation of sinusoidal wave forms, peak and RMS values, phasor representation, real power, reactive power, apparent power, power factor, Analysis of single-phase ac circuits consisting of RL, RC, RLC combinations (series only), resonance in series R-L-C circuit. Three-phase balanced circuits, voltage and current relations in star and delta connections.

UNIT-III: **12 Hrs**

Magnetic Circuits &Transformers:

Magnetic circuits: Faraday's laws of Electromagnetic Induction, Lenz's Law, and Fleming's Right-hand Rule and Left-hand Rule. Magnetic Circuits - concept of Self Inductance & Mutual Inductance, Coefficient of Coupling.

Transformers: Ideal and practical transformer, Equivalent circuit, Open Circuit and Short Circuit tests, losses in transformer, regulation and efficiency, Applications, Simple problems.

UNIT-IV: **12 Hrs**

Electrical Machines:

DC generators: Construction, Basic working principle, EMF equation, types of dc generators, Applications, simple problems.

DC Motors: Basic working principle, types of dc motors, torque equation, speed control of dc shunt motor, Applications, Simple problems.

Three Phase Induction Motors: Construction, Principle of operation, Applications, Simple problems.

UNIT-V: **11 Hrs**

Electrical Installations: Components of LT Switchgear: Switch Fuse Unit (SFU), MCB, ELCB, MCCB, Earthing, Types of Earthing. Types of Batteries-SOC Characteristics, Elementary calculations for single phase energy consumption.

Self-study content

UNIT-I	*UNIT- II*	*UNIT- III*	*UNIT-IV*	*UNIT-V*
Practical Applications of Kirchhoff's Laws in Electrical Networks	*Power Factor Improvement in Practical AC Systems*	*Applications of Transformers in Power Transmission and Distribution*	*Selection of DC Motors for Different Industrial Applications*	*Electrical Safety Practices in Domestic and Industrial Installations*

Text Books:

1. D.P. Kothari and I. J. Nagrath, "Basic Electrical Engineering", Tata McGraw Hill, 4th Edition, 2019.
2. P. Ramana, M. Suryakalavathi, G.T. Chandrasheker, "Basic Electrical Engineering", S. Chand, 2nd Edition, 2019.

Reference Books:

1.D. C. Kulshreshtha,"Basic Electrical Engineering", McGraw Hill, 2nd edition, 2009.
2. M. S. Sukhija, T. K. Nagsarkar, "Basic Electrical and Electronics Engineering", paperback Edition, 2017.

Course Outcomes: Upon successful completion of the course, the student will be able to,

CO-1. Understand the importance of DC circuits and basic laws.
CO-2. Understand the concept of AC circuits and resonance.
CO-3. Understand Concept of principle of operation of transformer.
CO-4. Analyze the performance of DC machines and Induction motors.
CO-5. Demonstrate the importance of electrical installation and Energy calculations.

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	2	-	-	2	-	-	-	-	-	-	2	-
CO-2	3	2	-	-	2	-	-	-	-	-	-	2	-
CO-3	3	3	2	-	2	-	-	-	-	-	-	3	-
CO-4	3	2	2	2	3	-	-	-	-	-	-	2	2
CO-5	2		3	-	2	2	1	2	-	-	**2**	-	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



I B.Tech. I Semester

		L	T	P	C
Course Name	**: ENGINEERING PHYSICS LABORATORY**				
Course Code	**: 2599107**	0	0	2	1
Course Category	: **Practical**				

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To provide hands-on experience in studying the electrical, optical, and magnetic properties of various materials.
- To understand the working principles and characteristics of semiconductor devices like PN junction diodes, LEDs, and solar cells.
- To explore advanced concepts in modern physics through experiments on lasers, photoelectric effect, and optical fibers.
- To develop experimental skills for measuring fundamental constants and material parameters.
- To improve abilities in scientific data collection, result analysis, and technical reporting.

List of experiments:

1. Determination of Energy gap of a semiconductor.
2. I-V Characteristics of P-N junction diode.
3. B-H curve of a ferromagnetic material.
4. (a) V-I Characteristics of LED diode.
   (b) V-I Characteristics of a solar cell.
5. Determination of wavelength of laser using diffraction grating.
6. Determination of Planck's constant using photo electric effect.
7. Determination of magnetic moment of a bar magnet and horizontal earth magnetic field.
8. (a) Determine the Numerical Aperture of an Optical Fiber
   (b) Determine the bending losses of an Optical Fiber
9. Determination of dielectric constant of a given material.
10. Determination of Hall coefficient and carrier concentration of a given semiconductor.

Note: Any 8 experiments are to be performed.

Self-study content

- *Input and output characteristics of BJT (CE, CB & CC configurations)*
- *Understanding the method of least squares–torsional pendulum as an example.*

References:

1. Dr. Y. Aparna & Dr. K. Venkateswara Rao, Laboratory Manual of Engineering Physics, V.G.S Publishers
2. G.L. Squires – *Practical Physics*, Cambridge University Press, 2012.
3. S. Balasubramanian, M.N. Srinivasan "A Text book of Practical Physics"- S Chand Publishers,2017

Course Outcomes:

After completing this laboratory course, the student will be able to:

1. **CO1:** Determine key semiconductor parameters such as energy gap, Hall coefficient, and carrier concentration.
2. **CO2:** Analyze the I–V characteristics of PN junction diodes, LEDs, and solar cells for different applications.
3. **CO3:** Determine key electrical, magnetic, and optical properties of semiconductors and other functional materials.

4. **CO4:** Demonstrate the working principles of lasers, optical fibers, and the photoelectric effect through precise measurements.
5. **CO5:** Apply scientific methods for accurate experimentation, data analysis, and preparation of technical reports.

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	2	2	1	-	2	2	2	-	2	-	2	2	2
CO-2	2	2	1	2	2	1	-	-	1	-	1	2	2
CO-3	3	3	2	-	-	-	2	-	1	-	2	2	1
CO-4	2	1	-	1	2	-	-	1	2	-	2	1	1
CO-5	2	2	-	1	-	2	2	-	1	1	-	2	2

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



I B.Tech. I Semester

		L	T	P	C
Course Name	: **C PROGRAMMING AND DATA STRUCTURES LAB**				
Course Code	:**2505104**	0	0	2	1
Course Category	: **Practical**				

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To apply the fundamentals of C programming to solve simple mathematical and logical problems.
- To develop programs using functions, recursion, arrays, and structures.
- To implement searching and sorting algorithms for computational problem-solving.
- To design and implement linked lists and perform insertion, deletion, and traversal operations.
- To implement stacks and queues using arrays and pointers for problem-solving.

Week-1 & 2

1)Write a C program to find the roots of a quadratic equation.
2)Write a C program, which takes two integer operands and one operator from the user, performs the operation and then prints the result. (Consider the operators +,-,*, /, % and use Switch Statement)
3) Write a C program to find the sum of the individual digits of a positive integer.

Week-3 & 4

4)Write a C program to find the factorial of a given integer.
5) Write a C program to find the GCD (greatest common divisor) of two given integers.
6) Write a C program to find both the largest and smallest number in a list of integers.
7)Write a C program that uses functions to perform the following:
i) Addition of Two Matrices ii) Multiplication of Two Matrices

Week-5 & 6

8)Write a C program to construct a pyramid of numbers.
9)Write a C program that uses functions to perform the following operations:
i) Reading a complex number ii) Writing a complex number
iii) Addition of two complex numbers
iv) Multiplication of two complex numbers (Note: represent complex number using a structure.)

Week- 7 & 8

10)Write a C program that implements the following sorting methods to sort a given list of integers in ascending order i) Bubble sort ii) Selection sort iii) Insertion sort
11)Write C programs that use both recursive and non-recursive functions to perform the following searching operations for a Key value in a given list of integers: i) Linear search ii) Binary search

Week- 9 & 10

12.Write a C program that uses functions to perform the following operations on singly linked list.
i) Creation ii) Insertion iii) Deletion iv) Traversal

Week-11 & 12

13.Write C programs that implement a stack (its operations) using i) Arrays ii) Pointers
14.Write C programs that implement Queue (its operations) using i) Arrays ii) Pointers

Self-study content

- *Student Record Management System (Array-based)*
- *Library Book Tracking System (Structures + Files)*
- *Simple Billing System*
- *Matrix Operations (Addition, Multiplication, Transpose)*

- ❖ *Bank Account Management (File Handling)*

Textbooks:

1. C Programming & Data Structures, B.A. Forouzan and R. F. Gilberg, Third Edition, Cengage Learning.
2. Let us C, Yeswanth Kanitkar
3. C Programming, Balaguruswamy.

Reference Books:

1. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language, Prentice Hall of India
2. E. Balagurusamy, Computer Fundamentals and C, 2nd Edition, McGraw-Hill

Course Outcomes:

After completion of the Course, the student should be able to

CO-1. Apply C programming constructs (operators, I/O, control) to solve mathematical problems.
CO-2. Develop solutions using functions, recursion, arrays, and structures.
CO-3. Implement searching and sorting algorithms using iterative/recursive methods.
CO-4. Design and implement linked list operations (creation, insertion, deletion, traversal).
CO-5. Create stack and queue applications using arrays and pointers.

Course Articulation Matrix:

CO	Program Outcomes (PO)											Program Specific Outcomes (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	2	1		2					1		3	1
CO-2	3	3	2		3					1		3	2
CO-3	3	3	3		3	1				2		3	3
CO-4	2	3	3		3	1				2		3	3
CO-5	2	3	3	1	3	2				2		3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



I B.Tech. I Semester

Course Name: BASIC ELECTRICAL ENGINEERING LAB
Course Code: 2502104
Course Category: Practical

L	T	P	C
0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:
The objectives of this course are:

- To analyze a given network by applying various electrical laws.
- To understand magnetic circuits.
- To study the concept of resonance.
- To determine the performance characteristics of DC and AC electrical machines.
- To determine the performance of Transformers through various testing methods.

PART- A (compulsory)

1. Verification of KVL and KCL.
2. Determination of self, mutual inductances and coefficient of coupling.
3. Resonance in series RLC circuit.
4. Calculations and Verification of Impedance and Current of RL, RC and RLC series circuits.
5. Measurement of Voltage, Current and Real Power in primary and Secondary Circuits of a Single-Phase Transformer.
6. Performance Characteristics of a DC Shunt Motor.
7. OC & SC Tests on Single phase transformer.
8. Speed control of a DC Shunt Motor.

PART-B (any two experiments from the given list)

1. Magnetization Characteristics of DC Shunt Generator.
2. Load Test on Single Phase Transformer (Calculate Efficiency and Regulation).
3. Torque-Speed Characteristics of a Three-phase Induction Motor.
4. Study of SFU, MCB, ELCB and MCCB.

Self-study content

- *Effect of Load Variation on Speed, Torque and Efficiency of DC Shunt Motors*
- *Selection of Protective Devices for Different Domestic and Industrial Loads*

Course Outcomes: Upon the successful completion of this course, the student will be able to:

CO-1. Apply the various procedures and techniques for the experiments.
CO-2. Verify the various electrical laws with DC Excitation.
CO-3. Understand Resonance concept.
CO-4. Determine the losses, efficiency and regulation of single-phase transformer.
CO-5. Obtain the performance of DC and AC electrical machines.

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	2	-	3	2	-	-	-	2	2	1	3	-
CO-2	3	2	-	3	1	-	-	-	2	2	1	3	-
CO-3	3	2	-	3	1	-	-	-	2	2	1	3	-
CO-4	3	2	-	3	2	-	-	-	2	2	1	3	-
CO-5	3	2	-	2	2	-	-	-	2	2	1	3	-

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



I B.Tech. I Semester

Course Name	:	**IT WORKSHOP**				
Course Code	:	**2505105**	L	T	P	C
Course Category	:	**Practical**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- Training on PC Hardware, troubleshooting
- Training Internet & World Wide Web and
- Training Productivity tools including Word, Excel, PowerPoint and Publisher.
- Training on Safeguard computer systems from viruses/worms
- Training on Document/ Presentation preparation

PC Hardware

Task 1: Identify the peripherals of a computer, components in a CPU and its functions. Draw the block diagram of the CPU along with the configuration of each peripheral and submit to your instructor.

Task 2: Every student should disassemble and assemble the PC back to working condition. Lab instructors should verify the work and follow it up with a Viva. Additionally, students should watch the video, which demonstrates the process of assembling a PC. A video would be given as part of the course content.

Task 3: Every student should individually install Microsoft Windows on their personal computer. The lab instructor should verify the installation and follow it up with a Viva.

Task 4: Every student should install Linux on the computer. This computer should have Windows installed. The system should be configured as a dual-boot with both Windows and Linux. Lab instructors should verify the installation and follow it up with a Viva

Internet & World Wide Web

Task1: Orientation & Connectivity Boot Camp: Students should get connected to their Local Area Network and access the Internet. In the process, they configure the TCP/IP settings. Finally, students should demonstrate to the instructor how to access the websites and email. If there is no internet connectivity, preparations need to be made by the instructors to simulate the WWW on the LAN.

Task 2: Web Browsers, Surfing the Web: Students customize their web browsers with the LAN proxy settings, bookmarks, search toolbars and pop up blockers. Also, plug-ins like Macromedia Flash and JRE for applets should be configured.

Task 3: Search Engines & Netiquette: Students should know what search engines are and how to use them. A few topics would be assigned to the students, for which they would need to conduct a Google search. This should be demonstrated to the instructors by the student.

Task 4: Cyber Hygiene: Students would be exposed to the various threats on the internet and would be asked to configure their computers to be safe on the internet. They need to customize their browsers to block pop-ups, block ActiveX downloads to avoid viruses and/or worms.

LaTeX and WORD

Task 1 – Word Orientation: The mentor needs to give an overview of LaTeX and Microsoft (MS) office or equivalent (FOSS) tool word: Importance of LaTeX and MS office or equivalent (FOSS) tool Word as word Processors, Details of the four tasks and features that would be covered in each, Using LaTeX and word – Accessing, overview of toolbars, saving files, Using help and resources, rulers, format painter in word.

Task 2: Using LaTeX and Word to create a project certificate. Features to be covered: - Formatting Fonts in Word, Drop Cap in Word, Applying Text effects, Using Character Spacing, Borders and Colours, Inserting Header and Footer, Using Date and Time option in both LaTeX and Word.

Task 3: Creating project abstract Features to be covered: -Formatting Styles, inserting table, Bullets and Numbering, Changing Text Direction, Cell alignment, Footnote, Hyperlink, Symbols, Spell Check, Track Changes.

Task 4: Creating a Newsletter: Features to be covered include Table of Contents, Newspaper columns, Images from files and clip art, the Drawing toolbar and Word Art, Formatting Images, Text Boxes, Paragraphs, and Mail Merge in Word.

Excel

Excel Orientation: The mentor needs to emphasize the importance of MS Office or an equivalent (FOSS) tool, such as Excel, as a Spreadsheet tool. They should provide details of the four tasks and features that will be covered in each. Using Excel – Accessing, overview of toolbars, saving Excel files, using help and resources.

Task 1: Creating a Scheduler - Features to be covered: Gridlines, Format Cells, Summation, auto fill, Formatting Text.

Task 2: Calculating GPA - Features to be covered: - Cell Referencing, Formulae in Excel – average, std. Deviation, Charts, Renaming and Inserting worksheets, Hyperlinking, Count function, LOOKUP/VLOOKUP.

Task 3: Split cells, freeze panes, group and outline, Sorting, Boolean and logical operators, and Conditional formatting.

PowerPoint

Task 1: Students will be working on basic PowerPoint utilities and tools, which help them create basic PowerPoint presentations. PPT Orientation, Slide Layouts, Inserting Text, Word Art, Formatting Text, Bullets and Numbering, Auto Shapes, Lines and Arrows in PowerPoint.

Task 2: Interactive Presentations - Hyperlinks, Inserting Images, Clip Art, Audio, Video, Objects, Tables, and Charts.

Task 3: Master Layouts (slide, template, and notes), Types of views (basic, presentation, slide slotter, notes etc), and Inserting – Background, textures, Design Templates, Hidden slides.

Self-study content

Reference Books:

1. Comdex Information Technology course tool kit Vikas Gupta, WILEY Dream tech
2. The Complete Computer upgrade and repair book, 3rd edition Cheryl A Schmidt, WILEY Dream tech
3. Introduction to Information Technology, ITL Education Solutions limited, Pearson Education.

4. PC Hardware - A Handbook – Kate J. Chase PHI (Microsoft)
5. LaTeX Companion – Leslie Lamport, PHI/Pearson.
6. IT Essentials PC Hardware and Software Companion Guide Third Edition by David Anfinson and Ken Quamme. – CISCO Press, Pearson Education.
7. IT Essentials PC Hardware and Software Labs and Study Guide Third Edition by Patrick Regan – CISCO Press, *Pearson Education.*

Course Outcomes:

After completion of the Course, the student should be able to

CO-1. Perform Hardware troubleshooting
CO-2. Understand Hardware components and inter dependencies
CO-3. Safeguard computer systems from viruses/worms
CO-4. Document/ Presentation preparation
CO-5. Perform calculations using spreadsheets

Course Articulation Matrix:

CO	Program Outcomes (PO)											Program Specific Outcomes (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	2	2	1	2						1	2	1
CO-2	3	2	2	2	3	1				1	1	3	2
CO-3	2	2	3	2	2	1				1	2	3	2
CO-4	2	1	2	2	3					2	2	2	3
CO-5	2	1	2	2	3	1				2	3	2	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



I B.Tech. II Semester

Course Name	: **ORDINARY DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS**	L	T	P	C
Course Code	: **2599201**	3	0	0	3
Course Category	: **Theory**				

CIE: 40 SEE: 60 Total: 100

Course Objectives: To learn

- Methods of solving the differential equations of first and higher order.
- Concept, properties of Laplace transforms.
- Solving ordinary differential equations using Laplace transforms techniques.
- The physical quantities involved in engineering field related to vector valued functions
- The basic properties of vector valued functions and their applications to line, surface and volume integrals

UNIT-I: First Order Ordinary Differential Equations **8Hrs**

Exact differential equations – Equations form: Homogeneous form and $yf(xy)dx + xg(xy)dy = 0$ reducible to exact differential equations – linear and Bernoulli's equations. Applications: Newton's law of cooling – Law of natural growth and decay.

UNIT-II: Ordinary Differential Equations of Higher Order **10Hrs**

Higher order linear differential equations with constant coefficients: Non-Homogeneous terms of the type $V(x): e^x,\ \sin ax, \cos ax,$ polynomials in x, $e^{ax}V(x)$ and $xV(x)$ – Method of variation of parameters.

UNIT-III: Laplace Transforms **10Hrs**

Laplace Transforms: Laplace Transform of standard functions – First shifting theorem – Second shifting theorem-Laplace transforms of functions multiplied by $'t'$ and divided by $'t'$ – Evaluation of integrals by Laplace transforms – Inverse Laplace transform by different methods, convolution theorem (without proof).
Applications: solving Initial value problems by Laplace Transform method**.**

UNIT-IV: Vector Differentiation **10Hrs**

Vector point functions and scalar point functions – Gradient – Divergence and Curl – Directional derivatives – Angle between two surfaces – Scalar potential functions – Solenoidal and Irrotational vectors.

UNIT-V: Vector Integration **10Hrs**

Line, Surface and Volume Integrals. Theorems of Green, Gauss and Stokes (without proofs) and their applications.

Self-study content

UNIT-I	***UNIT- II***	***UNIT- III***	***UNIT-IV***	***UNIT-V***
Existence & Uniqueness of Solutions	*Linear Independence & Wronskian*	*Transform as Operator Mapping*	*Physical interpretation of Gradient, Divergence and Curl*	*Applications of Line, Surface and Volume integrals in physics*

TEXT BOOKS:

1. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.
2. R.K. Jain and S.R.K. Iyengar, Advanced Engineering Mathematics, Narosa Publications, 5th Edition, 2016.

REFERENCE BOOKS:

1. Erwin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, 9thEdition, Pearson, Reprint,2002
3. N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint,2008.
4. H. K. Dass and Er. Rajnish Verma, Higher Engineering Mathematics, S Chand and Company Limited, New Delhi.

Course Outcomes: After learning the contents of this paper, the student must be able to

CO-1. Identify whether the given differential equation of first order is exact or not

CO-2. Solve higher differential equation and apply the concept of differential equation to real world problems.

CO-3. Use the Laplace Transforms techniques for solving Ordinary Differential Equations.

CO-4. Analyse and solve problems involving gradient, divergence, and curl in various fields such as physics and engineering

CO-5. Evaluate the Line, Surface and Volume integrals and converting them one to another

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	3	1	1	1	-	-	-	-	-	1	2	1
CO-2	3	3	2	1	1	-	-	-	-	-	1	2	2
CO-3	3	3	2	1	1	-	-	-	-	-	1	2	1
CO-4	3	3	2	2	1	-	-	-	-	-	1	2	2
CO-5	3	2	1	2	1	-	-	-	-	-	1	1	1

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



I B.Tech. II Semester

Course Name	**: ENGINEERING CHEMISTRY**	L	T	P	C
Course Code	**: 2599204**	3	0	0	3
Course Category	: **Theory**				

CIE: 40 SEE: 60 Total: 100

Course Objectives:

1. To develop adaptability to new advances in Engineering Chemistry and acquire the essential skills to become a competent engineering professional.
2. To understand the industrial significance of water treatment, fundamental principles of battery chemistry, and the impact of corrosion along with its control methods for structural protection.
3. To impart foundational knowledge of various energy sources and their practical applications in engineering.
4. To equip students with an understanding of smart materials, biosensors, and analytical techniques applicable in engineering, industrial, environmental, and biomedical fields.
5. To acquire required knowledge about engineering materials like cement, UV-Visible, IR spectroscopy

UNIT-I: Water and its treatment: **12 Hrs**

Introduction- Hardness, types, degree of hardness and units. Estimation of hardness of water by complexometric method - Numerical problems for Temporary and permanent hardness. Potable water and its specifications - Steps involved in the treatment of potable water - Disinfection of potable water by chlorination and break-point chlorination. DE fluoridation- Nalgonda technique.

Boiler troubles: Scales, sludges and Caustic embrittlement. Internal treatment of boiler feed water - Calgon conditioning, Phosphate conditioning, Colloidal conditioning. External treatment methods - Softening of water by ion- exchange processes. Desalination of brackish water – Reverse osmosis.

UNIT-II: Electrochemistry and Corrosion: **13 Hrs**

Introduction- Electrode potential, standard electrode potential, Nernst equation (no derivation), electrochemical cell - Galvanic cell, cell representation, EMF of cell - Numerical problems. Types of electrodes, reference electrodes - Primary reference electrode - Standard Hydrogen Electrode (SHE), Secondary reference electrode - Calomel electrode. Construction, working and determination of pH of unknown solution using SHE and Calomel electrode.

Corrosion: Introduction- Definition, causes and effects of corrosion – Theories of corrosion, chemical and electrochemical theories of corrosion, Types of corrosion: galvanic, water-line and pitting corrosion. Factors affecting rate of corrosion - Nature of the metal (Position of metal in galvanic series, Purity of metal, Nature of corrosion product), Nature of the corroding environment (Temperature, Humidity, and pH).. Corrosion control methods - Cathodic protection Methods - Sacrificial anode and impressed current methods.

UNIT–III: Energy sources: **13 Hrs**

Batteries: Introduction – Classification of batteries - Primary, secondary and reserve batteries with examples. Construction, working and applications of Zn-air and Lithium-ion battery. Fuel Cells – Differences between a battery and a fuel cell, Construction and applications of Direct Methanol Fuel Cell (DMFC).

Fuels: Introduction and characteristics of a good fuel, Calorific value – Units- HCV, LCV- Dulong formula - Numerical problems.

Fossil fuels: Introduction, Classification, petroleum- Refining of Crude oil, Cracking - Types of cracking - Moving bed catalytic cracking. LPG and CNG composition and use.

Synthetic Fuels: Fischer-Tropsch process, Introduction and applications of Hy thane and Green Hydrogen.

UNIT- IV: Polymers: 13 **Hrs**

Definition - Classification of polymers: Based on origin and tacticity with examples – Types of polymerizations - Addition and condensation polymerization.

Plastics, Elastomers and Fibers: Definition and applications (PVC, Buna-S, Nylon-6,6), preparation, properties and applications of Bakelite. Differences between thermoplastics and thermo setting plastics, Conducting polymers: Definition and Classification with examples - Mechanism of conduction in trans-poly-acetylene and applications of conducting polymers.

Biodegradable polymers: Polylactic acid and its applications.

UNIT- V: Advanced Functional Materials: **10 Hrs**

Cement: Portland cement, its composition, setting and hardening. Biosensor - Definition, Amperometric Glucose monitor sensor. Interpretative spectroscopic applications of UV-Visible spectroscopy for Analysis of pollutants in dye industry, IR spectroscopy in night vision-security, Pollution Under Control- CO sensor (Passive Infrared detection).

Self-study content

UNIT-I	*UNIT- II*	*UNIT- III*	*UNIT-IV*	*UNIT-V*
Permutit process, Boiler Corrosion	*Conductimetric Titrations*	*Lead-Acid storage battery*	*Preparation and applications of Vulcanizations and Teflon polymers*	*Lubricants*

TEXT BOOKS:

1. Engineering Chemistry by P.C. Jain and M. Jain, Dhanpatrai Publishing Company, 2010.
2. Engineering Chemistry by Rama Devi,Dr.P.Aparna and Rath, Cengage learning, 2025.

REFERENCE BOOKS:

1. Engineering Chemistry: by Thirumala Chary Laxminarayana& Shashikala, Pearson Publications (2020)
2. Engineering Chemistry by Shashi Chawla, Dhanpatrai and Company (P) Ltd. Delhi 2011.
3. Engineering Chemistry by Shikha Agarwal, Cambridge University Press, Delhi 2015.
4. Engineering Analysis of Smart Material Systems by Donald J. Leo, Wiley, 2007.
5. Challenges and Opportunities in Green Hydrogen by Editors: Paramvir Singh, Avinash Kumar Agarwal, Anupma Thakur, R.K Sinha.
6. Raman Spectroscopy in Human Health and Biomedicine, https://www.worldscientific.com/doi/epdf/10.1142/13094
7. E-Content- https://doi.org/10.1142/13094 | October 2023
8. E-books: https://archive.org/details/EngineeringChemistryByShashiChawla/page/n11/mode/2u

Course Out comes: After completion of the course ,the student should be able to

CO-1. Students will be able to understand the fundamental properties of water and its applications in both domestic and industrial purposes.

CO-2. Students will gain basic knowledge of electrochemical processes and their relevance to corrosion and its control methods.

CO-3. Students will comprehend the significance and practical applications of batteries and various energy sources, enhancing their potential as future engineers and entrepreneurs

CO-4. Students will learn the basic concepts and properties of polymers and other engineering materials

CO-5. Students will be able to apply the principles of UV-Visible, IR spectroscopy in analyzing pollutants in dye industries and biomedical applications

Course Articulation Matrix

CO	PROGRAM OUTCOMES(PO)											PROGRAMSPECIFIC OUTCOMES(PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	2	3	1		3							1
CO-2	2	3	3	2	2	2						2	2
CO-3			3		1							2	2
CO-4			1		2							2	2
CO-5	2	3	2			2						1	1

CorrelationofCourseOutcomeswithProgramOutcomesandProgramSpecificOutcomesusing ***mappinglevels1=Slight,2=Moderate and 3=Substantial)***

SDG'S Mapping:



I B. Tech II Semester

		L	T	P	C
Course Name	**: English For Skill Enhancement**				
Course Code	**: 2599205**	3	0	0	3
Course Category	**: Theory**				

CIE: 40 SEE: 60 Total: 100

OBJECTIVES:

a. Improve their vocabulary.
b. Use appropriate sentence structures in their oral and written communication.
c. Develop their reading and study skills.
d. Equip students to write paragraphs, essays, précis and draft letters.
e. Acquire skills for technical report writing.

UNIT –I **12hrs**

Theme: **Perspectives**
Lesson on 'The Generation Gap' by Benjamin M. Spock from the prescribed textbook titled *English for the Young in the Digital World* published by Orient Black Swan Pvt. Ltd.

Vocabulary: The Concept of Word Formation -The Use of Prefixes and Suffixes - Words Often Misspelt - Synonyms and Antonyms

Grammar: Identifying Common Errors in Writing with Reference to Parts of Speech particularly Articles and Prepositions – Degrees of Comparison

Reading: Reading and Its Importance- Sub Skills of Reading – Skimming and Scanning.

Writing: Sentence Structures and Types -Use of Phrases and Clauses in Sentences- Importance of Proper Punctuation- Techniques for Writing Precisely –Nature and Style of Formal Writing.

UNIT –II **12hrs**

Theme: **Digital Transformation**
Lesson on '*Emerging Technologies*' from the prescribed textbook titled *English for the Young in the Digital World* published by Orient BlackSwan Pvt. Ltd.

Vocabulary: Homophones, Homonyms and Homographs

Grammar: Identifying Common Errors in Writing with Reference to Noun-pronoun Agreement and Subject-verb Agreement.

Reading: Reading Strategies-Guessing Meaning from Context – Identifying Main Ideas – Exercises for Practice

Writing: Paragraph Writing – Types, Structures and Features of a Paragraph - Creating Coherence – Linkers and Connectives - Organizing Principles in a Paragraph – Defining- Describing People, Objects, Places and Events – Classifying- Providing Examples or Evidence - Essay Writing - Writing Introduction and Conclusion.

UNIT –III **13hrs**

Theme: **Attitude and Gratitude**
Poems on '*Leisure'* by William Henry Davies and '*Be Thankful'* - Unknown Author from the prescribed textbook titled *English for the Young in the Digital World* published by Orient Black Swan Pvt. Ltd.

Vocabulary: Words Often Confused - Words from Foreign Languages and their Use in English.

Grammar: Identifying Common Errors in Writing with Reference to Misplaced Modifiers and Tenses.

Reading: Sub-Skills of Reading – Identifying Topic Sentence and Providing Supporting Ideas - Exercises for Practice.

Writing: Format of a Formal Letter-Writing Formal Letters E.g.., Letter of Complaint, Letter of Requisition, Job Application with CV/Resume –Difference between Writing a Letter and an Email - Email Etiquette.

UNIT –IV **12hrs**

Theme: **Entrepreneurship**
Lesson on '*Why a Start-Up Needs to Find its Customers First*' by Pranav Jain from the prescribed textbook titled *English for the Young in the Digital World* published by Orient Black Swan Pvt. Ltd.

Vocabulary: Standard Abbreviations in English – Inferring Meanings of Words through Context – Phrasal Verbs – Idioms.

Grammar: Redundancies and Clichés in Written Communication – Converting Passive to Active Voice and Vice-Versa.

Reading: Prompt Engineering Techniques– Comprehending and Generating Appropriate Prompts - Exercises for Practice

Writing: Writing Practices- Note Making-Précis Writing.

UNIT –V **12hrs**

Theme: **Integrity and Professionalism**
Lesson on '*Professional Ethics*' from the prescribed textbook titled *English for the Young in the Digital World* published by Orient Black Swan Pvt. Ltd.

Vocabulary: Technical Vocabulary and their Usage– One Word Substitutes – Collocations.

Grammar: Direct and Indirect Speech **-** Common Errors in English (Covering all the other aspects of grammar which were not covered in the previous units)

Reading: Survey, Question, Read, Recite and Review (SQ3R Method) – Inferring the Meaning and Evaluating a Text- Exercises for Practice

Writing: ***Report Writing - Technical Reports- Introduction – Characteristics of a Report – Categories of Reports Formats- Structure of Reports (Manuscript Format) - Types of Reports - Writing a Technical Report.***

Self-Study Content:

UNIT-I	***UNIT- II***	***UNIT- III***	***UNIT-IV***	***UNIT-V***
Enhancing vocabulary and word usage through self-study and contextual learning	*Developing independent reading skills using skimming, scanning, and critical reading strateaies.*	*Improving writing skills through self-practice of paragraphs, essays, letters, precis, and reports.*	*Applying digital and professional communication skills, including email etiquette and technical writing.*	*Reviewing and editing written work to ensure clarity, coherence, and correctness.*

TEXT BOOK:

1. Board of Editors*. 2025. English for the Young in the Digital World.* Orient Black Swan Pvt. Ltd.

REFERENCE BOOKS:

1. Swan, Michael. (2016). *Practical English Usage.* Oxford University Press. New Edition.
2. *Karal, Rajeevan. 2023. English Grammar Just for You.* Oxford University Press. New Delhi
3. 2024*. Empowering with Language: Communicative English for Undergraduates.* Cengage Learning India Pvt. Ltd. New Delhi
4. Sanjay Kumar &Pushp Lata. 2022. *Communication Skills – A Workbook.* Oxford University Press. New Delhi
5. Wood, F.T. (2007). *Remedial English Grammar.* Macmillan.
6. Vishwa Mohan, Aysha. (2013). *English for Technical Communication for Engineering Students.* Mc Graw-Hill Education India Pvt. Ltd.

COURSE OUTCOMES: Students will be able to:

1. Choose appropriate vocabulary in their oral and written communication.
2. Demonstrate their understanding of the rules of functional grammar and sentence structures.
3. Develop comprehension skills from known and unknown passages.

4. Write paragraphs, essays, précis and draft letters.
5. Write abstracts and reports in various contexts.

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOME (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**1**	**2**
CO1						2		1	3	3			2
CO2						1		2	3	3			2
CO3						1		1	1	2			1
CO4						1		2	2	3			2
CO5						2	1	2	3	3			2

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1= Slight, 2= Moderate and 3= Substantial**)

SDG'S Mapping:



I B. Tech II Semester

Course Name	**: ELECTRONIC DEVICES AND CIRCUITS**	L	T	P	C
Course Code	**: 2504201**	3	0	0	3
Course Category	**: Theory**				

CIE: 40 SEE: 60 Total: 100

COURSE OBJECTIVES:

1.To introduce components such as diodes and know the applications of Diode.
2.To understand the characteristics of the BJT.
3.To Learn the concepts of, load line analysis, biasing techniques
4.To analyze the transistor amplifier using h-parameter model
5.To understand the characteristics of the JFET and MOSFET.

UNIT - I: **12Hrs**

Diode Characteristics and Applications: PN junction diode – I-V characteristics, Diode resistance and capacitance, Rectifiers — Half-wave, Full- wave (Center-tap and bridge), clippers and clampers, Zener diode – I-V characteristics and voltage regulation.

UNIT - II: **12Hrs**

Bipolar Junction Transistor (BJT): Structure and working principle of BJT, Current components and transistor action, Configurations: Common Base (CB), Common Emitter (CE), Common Collector (CC), Input and output characteristics, Determination of h-parameters from transistor characteristics.

UNIT - III: **12Hrs**

BJT Biasing: Need for biasing and stabilization, Load line and operating point, Biasing techniques: Fixed bias, Collector-to-base bias, Voltage divider bias, Stability factors and thermal runaway

UNIT - IV: **12Hrs**

Transistor Amplifiers: Transistor as a small-signal amplifier, h-parameter equivalent circuit, CE, CB, CC amplifier analysis using h-parameters, Approximate CE model – with and without emitter bypass capacitor.

UNIT - V: **12Hrs**

Special Purpose Diodes: Principle of Operation of — SCR, Tunnel Diode, Varactor Diode.
Field Effect Transistors and Advanced Devices: JFET: Structure, operation, and characteristics, MOSFET: Enhancement and Depletion modes — Structure, operation, and characteristics.

Self-Study Content:

UNIT-I	***UNIT- II***	***UNIT- III***	***UNIT-IV***	***UNIT-V***
Temperature-dependent diode behavior	*Transistor as a switch*	*Thermal runaway,*	*Frequency response of BJT amplifiers*	*Enhancement vs depletion MOSFETs, PIN diode*

TEXT BOOKS:

1. Millman, Jacob, and Christos C. Halkias. *Electronic Devices and Circuits*. Tata

McGraw-Hill, 1991.
2. Boylestad, Robert L., and Louis Nashelsky. *Electronic Devices and Circuit Theory*. Pearson, 11th ed., 2013.

REFERENCE BOOKS:

1. Sedra, Adel S., and Kenneth C. Smith. *Microelectronic Circuits*. Oxford University Press, 7th ed., 2014.
2. Bell, David A. *Electronic Devices and Circuits*. Oxford University Press, 5th ed., 2008.
3. Neamen, Donald A. *Electronic Circuit Analysis and Design*. McGraw-Hill, 2nd ed., 2001.
4. Salivahanan, S., and N. Suresh Kumar. *Electronic Devices and Circuits*. McGraw-Hill Education, 4th ed., 2017.
5. Razavi, Behzad. *Fundamentals of Microelectronics*. Wiley, 2nd ed., 2013.
6. Taur, Yuan, and Tak H. Ning. *Fundamentals of Modern VLSI Devices*. Cambridge University Press, 2nd ed., 2009.

COURSE OUTCOMES:

CO1: Analyze the PN Junction diode operation and its characteristics
CO2: Analyze the characteristics of BJT.
CO3: Design the amplifiers with various biasing techniques
CO4: Analyze single stage amplifiers using hybrid model.
CO5: Analyze the characteristics of FET.

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	2	2	1	2	-	-	-	-	1	3	2	2
CO-2	3	3	2	2	2	-	-	-	-	1	3	2	2
CO-3	3	3	3	2	2	-	-	-	-	2	3	2	2
CO-4	3	3	3	3	2	-	-	-	-	2	3	2	2
CO-5	3	2	2	2	2	-	-	-	-	2	3	2	2

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



I B. Tech II Semester

Course Name	**: NETWORK ANALYSIS AND SYNTHESIS**	L	T	P	C
Course Code	**: 2504202**	3	0	0	3
Course Category	**: Theory**				

CIE: 40 SEE: 60 Total: 100

COURSE OBJECTIVES:

1 To understand the basic concepts on network topology
2 To know the behavior of the steady state and transient states in RLC circuits
3 To understand the two port network parameters.
4 Learn the design concepts of various filters and attenuators
5 Learning of implementation of transfer function

UNIT - I **12Hrs**

Network Topology: Graph, tree, co-tree, twig, link & fundamental matrix, Basic cut set and tie set matrices for planar networks, magnetic circuits, self and mutual inductances, dot convention, coefficient of coupling, equivalent T for Magnetically coupled circuits.

UNIT - II **12Hrs**

Transient and Steady state analysis: RC, RL and RLC Circuits, Sinusoidal, Step. RC Circuits as integrator and differentiators. 2^{Nd} order series RLC Circuit, resonance frequency, quality factor and bandwidth for series and parallel resonance, applications of resonance.

UNIT - III **12Hrs**

Two port network parameters: Z, Y, ABCD, h and g parameters, Characteristic impedance, image and iterative impedance, network function. Standard T, π, L Sections, Characteristic impedance, image transfer constants, Design of Attenuators, impedance matching network.

UNIT-IV **13Hrs**

Filters: Classification of Filters, Filter Networks, Constant-K Filters-Low pass, high pass, Band pass, band-stop filters, M-derived Filters- T and π filters- Low pass, high pass.

Attenuators: Types — T, π, L, Bridge T and lattice, Asymmetrical Attenuators T, π, L Equalizers- Types- Series, Shunt, Constant resistance, bridge T attenuation, bridge T phase, Lattice attenuation, lattice Phase equalizers.

UNIT – V **11Hrs**

Network Synthesis: Driving point impedance and admittance, transfer impedance and admittance, network functions of Ladder and non-ladder networks, synthesis of LC, RC and RL Functions by foster and causer methods.

Self-Study Content:

UNIT-I	*UNIT- II*	*UNIT- III*	*UNIT-IV*	*UNIT-V*
Directed and Undirected graph and Incidence Matrix	*Laplace transform of Equations*	*Stability of poles and zeros in s domain*	*Applications of attenuators.*	*Comparison of Cauer and Faster Methods*

TEXT BOOKS:

1. Van Valkenburg -Network Analysis, 3rd Ed., Pearson, 216.
2. JD Ryder - Networks, Lines and Fields, 2nd Ed., PHI, 1999.

REFERENCE BOOKS:

1. J. Edminister and M. Nahvi - Electric Circuits, Schaum's Outlines, Mc Graw Hills Education, 1999.
2. A. Sudhakar and Shyammohan S Palli - Networks & Circuits, 4th Ed., Tata McGraw- Hill Publications
3. William Hayt and Jack E. Kimmerley - Engineering Circuit Analysis, 6th Ed., William Hayt and Jack E. Kimmerley, McGraw Hill Company

COURSE OUTCOMES:

CO1: Gain knowledge on basic networks.
CO2: Analyze the Steady state and transient analysis of RLC Circuits
CO3: Characterization of two port network parameters
CO4: Analyze the Design aspect of various filters and attenuators
CO5: Analysis and designing of transfer function.

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	2	1				1				1	1	
CO-2	2	3	2				1				1	1	
CO-3	3	2	1								1	2	1
CO-4	2	3	3				1				1	3	1
CO-5	3	3	3									3	1

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



I B. Tech II Semester

		L	T	P	C
Course Name	**: ENGINEERING CHEMISTRY LAB**				
Course Code	**: 2599206**	0	0	2	1
Course Category	**: Practical**				

CIE: 40 SEE: 60 Total: 100

Course Description: The course includes experiments based on fundamental principles of chemistry essential for engineering students, aiming to develop practical skills and reinforce theoretical concepts.

Course Objectives

1. Students will understand and perform experiments based on core chemical principles relevant to engineering applications.
2. Students will learn to estimate the hardness of water to assess its suitability for drinking purposes.
3. Students will acquire the ability to perform acid-base titrations using instrumental methods such as conductometry, potentiometry, and pH metry.
4. Students will gain hands-on experience in synthesizing polymers like Bakelite and Nylon – 6, 6 in the laboratory.
5. Students will learn to determine the unknown concentration of potassium permanganate (KMnO4) using a calibration curve.

List of Experiments:

1. Estimation of Hardness of water by EDTA Complexometric method
2. Estimation of the concentration of strong acid by Conductometry.
3. Estimation of the concentration of strong and weak acid in an acid mixture by Conductometry.
4. Estimation of concentration of Fe^{+2}ion by Potentiometry using $KMnO_4$.
5. Estimation of concentration of strong acid with strong base by Potentiometry using quinhydrone
6. Determination of an acid concentration using pH meter.
7. Preparation of Bakelite.
8. Preparation Nylon – 6, 6.
9. Corrosion: Determination of rate of corrosion of mild steel in the presence and absence of inhibitor.
10. Verification of Lambert-Beer's law using $KMnO_4$.
11. Smart materials for Biomedical applications (Virtual lab)
12. Functioning of solar cell and its applications. (Virtual lab)

Self-Study Content :

- *Estimation of permanent Hardness of water by EDTA method.*
- *Estimation of the concentration of weak acid by Conductometry*
- *Determination of Weak acid strength by Strong base(potentiometer)*
- *Estimation of PH of different water samples.*
- *Determination of Acidity of water sample.*

REFERENCE BOOKS:

1. Lab manual for Engineering chemistry by B. Ramadevi and P. Aparna, S Chand Publications, New Delhi (2022)
2. Vogel's text book of practical organic chemistry 5th edition
3. Inorganic Quantitative analysis by A.I. Vogel, ELBS Publications.
4. College Practical Chemistry by V.K. Ahluwalia, Narosa Publications Ltd. New Delhi (2007

Course Outcomes:

CO-1. Students will develop practical skills through hands-on chemistry experiments relevant to engineering.

CO-2. Students will learn to determine important parameters such as water hardness and the corrosion rate of mild steel under various conditions.

CO-3. Students will be able to apply techniques like conductometry, potentiometry, and pH metry to

determine concentrations or equivalence points in acid-base reactions.

CO-4. Students will gain experience in synthesizing polymers such as Bakelite.

CO-5. Students will understand the working principle of colorimetry and the relationship between absorbance and concentration (Beer-Lambert Law).

Course Articulation Matrix

CO	PROGRAM OUTCOMES(PO)											PROGRAMSPECIFIC OUTCOMES(PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	2		3	1		3	1	1	2	2		3	
CO-2		2		2	2		2		2		1	2	
CO-3		2			1	3	1	3					2
CO-4			1		2	2					1		
CO-5	2		2			2	2	1				3	3

(CorrelationofCourseOutcomeswithProgramOutcomesandProgramSpecificOutcomesusing ***mappinglevels1=Slight,2=Moderate and 3=Substantial****)*

SDG'S Mapping:



I B.Tech. II Semester

		L	T	P	C
Course Name	**: ENGINEERING WORKSHOP**				
Course Code	**: 2503202**	0	0	2	1
Course Category	**: Practical**				

CIE: 40 SEE: 60 Total: 100

Course Objectives

- To introduce students to basic manufacturing processes and workshop practices.
- To provide hands-on training in carpentry, fitting, welding, sheet metal, and machining
- To develop skills in using and tools and measuring instruments.
- To enhance safety awareness and proper handling of workshop equipment.
- To build a foundational understanding of industrial production and fabrication.

1. TRADES FOR EXERCISES:

At least two exercises from each trade:

i. **Carpentry:** (T-Lap Joint, Dovetail Joint, Mortise & Tenon Joint).
ii. **Fitting:** (V-Fit, Dovetail Fit & Semi-circular fit).
iii. **Tin-Smithy:** (Square Tin, Rectangular Tray & Conical Funnel).
iv. **Foundry:** (Preparation of Green Sand Mould using Single Piece and Split Pattern).
v. **Welding Practice:** (Arc Welding & Gas Welding).
vi. **House-wiring:** (Parallel & Series, Two-way Switch and Tube Light).
vii. **Black Smithy:** (Round to Square, Fan Hook and S-Hook).

2. TRADES FOR DEMONSTRATION & EXPOSURE:

I. Plumbing, Machine Shop, Metal Cutting, Power tools in construction and Wood Working.

Self-Study Content :

- *Square Fit*
- *Elbow Pipe (Sheet Metal)*
- *Casting of Simple Aluminum Object*
- *Calling Bell Indicator Circuit*
- *U-Clamp Preparation*
- *Wooden stool/small Table preparation*

Textbooks:

1. Workshop Practice, B. L. Juneja, Cengage Learning India,1st edition,2015.
2. Workshop Manual, K.Venkata Reddy, BS Publication, 6thEdition, Rpt.2025.

References:

1. Workshop Manual, K.Venugopal, Anuradha Publications,2012thedition,2012.

Course Outcomes: After completion of the lab, the student should be able to

CO-1. Understand the basic manufacturing processes and operations.
CO-2. Use hand tools and equipment safely and efficiently.
CO-3. Perform basic operations in carpentry, fitting, welding, sheet metal work and machining

CO-4. Read and interpret work shop drawings
CO-5. Develop teamwork, time management and quality awareness in a workshop environment

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	1	2
CO-1	1	2	-	-	-	-	-	-	-	-	1	-	1
CO-2	1	2	-	-	-	-	-	-	-	-	1	-	1
CO-3	1	-	-	-	1	-	-	-	-	-	-	-	1
CO-4	1	-	-	-	1	-	-	-	-	-	-	-	-
CO-5	1	-	-	-	2	-	-	-	-	-	1	-	-

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



ANURAG
Engineering Engineers

I B. Tech II Semester

Course Name	: ENGLISH LANGUAGE AND COMMUNICATION SKILLS LAB	L	T	P	C
Course Code	:2599207	0	0	2	1
Course Category	: **Practical**				

CIE: 40 SEE: 60 Total: 100

Objectives

1. To enable students, develop their active listening skills
2. To equip students with necessary training in listening, so that they can comprehend the speech of people from different linguistic backgrounds
3. To improve their pronunciation and neutralize accent
4. To enable students express themselves fluently and appropriate
5. To practice speaking in social and professional contexts

Exercise – I **12 hrs**

CALL Lab:

Instruction: Speech Sounds-Listening Skill - Importance – Purpose - Types- Barriers- Active Listening

Practice: Listening to Distinguish Speech Sounds (Minimal Pairs) **-** *Testing Exercises*

ICS Lab:

❖ **Diagnostic Test – Activity titled '*Express Your View*'**

Instruction: Spoken and Written language - Formal and Informal English - Greetings - Introducing Oneself and Others

Practice: Any Ice-Breaking Activity

Exercise – II **12 hrs**

CALL Lab:

Instruction: Listening vs. Hearing – Listening for Sequence

Practice: Listening for General Information - Multiple Choice Questions - *Listening Comprehension Exercises*

ICS Lab:

Instruction: Features of Good Conversation – Strategies for Effective Communication- Understanding Non-Verbal Communication

Practice: Role Play Activity -Situational Dialogues –Expressions used in Various Situations –Making Requests and Seeking Permissions – Taking Leave - Telephone Etiquette- Silent Speech

Exercise – III **12 hrs**

CALL Lab:

Instruction: Past Tense and Plural Markers – Tips for Neutralizing Mother Tongue Influence (MTI)

Practice: Differences between British and American Pronunciation –*Listening Comprehension Exercises*

ICS Lab:

Instruction: Describing Objects, Situations, Places, People and Events

Practice: Picture Description Activity – Looking at a Picture and Describing Objects, Situations, Places, People and Events

Exercise – IV **12 hrs**

CALL Lab:

Instruction: Techniques for *Effective* Listening

Practice: Listening for Specific Details - Gap Fill Exercises - Past Tense and Plural Markers words- *Listening Exercise*

ICS Lab:
Instruction: How to Tell a Good Story - Story Star- Sequencing-Creativity
Practice: Activity on Telling and Retelling Stories - Collage

Exercise – V **13 hrs**
CALL Lab:
Instruction: Identifying the literal and implied meaning
Practice: Listening for Evaluation - Write the Summary – Listening Comprehension Exercises

ICS Lab:
Instruction: Group Discussion Skills
Practice: Mock Group Discussions-Debates

❖ **Post-Assessment Test on 'Express Your View'**

Minimum Requirement of infrastructural facilities for ELCS Lab:

1. Computer Assisted Language Learning (CALL) Lab:
The Computer Assisted Language Learning Lab has to accommodate 40 students with 40 systems, with one Master Console, LAN facility and English language learning software for self- study by students.
System Requirement (Hardware component):
Computer network with LAN facility (minimum 40 systems with multimedia) with the following specifications:

i) Computers with Suitable Configuration
ii) High Fidelity Headphones

2. Interactive Communication Skills (ICS) Lab:
The Interactive Communication Skills Lab: A Spacious room with movable chairs and audio-visual aids with a Public Address System, LCD etc.

Suggested Software:

- Cambridge Advanced Learners' English Dictionary with CD.
- Grammar Made Easy by Darling Kindersley.
- Punctuation Made Easy by Darling Kindersley.
- Oxford Advanced Learner's Compass, 10th Edition.
- English in Mind (Series 1-4), Herbert Puchta and Jeff Stranks with Meredith Levy, Cambridge.
- English Pronunciation in Use (Elementary, Intermediate, Advanced) Cambridge University Press.
- English Vocabulary in Use (Elementary, Intermediate, Advanced) Cambridge University Press.
- TOEFL & GRE (KAPLAN, AARCO & BARRONS, USA, Cracking GRE by CLIFFS).

Self-Learning Concepts:

- ❖ *Fundamentals of listening and speaking skills in English.*
- ❖ *Basics of English phonetics, pronunciation, stress, and intonation.*
- ❖ *Accent development and neutralization of mother tongue influence.*
- ❖ *Effective communication in formal, informal, and professional contexts.*
- ❖ *Public speaking, group discussion, and interview skills.*

REFERENCE BOOKS:

1. Shobha, KN &Rayen, J. Lourdes. (2019). *Communicative English – A workbook.* Cambridge University Press
2. Board of Editors. (2016). *ELCS Lab Manual: A Workbook for CALL and ICS Lab Activities.* Orient BlackSwanPvt. Ltd.
3. Mishra, Veerendra et al. (2020). *English Language Skills: A Practical Approach.* Cambridge University Press

4. (2022). *English Language Communication Skills – Lab Manual cum Workbook.* Cengage Learning India Pvt. Ltd.
5. Ur, Penny and Wright, Andrew. 2022. *Five Minute Activities – A Resource Book for Language Teachers.* Cambridge University Press.

Learning Outcomes: Students will be able to:

1. Listen actively and identify important information in spoken texts
2. Interpret the speech and infer the intention of the speaker
3. Improve their accent for intelligibility
4. Speak fluently with clarity and confidence
5. Use the language in real life situations

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOME (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO1							2	1	2	2	3	-	1
CO2							1	3	3	2	3	-	1
CO3							2	1	2	1	2	-	1
CO4							1	3	3	2	1	-	1
CO5							2	3	3	2	3	-	1

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1= Slight, 2= Moderate and 3= Substantial**)

SDG'S Mapping:



I B. Tech II Semester

		L	T	P	C
Course Name	**: ELECTRONIC DEVICES AND CIRCUITS LAB**				
Course Code	**: 2504203**	0	0	2	1
Course Category	**: Practical**				

CIE: 40 SEE: 60 Total: 100

COURSE OBJECTIVES:

1.Understand the working principles of semiconductor devices like diodes, BJTs, and FETs.
2.Build and test basic rectifier, clipper, clamper, and voltage regulator circuits.
3.Learn transistor biasing techniques and operating point determination.
4.Analyze and design transistor amplifier circuits in various configurations.
5.Use simulation software to design and validate electronic circuits.

List of Experiments

A. Hardware-Based Experiments (7):

1. Study the I–V characteristics of a PN junction diode in forward and reverse bias to determine cut-in voltage and dynamic resistance.
2. Examine the reverse bias characteristics of a Zener diode and demonstrate its application as a voltage regulator under varying conditions.
3. Design and analyze half-wave and full-wave rectifiers (center-tap and bridge) with and without capacitor filters to evaluate ripple factor and output voltage.
4. Implement clipper and clamper circuits to observe waveform shaping through positive, negative, and biased configurations.
5. Plot the input and output characteristics of a BJT in common emitter configuration to determine input/output resistance and current gain.
6. Design and test fixed bias and voltage divider bias circuits to establish a stable operating point for a BJT amplifier and study DC load line behavior.
7. Construct and analyze a Common Base (CB) configuration of a BJT to study input-output characteristics and determine current gain (α) and input/output resistance.

B. Software-Based Simulation Experiments (7):

1. Simulate a full-wave bridge rectifier with capacitor filter to analyze waveform smoothing and ripple reduction in DC power supply design.
2. Simulate a Zener diode-based voltage regulator to study voltage stabilization against varying supply voltages and load resistances.
3. Simulate a common emitter amplifier with and without emitter bypass capacitor to analyze the effect on voltage gain and signal amplification.
4. Simulate BJT operation as a switch and small-signal amplifier to understand its dual functionality in digital and analog applications.
5. Simulate the Drain and transfer characteristics of a JFET to determine parameters such as pinch-off voltage, drain resistance, and transconductance.
6. Implement clipper circuits to observe waveform shaping through positive, negative, and biased configurations.
7. Implement clamper circuits to observe waveform shaping through positive, negative, and biased configurations.

Hardware Requirements:

1. Regulated DC Power Supply (0–30V)
2. Function Generator
3. Digital Multimeter

4. Cathode Ray Oscilloscope (CRO) or DSO
5. Breadboards and Connecting Wires
6. Resistors, Capacitors, Diodes (1N4007, Zener Diodes)
7. BJTs (e.g., BC107, 2N2222), JFETs (e.g., J201), MOSFETs (e.g., IRF540N)
8. Trainer Kits (optional but preferred for ease)

Software Requirements (Any one of the listed tools or equivalent):

1. LT Spice (Free from Analog Devices)
2. NI Multisim (Academic License or Student Version)
3. Proteus Design Suite (Simulation and PCB Design)
4. TINA-TI (Free from Texas Instruments)
5. PSPICE for TI or OrCAD Lite
6. Windows PC or Laptop with minimum 4GB RAM and i3 processor or better

Self-study content

- *Advanced waveform-shaping and signal-conditioning circuits*
- *High-frequency applications of common-base configuration*

COURSE OUTCOMES:

CO1: Analyze the I–V characteristics of semiconductor devices such as diodes, BJTs, and FETs.
CO2: Design and evaluate basic rectifier, clipper, clamper, and voltage regulation circuits.
CO3: Demonstrate biasing techniques for BJTs and determine their operating point using DC load line analysis.
CO4: Design and analyze transistor amplifier circuits in various configurations using h-parameter models.
CO5: Simulate and interpret electronic circuits using appropriate simulation tools.

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	3	2	2	2	-	-	-	-	-	-	-	-
CO-2	3	3	2	2	2	1	-	-	-	-	-	1	-
CO-3	3	2	2	2	2	-	-	-	-	-	-	2	1
CO-4	2	2	2	3	3	-	-	-	-	-	-	2	2
CO-5	2	2	2	3	3	-	-	-	-	-	-	1	2

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



II B. Tech I Semester

Course Name	**: PROBABILITY THEORY AND STOCHASTIC PROCESSES**	L	T	P	C
Course Code	**: 2504303**	3	0	0	3
Course Category	**: Theory**				

CIE: 40 SEE: 60 Total: 100

Pre-requisite: Mathematics

Course Objectives:

1. This gives basic understanding of random variables
2. This gives basic understanding of operations that can be performed on them
3. To know the temporal characteristics of Random Process.
4. To know the Spectral characteristics of Random Process
5. To Learn the Basic concepts of Information theory Noise sources and its representation forunderstanding its characteristics.

UNIT - I **13Hrs**

Probability: Probability introduced through Sets and Relative Frequency: Experiments and Sample Spaces, Discrete and Continuous Sample Spaces, Events, Probability Definitions and Axioms, Joint Probability, Conditional Probability, Total Probability, Bay's Theorem, Independent Events.

Random Variables- Definition, Conditions for a Function to be a Random Variable, Discrete, Continuous and Mixed Random Variable, Distribution and Density functions, Properties, Binomial, Poisson, Uniform, Gaussian, Exponential, Conditional Distribution, Conditional Density and their Properties.

UNIT - II

Operations on single Random Variable **10Hrs**

Expected Value of a Random Variable, Function of a Random Variable, Moments about the Origin, Central Moments, Variance and Skew, Characteristic Function, Moment Generating Function, Transformations of a Random Variable - Monotonic and Non-monotonic Transformations of Continuous and Discrete Random Variable.

UNIT - III **13Hrs**

Multiple random variables and Operations on Multiple random variables: Vector Random Variables, Joint Distribution Function and its Properties, Marginal Distribution Functions, Conditional Distribution and Density– Point and Interval conditioning, Statistical Independence, Sum of Two and more Random Variables, Central Limit Theorem, Equal and Unequal Distribution (Proof not expected). Expected Value of a Function of Random Variables- Joint Moments about the Origin, Joint Central Moments, Joint Characteristic Functions, Jointly Gaussian Random Variables: Two Random Variables case, N Random Variable case, Properties'

UNIT IV **12Hrs**

Random processes – Temporal characteristics: The Random Process Concept, Classification Of Processes, Deterministic and Nondeterministic Processes, Distribution and Density Functions, concept of Stationarity and Statistical Independence. First-Order Stationary Processes, Second- Order and Wide- Sense Stationarity, (N-Order) and Strict-Sense Stationarity, Time Averages and Ergodicity, Mean- Ergodic Processes, Correlation-Ergodic

Processes, Autocorrelation Function and Its Properties, Cross- Correlation Function and Its Properties, Covariance Functions, Gaussian Random Processes, Poisson Random Process.

UNIT V **12Hrs**

Random processes – Spectral characteristics: The Power Spectrum: Properties, Relationship between Power Spectrum and Autocorrelation Function, The Cross-Power Density Spectrum, Properties, Relationship between Cross-Power Spectrum and Cross-Correlation Function.

Noise sources: Resistive / Thermal Noise Source, Arbitrary Noise Sources, Effective Noise Temperature, Noise equivalent bandwidth, Average Noise Figures, Average Noise Figure of cascaded networks, Narrow Band noise, Quadrature representation of narrow band noise & its properties.

Self-study content

UNIT-I	*UNIT- II*	*UNIT- III*	*UNIT-IV*	*UNIT-V*
Random variables in real communication system modelling	*Joint moment generating function and its properties*	*Real-time examples of ergodicity in communication signals*	*Practical PSD estimation using FFT*	*Noise figure measurement in RF systems*

TEXT BOOKS:

1. Peyton Z. Peebles - Probability, Random Variables & Random Signal Principles - TMH, 4th Edition
2. Murray R Spiegel, John Schiller, R Alu Srinivasan. – Probability and Statistics – Schaum's Outlines, 2nd Edition, TMH

REFERENCES:

1. P Ramesh Babu - Probability Theory and Random Processes – McGraw Hill Education
2. Athanasios Papoulis and S. Unnikrishna Pillai - Probability, Random Variables and Stochastic Processes — McGraw Hill Education, 4th Edition
3. K. N. Hari Bhat, K. Anitha Sheela and Jayant Ganguly - Probability Theory and Stochastic Processes for Engineers - Pearson, 1st Edition, 2011
4. Taub and Schilling - Principles of Communication systems by (TMH), 2008
5. Y Mallikarjuna Reddy - Probability Theory and Stochastic Processes, 4th Edition, University Press

Course Outcomes: Upon completing this course, the students will be able to:

1. Perform the basic operations on different types of probability distribution and density functions.
2. Perform operations on single and multiple Random variables
3. Perform operations on multiple Random variables
4. Determine the temporal characteristics of Random Signals
5. Understand the concepts of Noise and Determine the Spectral characteristics of Random Signals and Characterize Driven by stationary random process by using ACFs and PSDs

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	3	-	2	-	-	-	-	-	-	2	1	-
CO-2	3	3	-	2	-	-	-	-	-	-	-	1	-
CO-3	3	3	3	2	-	-	-	-	-	-	-	-	-
CO-4	3	3	3	2	-	-	-	-	-	-	2	2	1
CO-5	3	3	3	2	-	-	-	-	-	-	-	2	-

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



II B. Tech I Semester

Course Name	**: SIGNALS AND SYSTEMS**	L	T	P	C
Course Code	**: 2504304**	3	0	0	3
Course Category	**: Theory**				

CIE: 40 SEE: 60 Total: 100

Pre-Requisites: Mathematics

COURSE OBJECTIVES:

1.Classify signals and systems and their analysis in time and frequency domains.
2.Understand representation of signals using Fourier series and transform on Standard signals
3.Study the concepts of Linearity, Time Invariance and distortion less transmission through LTI systems,
4.Understand the concepts Laplace and their properties for analysis of Signals and systems.
5.Identify the need for sampling of CT signals, types and merits and demerits of Each type.

UNIT - I

Signal Analysis **10Hrs**

Classification of Signals and systems, Exponential and Sinusoidal signals, Concepts of Impulse function, Unit Step function, Signum function. Analogy between Vectors and Signals, Orthogonal Signal Space, Signal approximation using Orthogonal functions, Mean Square Error.

UNIT - II **13Hrs**

Fourier series: Representation of Fourier series, Continuous time periodic signals, Dirichlet's conditions, Trigonometric Fourier Series and Exponential Fourier Series, Complex Fourier spectrum.

Fourier Transforms: Deriving Fourier Transform from Fourier series, Fourier Transform of standard signals, Fourier Transform of Periodic Signals, Properties of Fourier Transform, Fourier Transforms involving Impulse function and Signum function.

UNIT - III **12Hrs**

Signal Transmission through Linear Systems: Linear System, Impulse response, Response of a Linear System. Linear Time Invariant (LTI) System, Linear Time Variant (LTV) System, Transfer function of a LTI System, Filter characteristic of Linear System, Distortion less transmission through a system, Signal bandwidth, System Bandwidth, Causality and Paley-Wiener criterion for physical realization, Relationship between Bandwidth and risetime.

UNIT - IV **13Hrs**

Laplace Transforms: Laplace Transforms (L.T), Inverse Laplace Transform, Concept of Region of Convergence (ROC) for Laplace Transforms, Properties of L.T, Relation between L.T and F.T of a signal.

Correlation: Auto Correlation and Cross Correlation Functions, Relation between Convolution and Correlation, Properties of Correlation Functions, Energy Density Spectrum, Power Density Spectrum, Relation between Autocorrelation Function and Energy/Power Spectral Density Function, Parseval's Theorem.

UNIT - V **12Hrs**

Sampling theorem: Graphical and analytical proof of Sampling Theorem for Base band/Band Limited and Band Pass Signals, Types of Sampling: Impulse Sampling, Natural and Flattop

Sampling, Reconstruction of signal from its samples, Effect of under sampling — Aliasing, **Z–Transforms:** Concept of Z- Transform of a Discrete Sequence, Distinction between Laplace, Fourier and Z Transforms, Region of Convergence in Z-Transform, Constraints on ROC for various classes of signals, Inverse Z-transform, Properties of Z-transforms.

Self-study content

UNIT-I	UNIT- II	UNIT- III	UNIT-IV	UNIT-V
Energy vs. power signals in real communication systems	*Introduction to Hilbert Transform*	*Digital vs. analog filter characteristics*	*Correlation of Random Processes*	*Z-transform applications in DSP algorithms*

TEXT BOOKS

1. Signals, Systems & Communications -B.P. Lathi, BS Publications.
2. Signals and Systems – Allan. V. Oppenheim, Allan. S. Willsky with S. Hamid. Nawab, 2nd Ed. Pearson.

REFERENCE BOOKS

1. Signals and Systems–Simon Haykin, Barry Van Veen, 2nd Ed., Wiley.
2. Signals and Systems – A. Rama Krishna Rao, 2008, TMH.
3. Fundamentals of Signals and Systems – Michel J. Roberts, Govind Sharma, 2nd Ed., MGH.
4. Signals, Systems and Transforms - Charles. L. Philips, John M. Parr and Eve A. Riskin, 4th Ed., 2004, Pearson, Prentice Hall.

COURSE OUTCOMES:

CO1: Characterize various signals, system and approximate a signal in terms of another
CO2: Apply the transform on standard and arbitrary signals.
CO3: Identify the conditions for transmission of signals through systems and conditions for physical realization of systems.
CO4: Interpret the concepts of Laplace for analysis of signals and systems.
CO5: Use sampling theorem for baseband and band pass signals for various types of sampling

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	2	1	1	1						1	2	
CO-2	3	3	1	2	2						1	3	1
CO-3	3	3	2	2	2	1					1	3	1
CO-4	3	2	1	2	2						1	3	
CO-5	3	3	2	2	2	1					1	3	2

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



II B. Tech I Semester

Course Name	**: DIGITAL LOGIC DESIGN**	L	T	P	C
Course Code	**: 2504305**	3	0	0	3
Course Category	**: Theory**				

CIE: 40 SEE: 60 Total: 100

COURSE OBJECTIVES:

1.To understand common forms of number representation in logic circuits.
2. To learn basic techniques for the design of digital circuits and fundamental concepts used in the design of digital systems.
3. To understand the concepts of combinational logic circuits and sequential circuits.
4. To understand the Realization of Logic Gates Using Diodes & Transistors.
5. To understand the concept of FSM and ASM charts.

UNIT – I: **14Hrs**

Number Systems: Binary, Octal, Decimal, Hexadecimal, Fixed-point and Floating-point Number Representations, Complements of Numbers: 1's and 2's Complement, Error Detection and Correction Codes: Parity Check, Hamming Code.

Boolean Algebra and Logic Gates: Axiomatic definitions, basic theorems and properties, Boolean Functions: Canonical and standard forms, Digital Logic Gates Overview.

UNIT - II: **10Hrs**

Gate-Level Minimization Techniques: Karnaugh maps: 2, 3, and 4 variables, Sum-of-products (SOP) and product-of-sums (POS) simplification, don't care conditions, Implementation using NAND and NOR gates.

UNIT - III: **10Hrs**

Combinational Logic Circuits: Analysis and design procedures, Binary adder-subtractor and BCD adder, magnitude comparator, decoders, encoders, multiplexers and demultiplexers.

UNIT - IV: **12Hrs**

Sequential Logic Circuits: Gated latches, Flip-flops: Clocked S-R, D, T, JK, Master-Slave JK, Design of synchronous and asynchronous counters, Shift registers: types and applications.

UNIT - V: **14Hrs**

Synchronous Sequential Logic Moore and Mealy state machines, State diagrams, state tables, and state reduction, Case studies: sequence detector, **Programmable Logic Devices:** Memory devices - RAM, ROM, Programmable Logic Arrays (PLA), Programmable Array Logic (PAL)

Self-study content

UNIT-I	***UNIT- II***	***UNIT- III***	***UNIT-IV***	***UNIT-V***
Why Binary Dominates Digital System, Self-Complementing Codes and Trade-off Between NAND and NOR Gates	*Optimization of Logics Gates with Constraints*	*Faults in Combinational Circuits and Optimization Goals*	*Designing of Sequence Generators*	*Analysis of FSM Sequential Behaviour.*

TEXT BOOK:

1. M. Morris Mano, Michael D. Ciletti, *Digital Design with an Introduction to the Verilog*

HDL, 6th Edition, Pearson Education/PHI, 2017.

REFERENCE BOOKS:

1. Ronald J. Tocci, Neal S. Widmer, Gregory L. Moss, *Digital Systems: Principles and Applications*, 10th Edition, Pearson Education.
2. Charles H. Roth Jr., Larry L. Kinney, *Fundamentals of Logic Design*, 6th Edition, Cengage Learning.

COURSE OUTCOMES:

CO1. Acquire the knowledge on numerical information in different forms and Boolean algebra theorems.
CO2. Define Postulates of Boolean algebra and to minimize combinational functions, and design the combinational circuits.
CO3. Design and analyze sequential circuits for various cyclic functions.
CO4. Characterize logic families and analyze them for the purpose of AC and DC Parameters.
CO5. Understand the concept of FSM and ASM charts

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**1**	**2**
CO-1	3	2	3	1	2	1	-	-	-	-	2	2	-
CO-2	3	2	2	1	2	1	-	-	-	-	2	2	-
CO-3	2	3	3	2	2	1	-	-	-	-	1	2	-
CO-4	3	2	1	1	1	1	-	-	-	-	-	1	-
CO-5	2	-	2	3	-	-	-	-	-	-	-	2	-

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



II B. Tech I Semester

Course Name	**: ELECTRONIC CIRCUIT ANALYSIS**	L	T	P	C
Course Code	**: 2504306**	3	0	0	3
Course Category	**: Theory**				

CIE: 40 SEE: 60 Total: 100

COURSE OBJECTIVES:

1 To give understanding of various types of amplifier circuits
2 Learn the concepts of Feedback amplifiers
3 To familiarize the Concept of feedback in amplifiers so as to differentiate between negative and positive feedback
4 Learn the concepts of Power Amplifiers.
5 Understand various multivibrators and time base generators

UNIT - I: **13Hrs**

Multistage Amplifiers: Classification of Amplifiers, Distortion in Amplifiers, Coupling schemes: RC, Transformer, Direct coupling, Frequency response of multistage amplifiers, Transistor configuration choice in cascade amplifiers, Cascade and Cascode amplifiers, Darlington pair amplifier.

High-Frequency Transistor Model: Hybrid-π model, Hybrid-π parameters: CE short-circuit current gain, Gain with resistive load.

UNIT - II: **10Hrs**

Feedback Amplifiers: Concept and need for feedback in amplifiers, Types and classification of feedback amplifiers, Characteristics of negative feedback: Gain stability, bandwidth, noise, distortion, Voltage series, Voltage shunt, Current series, Current shunt configurations.

UNIT - III: **12Hrs**

Oscillators: Principle of positive feedback, Barkhausen Criterion for oscillations, LC Oscillators: Generalized analysis, Hartley, Colpitts, RC Oscillators: RC phase shift, Wien bridge, Crystal oscillator: Working and advantages

UNIT - IV: **12Hrs**

Power Amplifiers: Classification: Class A, B, AB, C, Series-fed Class A amplifier, Transformer-coupled Class A amplifier, Class B amplifier: Push-pull, Complementary symmetry, Efficiency calculations and Crossover distortion.

UNIT - V: **13Hrs**

Multivibrators: Analysis and design of Bistable, Monostable and Astable multivibrators and Schmitt Trigger using transistors. **Time Base Generators**: General features of a time base signal, Miller and Bootstrap time base generators.

Self-study content

UNIT-I	*UNIT- II*	*UNIT- III*	*UNIT-IV*	*UNIT-V*
Miller Effect	*Nyquist Stability Criterion*	*Relaxation Oscillators*	*Thermal Runaway in Power Transistors*	*Sawtooth and Triangular Wave Generators*

TEXT BOOKS:

1. Millman, Jacob, and Christos C. Halkias. *Electronic Devices and Circuits*. McGraw-Hill Education, 2008.

2. Bell, David A. *Electronic Devices and Circuits*. Oxford University Press, 2008.

REFERENCE BOOKS:

1. Boylestad, Robert L., and Louis Nashelsky. *Electronic Devices and Circuit Theory*. 11th ed., Pearson Education, 2013.
2. Millman, Jacob, and Arvin Grabel. *Microelectronics*. 2nd ed., McGraw-Hill, 1987.
3. Malvino, Albert Paul. *Electronic Principles*. 7th ed., McGraw-Hill Education, 2007.
4. Millman, Jacob, and Herbert Taub. *Pulse, Digital, and Switching Waveforms*. McGraw-Hill Education, 1991.
5. Sedra, Adel S., and Kenneth C. Smith. *Microelectronic Circuits*. 7th ed., Oxford University Press, 2015.

COURSE OUTCOMES:

CO1: Design multistage amplifiers and understand the concepts of High Frequency Analysis of BJT.
CO2: Utilize the Concepts of negative feedback to improve the stability of amplifiers and positive feedback to sustained oscillations
CO3: Utilize the Concepts of positive feedback to improve the stability of Oscillators to sustained oscillations
CO4: Design the power amplifiers
CO5: Design Multivibrators and Time Base Generators for various applications.

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	2	3	3	2							1		
CO-2	2	3	3	2							1		1
CO-3	2	2	3	1							1	1	
CO-4	3	3	3	1		3	2				1	3	2
CO-5	3	3	3	1		3	2				1	3	2

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



II B.Tech. I Semester

Course Name	:	**PYTHON PROGRAMMING**				
Course Code	:	**25005301**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To introduce programming languages, problem-solving methods, algorithms, flowcharts, and the basics of Python.
- To understand Python data types, variables, operators, and input/output operations.
- To apply control statements and arrays for solving computational problems.
- To develop modular programs using functions, recursion, and scope of variables.
- To implement exception handling and file operations for robust Python applications.

UNIT – I **12 Hrs**

INTRODUCTION TO PROGRAMMING LANGUAGES: Computer definition, Computer system, Programming languages: Types, Uses. Program development steps, Algorithm, Flowchart, Solving simple problems.
INTRODUCTION TO PYTHON: History of Python, Features of Python, Flavors of Python, Execution of a Python Program, Python Virtual Machine, Viewing the byte code, Comparisons between C and Python, Comments in Python, Docstrings.

UNIT – II **12 Hrs**

DATATYPES IN PYTHON: Built-in data types: None, Numeric and Boolean data type, Sequences in Python: strings, bytes, bytearrays, list, tuple, sets, frozenset, dictionaries. User-defined data types, constants in Python, identifiers and reserved words, naming conventions in Python.
VARIABLES AND OPERATORS: Understanding Python variables, multiple variable declarations; Operators in Python - Arithmetic operators, Assignment operators, Relational Operators, Logical operators, Boolean Operators, Bitwise operators, Membership operators, Identity operators. Operator Precedence and Associativity.
Input Output Statements: Output statements, Input Statements and Command Line Arguments, A word on Indentation.

UNIT – III **12 Hrs**

CONTROL STATEMENTS: The if Statement, the if else statements, the if…Elif…else statements, while loop, for loop, Infinite loop, Nested Loops, else suite, break, continue, pass statement, assert statement.
ARRAYS: Introduction to Arrays, Creating an Array, Types of arrays, Importing the array Module and NumPy, Indexing and Slicing on NumPy Arrays, Mathematical Operations on Arrays, Comparison of Arrays.

UNIT – IV **12 Hrs**

FUNCTIONS: Introduction, defining a Function, calling a function, Formal and Actual Arguments, Positional Arguments, keyword Arguments, Default Arguments, variable length arguments, local and global variables, global keyword, nesting of functions and Recursion.

UNIT – V **12 Hrs**

EXCEPTIONS IN PYTHON: Errors in Python Program - Compile-Time Errors, Runtime Errors, Logical

Errors; Exceptions, Exception Handling, Types of Exceptions, and Keywords in Exception Handling.
FILES IN PYTHON: Types of Files in Python, opening a File, closing a File, Random File accessing methods- seek () and tell ().

Self-study content

UNIT-I	UNIT- II	UNIT- III	UNIT-IV	UNIT-V
Python Integrated Development Environment (IDE)	*Type Conversion and Type Casting in Python*	*List Comprehension in Python*	*Lambda Functions in Python*	*CSV File Handling in Python*

Textbooks:
1. R. Nageswara Rao, "Core Python Programming", Dream tech
2. Fundamentals of Python First Programs, Kenneth A. Lambert, 2nd Edition, Cengage.

Reference Books:
1. Allen B. Downey, "Think Python: How to Think like a Computer Scientist", 2nd edition, Updated for Python 3, Shroff/O 'Reilly Publishers, 2016.
2. Core Python Programming, W. Chun, Pearson.

Course Outcomes:

After completion of the Course, the student should be able to
CO-1. Understand problem-solving strategies, algorithms, flowcharts, and basics of Python programming.
CO-2. Apply Python data types, variables, operators, and I/O operations to solve simple problems.
CO-3. Use control structures and arrays to implement Python programs.
CO-4. Develop modular solutions using functions, recursion, and scope management.
CO-5. Implement exception handling and file operations for building reliable Python applications.

Course Articulation Matrix:

CO	Program Outcomes (PO)											Program Specific Outcomes (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	2	1		1					1		3	1
CO-2	3	3	2		2					1		3	2
CO-3	2	3	3		3	1				2	1	3	2
CO-4	2	2	2		2	1				2	2	2	2
CO-5	2	3	3		3	2		1		2	2	3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



II B. Tech I Semester

		L	T	P	C
Course Name	**: Modelling and Simulation Lab**				
Course Code	**: 2504307**	0	0	2	1
Course Category	**: Practical**				

CIE: 40 SEE: 60 Total: 100

Note:

- All the experiments are to be simulated using MATLAB or equivalent software
- Minimum of 12 experiments are to be completed / simulated.

COURSE OBJECTIVES:

1.To familiarize students with simulation tools such as MATLAB for modeling and analysis of signals and systems.
2.To develop skills in analyzing the properties of continuous and discrete-time signals and systems.
3.To implement Fourier, Laplace, and Z-domain analysis using computational tools.
4.To study the impact of random processes and noise in signal modeling.
5.To design and simulate applications involving real-time signals and communication systems.

List of Experiments:

Signals and Systems (Minimum 7 Experiments)

1. Write the code / script for generating various standard viz: Periodic and Aperiodic, Unit Impulse, Unit Step, Square, saw tooth, Triangular, Sinusoidal, Ramp, Sinc and Nonstandard Signals and Sequences generated from these standard signals /sequences using Waveform synthesis. Also, for perform different operations viz: Addition, Multiplication, Scaling, Shifting, Folding, Computation of Energy and Average Power on them.
2. Write the code / script for finding the Even and Odd parts of Signal / Sequence and Real and Imaginary parts of Signal.
3. Write the code / script for finding the output of a System for a given input and Impulse Response and finding Auto Correlation and Cross Correlation of Signals / sequences
4. Write the code / script for Verifying whether a given Continuous/Discrete System is Linear, Time Invariant, Stable and Physically Realizable
5. Write the code / script for obtaining Sinusoidal response and Impulse response of a given Continuous / Discrete LTI System.
   a) Plot the Real and Imaginary part and
   b) Magnitude and Phase Plot of the response
6. Write the code / script for finding and plotting the Magnitude and Phase Spectrum of any given Signal by finding its Fourier Transform by using the properties where ever required.
7. Write the code / script for finding and plotting the Magnitude and Phase Spectrum of any given Signal by finding its Laplace Transform by using the properties where ever required. Also plot pole-zero diagram in S-plane
8. Write the code/ script for finding and plotting the Magnitude and Phase Spectrum of any given Sequence by finding its Z-Transform by using the properties wherever required. Also plot pole — zero diagram in Z-plane
9. Gibbs Phenomenon and waveform synthesis

Probability Theory and Stochastic Processes (Minimum 3 Experiments)

10. Write the code / script for generating various Random Variables with different CDFs/ PDFs
11. Write the code / script for generating Gaussian noise and for finding its mean, Skewness, Kurtosis, PDF and PSD.
12. Write the code / script for Verifying Sampling theorem for different sampling rates,

Sampling types and Duty Cycles and for plotting the sampled and reconstructed Signals.

13. Write the code / script for Removal of noise from the signal using Cross correlation.
14. Write the code / script for Extraction of Periodic Signal masked by noise using Auto Correlation

Note: For the experiments with code/scripts written in MATLAB or equivalent (1-8, 10-14), the student can design a user interface or app using MATLAB App Designer or equivalent.

Application on Real Time signals

1. Application of Autocorrelation: GPS Synchronization Satellite communication toolbox is required for this experiment.

Generate the GPS signal. Visualize the GPS signal. Plot of autocorrelation of C/A code and visualize the spectrum of GPS signals. For exact steps, go through the following page: https://www.mathworks.com/help/satcom/ug/gps-waveform-generation.html

2. Sampling of Speech Signals

Record and play speech in MATLAB. For steps, go through the following page: https://in.mathworks.com/help/matlab/import_export/record-and-play-audio.html

Change the sampling rate of the recorded speech signal and play back to see the effect of aliasing. For steps, go through the following page: https://in.mathworks.com/help/signal/ug/changing-signal-sample-rate.html

Self-study content

- ❖ Introduction to MATLAB/Simulink and its Applications in Modelling and Simulation
- ❖ Monte Carlo Simulation and Random Number Generation Technique

COURSE OUTCOMES:

CO1: Generate and analyze standard and derived signals/sequences in time and frequency domains.
CO2: Apply MATLAB or equivalent tools for system analysis including LTI properties, stability, and realizability.
CO3: Evaluate spectral characteristics of signals using Fourier, Laplace, and Z-transform techniques.
CO4: Analyze the effect of noise and random processes using probability and statistical tools.
CO5: Apply autocorrelation and cross-correlation methods for signal extraction and synchronization in real-time applications.

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	2	2	3	2	-	-	-	-	1	1	3	2
CO-2	3	3	3	3	2	-	-	-	-	1	1	3	2
CO-3	3	3	3	3	2	-	-	-	-	2	1	3	2
CO-4	3	3	2	3	-	-	-	-	-	2	1	2	3
CO-5	3	2	3	3	3	-	*	-	-	2	1	3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



II B. Tech I Semester

Course Name	**: DIGITAL LOGIC DESIGN LAB**	L	T	P	C
Course Code	**: 2504308**	0	0	2	1
Course Category	**: Practical**				

CIE: 40 SEE: 60 Total: 100

List of Experiments

A. Realization in Hardware Laboratory (Using Logic ICs)

These are fundamental hands-on experiments conducted using logic ICs such as AND, OR, NOT, NAND, NOR, XOR gates, flip-flops, multiplexers, and decoders.

1. Realize and minimize Boolean functions using basic gates and universal gates (NAND/NOR) in SOP/POS'S form.
2. Design and implement Half Adder, Full Adder, Half Subtractor, and Full Subtractor using logic gates.
3. Construct and analyze basic logic gates (AND, OR, NOT, XOR, XNOR) using only NAND and NOR gates.
4. Design and implement parity bit generators (even and odd) and a 4-input majority logic circuit.
5. Design and implement code converters such as Binary to Gray, Gray to Binary, and BCD to Excess-3 using gates.
6. Design and implement simple combinational circuits: 2-to-1 multiplexer, 1-bit comparator, and 7-segment decoder logic.

B. Verilog HDL-Based Digital Design Experiments (Simulation-Based)

These experiments are implemented using **Verilog HDL** with different modeling styles (dataflow, behavioral, structural) and simulated using tools like **Viv ado, Model Sim, or Xilinx ISE**.

1. Design and simulate a 2-bit comparator using dataflow modeling; extend it to 4-bit using structural modeling.
2. Implement a 2:1 multiplexer using dataflow modeling and design an 8:1 multiplexer using structural modeling.
3. Design a 2-to-4 decoder using dataflow modeling and realize a 3-to-8 decoder using structural modeling.
4. Implement a given Boolean function using a decoder-based approach in behavioral modeling.
5. Design and simulate a universal n-bit shift register (left, right, hold, parallel load) using behavioral modeling.
6. Design a synchronous MOD-n counter using behavioral modeling with D or JK flip-flops.
7. Design and simulate an asynchronous (ripple) counter for a custom sequence using structural modeling.
8. Implement a sequence detector for a given binary pattern using FSM (Moore/Mealy) in behavioral modeling.

Self-study content

- *Realize all logic gates using TTL Logic family.*
- *Realize 8 to 3 encoders using logic gates and verify the truth table.*

Required Hardware (for Hardware Lab Experiments)

Component	Description
Digital Trainer Kit	Breadboard with power supply and clock generator
Logic ICs	7400 (NAND), 7402 (NOR), 7408 (AND), 7432 (OR), 7486 (XOR), 7404 (NOT), etc.
Flip-Flop ICs	7474 (D Flip-Flop), 7476 (JK Flip-Flop)
MUX/Decoder ICs	74153, 74138, 74139
LEDs, switches, connecting wires	For I/O interface and testing

Required Software Tools (for Verilog HDL Experiments) (Any one of the tool below)

Software	Purpose
Xilinx Vivado	HDL simulation and synthesis (preferred tool)
ModelSim	Verilog simulation and waveform analysis
Xilinx ISE	Legacy support for simulation and FPGA design

Course Outcomes: Upon completing this course, the students will be able to

1. Acquire the knowledge on numerical information in different forms and Boolean Algebra theorems.
2. Define Postulates of Boolean algebra and to minimize combinational functions, and designthe combinational circuits.
3. Design and analyze sequential circuits for various cyclic functions.
4. Characterize logic families and analyze them for the purpose of AC and DC parameters.
5. Design logic circuits using different logic families.

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	2	3	1	2	1			1		2	2	3
CO-2	3	2	2	1	2	1			1		2	1	3
CO-3	2	3	3	2	2	1			1		1	2	3
CO-4	3	2	1	1	1							1	3
CO-5	3	3	3	2	1					2	1	1	2

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



II B. Tech I Semester

Course Name	**: ELECTRONIC CIRCUIT ANALYSIS LAB**	L	T	P	C
Course Code	**: 2504309**	0	0	2	1
Course Category	**: Practical**				

CIE: 40 SEE: 60 Total: 100

COURSE OBJECTIVES:

1 To observe the frequency response of various amplifiers
2 To examine the effect of negative feedback amplifiers
3 To observe the output signal with respect to oscillation.
4 To observe the output signal and calculate the efficiency and distortion.
5 To observe the waveforms for various multivibrators and time base generators

List of Experiments:

A. Hardware Experiments (7):

Perform practical design, implementation, and waveform analysis of amplifiers, oscillators, power stages, and multivibrators to validate theoretical concepts and observe real-world circuit behavior.

1. Design and analyze a two-stage RC coupled amplifier to demonstrate gain enhancement and study coupling capacitance effects.
2. Design Hartley and Colpitts oscillators for a specified frequency and observe their output waveforms.
3. Design an RC phase shift oscillator and derive the practical gain condition for oscillations at a given frequency.
4. Design a transformer-coupled class A power amplifier, observe input/output waveforms, and calculate efficiency.
5. Design a class B power amplifier, analyze input/output waveforms, and evaluate harmonic distortion.
6. Design a bistable multivibrator, analyze commutating capacitor effects, and record transistor waveforms.
7. Design an astable multivibrator and observe transistor base and collector waveforms.

B. Software Simulations (7):

Use circuit simulation software to design, analyze, and verify the performance of feedback amplifiers, waveform generators, and power amplifier circuits through virtual experimentation and frequency response evaluation.

1.Design two stage CE Amplifier and observe the frequency response
2.Simulate four feedback amplifier topologies and compare their frequency responses with and without feedback.
3.Simulate a monostable multivibrator and analyze its input/output waveforms.
4.Simulate a Schmitt trigger for gain values greater than and less than one and analyze response behavior.
5.Simulate a bootstrap time base generator using BJT and observe the output sweep waveform.
6.Simulate a Miller sweep circuit using BJT and observe the time base output waveform.
7.Simulate a complementary symmetry push-pull amplifier and verify elimination of crossover distortion.
8.Simulate Astable Multivibrator and analyze its input and output waveform
9.Simulate Wein bridge oscillator and analyze output waveform

Self-study content

- *Execute bistable multivibrator as binary circuit.*

- *Execute delay time for monostable multivibrator theoretically and practically.*

Software Requirements:
Simulation Tools: LTspice / Multisim / PSpice / Proteus / NI Multisim Live or equivalent
Operating System: Windows 10/11 or Linux (Ubuntu preferred)
Hardware Requirements:

1. Dual Power Supply (±15V, 0–30V)
2. Function Generator (up to 1 MHz)
3. CRO / DSO (Dual Channel, 20 MHz or more)
4. Digital Multimeters
5. Breadboards and Connecting Wires
6. BJTs: BC107, BC547, BC557, 2N2222, etc.
7. Resistors, Capacitors (Wide range of values)
8. Transformers (for power amplifiers)
9. Inductors, Crystals (1 MHz, 4 MHz, etc.)
   Heat sinks, transistors for power stages (e.g., TIP41, TIP42 etc.)

COURSE OUTCOMES:
CO1: Examine the effect multistage amplification on frequency response
CO2: Investigate negative feedback concept in amplifiers.
CO3: Investigate positive feedback concept in Oscillators
CO4: Design power amplifiers and find its efficiency
CO5: Design various multivibrators and sweep circuits. Understand the necessity of linearity

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	1	-	2	3	-	-	-	3	3	-	1	2	2
CO-2	1	-	2	3	-	-	-	3	3	-	1	2	2
CO-3	1	-	2	2	-	-	-	2	2	-	1	1	1
CO-4	1	-	3	3	3	-	-	-	3	3	1	3	3
CO-5	1	-	3	3	3	-	-	-	3	3	1	2	2

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



II B.Tech. I Semester

Course Name	:	**PYTHON PROGRAMMING LAB**				
Course Code	:	**2505302**	L	T	P	C
Course Category	:	**Practical**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To familiarize students with Python environment setup, syntax, operators, and basic input/output programs.
- To develop problem-solving skills using decision-making, control structures, and iterative constructs.
- To implement and manipulate Python data structures such as lists, tuples, dictionaries, and arrays (including NumPy).
- To design modular programs using user-defined functions and recursion.
- To implement file handling, exception handling, and command-line arguments for robust applications.

List of Experiments:

Week – 1:

1. Demonstrate the installation of Python for windows
2. Demonstrate the execution of Python program in Jupyter notebook and IDLE
3. Demonstrate the importing of NumPy, pandas, matplotlib and mention briefly about their purpose.
4. Write a Python program to assign a value to a variable and print the value of the variable.

Week – 2:

5. Write a Python program to assign different values to a single variable and print the value of the variable.
6. Write a Python program to assign multiple values to multiple variables and print the value of the variable.
7. Write a python program for printing values using format() method, % operator and format specifier.
8. Write a Python program to read two numbers from the user and print their sum and difference.

Week – 3:

9. Write a Python program implementing a simple calculator using all arithmetic operators.
10. Write a Python program illustrating the relation between the given two numbers using relational operators.
11. Write a Python program illustrating all logical operators.
12. Write a Python program illustrating all bitwise operators.

Week – 4:

13. Write a Python program to perform the swapping of the given two numbers:
    i) Using a temporary variable ii) Without using a temporary variable
14. Write a Python program to evaluate the following expressions:
    i) r // a + b * c – d / e ii) r //((a+b) * ((c- d) / e))
15. Write a Python program to calculate Simple Interest: SI = (P x T x R) /100

Week – 5:

16. Write a Python program to check whether the given number is positive or negative using a simple if statement.
17. Write a Python program to find the largest of the given three numbers using a selection statement.
18. Write a Python program to display the corresponding day of a given number in a week.
19. Write a Python program to check whether the given year is a leap year or not.

Week – 6:

20. Write a Python program to print natural numbers up to a given range.
21. Write a Python program to print the factors of a given number.
22. Write a Python program to check whether the given number is Armstrong or not.

Week – 7:

23. Write a Python program to print numbers that are divisible by a given number up to a given range

using unconditional statements.

24. Write a Python program to create a list and print all elements of the list using the list method.

Week – 8:

25. Write a Python program demonstrating the following operations on a tuple:
    i) Create a tuple and copy the contents of the list.
    ii) Delete elements from the tuple after displaying the contents.
26. Write a Python program demonstrating the following operations on a dictionary:
    i) Create a dictionary and display the contents.
    ii) Create a dictionary having multiple values for the same key.
    iii) Concatenate two lists and convert them into a dictionary

Week – 9:

27. Write a Python program to create a single-dimensional array and display its contents.
28. Write a Python program implementing the following operations on arrays using the array module:
    i) Print the first n elements
    ii) Print the elements with the given stride value
    iii) Print the elements in reverse order.
    iv) Copy the contents into another array.

Week – 10:

29. Write a Python program implementing the following operations on arrays using the NumPy module:
    i) Create a 2x3 array and add elements
    ii) Convert the dimensions of the array into 3x2.
    iii) Convert the dimensions of the array into 1x6.
    iv) Copy the contents into another array.
30. Write a Python program to perform Matrix addition and Matrix multiplication.

Week – 11:

31. Write a Python program to find the sum of n natural numbers using user-defined functions.
32. Write a Python program having a function that takes a list and returns a sorted list.

Week – 12:

33. Write a Python program to print the factorial of a given number using recursive function.
34. Write a Python program to display the Fibonacci series up to a given range using functions:

i) Without using recursion ii) Using recursion

Week – 13:

35. Write a Python program illustrating the occurrence of an exception.
36. Write a Python program to create a new file and add the given text to the file.

Week – 14:

37. Write a Python program demonstrating command-line arguments.
38. Write a Python program to add the contents into an existing file and display the contents of a file from the specified position.

Self-study content

- ❖ Write a Python program to plot a line graph using matplotlib.
- ❖ Write a Python program to implement a simple student record management system

Textbooks:

1. **R. Nageswara Rao, "Core Python Programming", Dream tech**
2. **Fundamentals of Python First Programs, Kenneth A. Lambert, 2nd Edition, Cengage.**

Reference Books:

1. **Allen B. Downey, "Think Python: How to Think like a Computer Scientist", 2nd edition, Updated for Python 3, Shroff/O 'Reilly Publishers, 2016.**
2. **Core Python Programming, W. Chun, Pearson.**

Course Outcomes:

After completion of the Course, the student should be able to

- **CO-1.** Apply Python syntax, operators, and I/O features to write and execute programs.
- **CO-2.** Apply operators, control statements, and loops to solve computational problems.
- **CO-3.** Use data structures like lists, tuples, dictionaries, and arrays to develop Python programs.

CO-4. Develop modular solutions using functions and recursion.
CO-5. Implement file operations, exception handling, and command-line arguments in Python.

Course Articulation Matrix:

CO	Program Outcomes (PO)											Program Specific Outcomes (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	1	2	3	2	1							1	2
CO-2	2	2	1	2	1							2	1
CO-3	2	2	2	1	2							3	2
CO-4	2	1	3	2	2							2	1
CO-5	2	3	1	2	3							1	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



ANURAG
Engineering Engineers

II B.Tech. I Semester

		L	T	P	C
Course Name	**: LINUX AND SHELL SCRIPTING**				
Course Code	**:2505303**	0	0	2	1
Course Category	: **Skill Development Course**				

CIE: 40 SEE: 60 Total: 100

Course Objectives:

- To familiarize students with Linux operating system commands, file handling, and process management.
- To enable students to write shell scripts using decision-making and conditional statements.
- To apply looping constructs in shell scripts for solving mathematical and computational problems.
- To design modular scripts using functions and arrays for automation tasks.
- To develop robust shell scripts for text processing, error handling, and practical problem solving.

List of Experiments

Fundamental LINUX Commands:

1. Demonstrate the basic Linux Commands
   i. ls ii. cd iii. pwd iv. cp v.mv vi. rm vii.mkdir viii. rmdir
2. Demonstrate the following
   i. File permissions and ownership in Linux (chmod, chown)
   ii. Process management (ps, kill)
3. Exercise on Text processing utilities (e.g., grep, awk, sed, cut, sort).

Shell Scripting Fundamentals:

Conditional statements (if-else, case):

4. i. Write a Shell Script to check the given number is Even or Odd.
   ii. Write a Shell Script to find the Biggest of 3 numbers.
5. Write a Menu Driven Shell Script using **case** statement.

Looping constructs (for, while, until):

6. i. Write a Shell Script to find Factorial of a given Number.
   ii. Write a Shell Script to find the Sum of Digits of a number.
7. i. Write a Shell Script to Reverse a number and check whether it is Palindrome or not.
   ii. Write a Shell script to print Multiplication Table of given number.
8. i. Write a Shell script to print all Prime numbers between the range 1 to n where n>1.
   ii. Write a Shell script to print the sum of 1 to n numbers where n>1.

Functions and Arrays in shell scripts:

9. Write a Shell script to demonstrate functions.
10. Write a Shell script to demonstrate Arrays.

Self-study content

- *Linux System Administration and Process Management*
- *Advanced Shell Scripting: Functions, Arrays, and File Handling*

Textbooks:

1. Sumit Abha Das, Unix Concepts and Applications, 4th edition, TMH, 2006.
2. Unix Network Programming, W.R. Stevens, PHI.

References:

1. N. Mathew, R. Stones, Wrox, Beginning Linux Programming, 4th edition, Wiley India Edition.
2. W.R. Stevens, Unix Network Programming, PHI, 2001.
3. Graham Glass, King Ables, Unix for programmers and users, 3rd Edition, Pearson Education, 2003.

4. B.A. Forouzan and R.F. Gilberg, Unix and Shell programming, Cengage Learning, 2009.
5. W.R. Stevens, Advanced Programming in the Unix Environment, 2nd Edition, Pearson Education, 2001.

Course Outcomes:

After completion of the Course, the student should be able to

CO-1. Apply fundamental Linux commands for file management, permissions, and process handling.
CO-2. Develop shell scripts using conditional statements for decision-making tasks.
CO-3. Implement shell scripts using looping constructs to solve computational problems.
CO-4. Design modular shell scripts using functions and arrays to handle automation tasks.
CO-5. Create robust shell scripts for text processing, error handling, and problem solving.

Course Articulation Matrix:

CO	Program Outcomes (PO)											Program Specific Outcomes (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	2	1		3					1		3	1
CO-2	3	3	2		3					1		3	2
CO-3	3	3	3		3	1				2		3	3
CO-4	2	3	3		3	1				2		3	3
CO-5	2	3	3	1	3	2				2		3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



II B.Tech. I Semester

		L	T	P	C
Course Name	**: Environmental Science /IKS**				
Course Code	**: 2599304**	1	0	0	1
Course Category	: Theory				

CIE: 0 SEE: 100 Total: 100

INDIAN KNOWLEDGE SYSTEM

UNIT-I:

Introduction to Indian Knowledge Systems: Meaning, Nature, Scope and Salient Aspects of Bharatiya Jnana Parampara - Introduction to Vedas, Upanishads, Vidya, Kala, Jnana, Shastra - Practices and Continuity of Tradition. Concept of Wellness – Yoga System - Ayurveda System - Ancient Indian Aesthetics

UNIT-II:

Overview of History of Indian Education and Scientific Literature: Gurukul System - Role of Sanskrit in Natural Language Processing - Scientific Literature – Vedic Literature - Available Scientific Treatises - Interlinking's

UNIT-III:

Development of Science & Technology in India: Superposition theorem, Thevenin's theorem, Norton's theorems (AC & DC), Maximum power transfer theorem, Tellegen's theorem, Compensation theorem, Millman's theorem and Reciprocity theorem (DC only).

1. Science and technology heritage of India
2. Science and technology heritage: Literary Sources
3. Science and technology heritage: Physical structures
4. Science and technology heritage: Temples
5. Science and technology heritage: Watershed management
6. Dyes, arts and perfume production.
7. Shipbuilding
8. Status of Indigenous S & T

Suggested Readings:

1. B. Mahadevan, Bhat Vinayak and Nagendra Pavan R.N., (2022) 'Introduction to Indian
2. Knowledge Systems: Concepts and Applications' PHI learning PVT, New Delhi
3. Dharmapal (1971) 'Indian Science and Technology in the Eighteenth Century'. Other India
4. Press, Goa.
5. Kapil Kapoor, Singh Avdhesh Kumar, (2005) 'Indian Knowledge Systems' D.K. Printworld (P)Ltd.
6. Chakradeo, Ujwala, Temples of Bharat, Aayu Publications, New Delhi, 2024.
7. D.N. Bose, S.N. Sen and B. V. Subbarayappa, A Concise History of Science in India, Indian National Science Academy, New Delhi, 2009.
8. Datta B. and A. N. Singh, History of Hindu Mathematics: Parts I and II, Asia Publishing House, Bombay, 1962.
9. Kapoor, K. (2021), Indian Knowledge System: Nature, Philosophy, Character in Indian
10. Knowledge System, vol. 1, Pub. Indian Institute of Advanced Studies, Shimla
11. Mahadevan, B., Bhat, V.R., Pavana, N. (2022), Philosophical Systems, in Introduction to Indian Knowledge System, Pub. PHI Learning, New Delhi.
12. Mahadevan, B., Bhat, V.R., Pavana, N. (2022), Knowledge: Framework and Classification, in Introduction to Indian Knowledge System, Pub. PHI Learning, New Delhi.

ENVIRONMENTAL SCIENCE

UNIT – IV:

Ecosystems & Natural Resources: Definition, Scope, and Importance of ecosystem. Classification, structure, and function of an ecosystem, Food chains, food webs, and ecological pyramids. Flow of energy, Bioaccumulation, ecosystem value.

Classification of Resources: Living and Non-Living resources, water resources: use and over utilization of surface and ground water, floods and droughts, Dams: benefits and problems. Mineral resources: use and exploitation, environmental effects of extracting and using mineral resources, Land resources: Forest resources, Energy resources: growing energy needs, renewable and non-renewable energy sources, use of alternate energy source.

UNIT – V:

Environmental Pollution and Control Technologies: Environmental Pollution: Classification of pollution, Ambient air quality standards.

Overview of air pollution control technologies, Concepts of bioremediation. Global Environmental Issues and Global Efforts: Climate change and impacts on human environment. Ozone depletion and Ozone depleting substances (ODS). Deforestation and desertification. International conventions / Protocols: Earth Summit, Kyoto Protocol, and Montréal Protocol. NAPCC-GoI Initiatives. Sustainable development goals (SDGs); Global environmental challenges; Environmental policies and international agreements.

Text Books:

1. Introduction to Environmental Science by Y. Anjaneyulu, BS. Publications.
2. Textbook of Environmental Studies for Undergraduate Courses by Erach Bharucha for University Grants Commission.
3. Environmental Studies by R. Rajagopalan, Oxford University Press.

Reference Books:

1. Environmental Science: towards a sustainable future by Richard T. Wright. 2008 PHL Learning Private Ltd. New Delhi.
2. Environmental Engineering and science by Gilbert M. Masters and Wendell P. Ela. 2008 PHI Learning Pvt. Ltd.
3. Environmental Science by Daniel B. Botkin & Edward A. Keller, Wiley INDIA edition.
4. Environmental Studies by Anubha Kaushik, 4th Edition, New age international publishers.

Text book of Environmental Science and Technology - Dr. M. Anji Reddy 2007, BS Publications

Course Outcomes:

After the completion of the course, the student will be able to understand

1. The multidisciplinary nature of environment, essence & function of environment
2. About the natural resource's utilization and their conservation
3. The importance of Biodiversity, reasons for its dwindling and its Conservation
4. About the causes and effects of environmental pollution and its management as well as environmental issues and the agreements and protocols the world and India is implementing
5. About the environmental wastes management rules, regulations and EIA for the protection of environment and to achieve sustainable development.

Course Articulation Matrix:

CO	PROGRAM OUTCOMES(PO)											PROGRAM SPECIFIC OUTCOMES(PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1			1			1		1					1
CO-2	1		1			2	2	1	3	2	1		1
CO-3			1			1		1	2				1
CO-4	1		1			2	3	1	3	2	2		1
CO-5			1			1	3	1	3	2	2		1

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



ANURAG
Engineering Engineers

II B.Tech. II Semester

Course Name	:	**NUMERICAL METHODS AND COMPLEX VARIABLES**				
Course Code	:	**2599402**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Course Objectives: To learn

- Expressing periodic function by Fourier series and a non-periodic function by Fourier transforms
- Various numerical methods to find roots of polynomial and transcendental equations.
- Concept of finite differences and to estimate the value for the given data using interpolation.
- Evaluation of integrals using numerical techniques
- Solving ordinary differential equations of first order using numerical techniques.
- Differentiation and integration of complex valued functions.
- Evaluation of integrals using Cauchy's integral formula and Cauchy's residue theorem.
- Expansion of complex functions using Laurent's series.

UNIT-I: Fourier Series & Fourier Transforms **10 Hrs**

Fourier series: Dirichlet's Conditions – Half-range Fourier series

Fourier Transforms: Fourier Integral Theorem (Only statements), Fourier Sine and Cosine transforms (Elementary illustrations)

UNIT-II: Numerical Methods-I **10Hrs**

Solution of polynomial and transcendental equations: Bisection method – Iteration Method – Newton-Raphson method and Regula-Falsi method.

Finite differences: forward differences – backward differences – Interpolation using Newton's forward and backward difference formulae – Lagrange's method of interpolation.

UNIT-III: Numerical Methods-II **10 Hrs**

Numerical integration: Trapezoidal rule - Simpson's 1/3rd and 3/8th rules.

Ordinary differential equations: Taylor's series – Euler's method – Runge-Kutta method of fourth order for first order ODE.

UNIT-IV: Complex Differentiation **10 Hrs**

Differentiation of Complex functions – Analyticity – Cauchy-Riemann equations (without proof) –Harmonic Functions – Finding harmonic conjugate – Milne-Thomson method – Elementary analytic functions (exponential, trigonometric, logarithm).

UNIT-V: Complex Integration **10 Hrs**

Line integral – Cauchy's theorem – Cauchy's Integral formula – Zeros of analytic functions – Singularities – Laurent's series. Residues – Cauchy Residue theorem (All theorems without Proof).

Self-study content

UNIT-I	***UNIT- II***	***UNIT- III***	***UNIT-IV***	***UNIT-V***
Orthogonal Function Systems	*Polynomial Approximation Theory*	*Consistency, Stability & Convergence*	*Analytic Functions are Power Series*	*Classification of Singularities*

TEXT BOOKS:

1. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010.
2. S.S. Sastry, Introductory methods of numerical analysis, PHI, 4th Edition, 2005.

REFERENCES:

1. Murray R. Spiegel, Ph.D., Seymour Lipschutz, Ph.D., John J. Schiller, Ph.D., Dennis Spellman, Ph.D., Complex Variables (Schaum's outline).
2. M. K. Jain, S.R.K. Iyengar, R.K. Jain, Numerical methods for Scientific and Engineering Computations, New Age International publishers.
3. Erwin Kreyszig, Advanced Engineering Mathematics, 9thEdition, John Wiley &Sons,2006.
4. J. W. Brown and R. V. Churchill, Complex Variables and Applications, 7th Edition, Mc Graw Hill, 2004.

Course Outcomes: After learning the contents of this paper, the student must be able to

CO-1. Express any periodic function in terms of sine and cosine.

CO-2. Find the root of a given polynomial and transcendental equations. Estimate the value for the given data using interpolation.

CO-3. Find the numerical solutions for a given first order ODE's

CO-4. Analyse the complex function with reference to their analyticity, integration using Cauchy's integral and residue theorems.

CO-5. Find the integrals using Cauchy's integral formula and Cauchy's residue theorem. Expansions of Laurent's series in complex function.

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**1**	**2**
CO-1	3	2	1	2	1	-	-	-	-	-	1	1	1
CO-2	3	3	2	2	2	-	-	-	-	-	1	2	2
CO-3	3	3	2	3	2	-	-	-	-	-	1	2	2
CO-4	3	2	1	2	1	-	-	-	-	-	-	2	1
CO-5	3	2	1	1	1	-	-	-	-	-	-	1	1

CO-1. *(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using*
CO-2. ***mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



II B.Tech. II Semester

Course Name	:	**ELECTROMAGNETIC FIELDS AND TRANSMISSION LINES**				
Course Code	:	**2504403**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Pre-requisite: Mathematics

COURSE OBJECTIVES:

1To learn the Basic Laws, Concepts and proofs related to Electrostatic Fields
2 To learn the Basic Laws, Concepts and proofs related to magnetostatic Fields
3 To distinguish between static and time-varying fields, and understand the significance and utility of Maxwell's Equations
4 To Analyze the characteristics of uniform plane waves, determine their propagation parameters
5 To study the propagation, reflection and transmission of plane waves inbounded and unbounded media

UNIT I – Electrostatics **13Hrs**

Review of Coordinate Systems & Vector Calculus, Coulomb's Law, Electric Field Intensity — Fields due to Different Charge Distributions, Electric Flux Density, Gauss Law and its applications, Electric Potential, Relation between E and V, Maxwell's Equations for Electrostatic Fields, Convection and Conduction Currents, Dielectric Constant, Isotropic and Homogeneous Dielectrics, Continuity Equation, Relaxation Time, Poisson's and Laplace's Equations.

UNIT II - Magnetostatics **10Hrs**

Biot-Savart's Law, Ampere's Circuit Law and its applications, Magnetic Flux Density, Maxwell's equations for Magnetostatic Fields, Magnetic Scalar and Vector Potentials, Forces due to Magnetic Fields, Ampere's Force Law.

UNIT III - Maxwell's Equations (Time Varying Fields) **12Hrs**

Faraday's Law, Transformer and Motional EMF, Inconsistency in Ampere's Law and Displacement Current Density, Maxwell's Equations in Differential, Integral and Phasor form. Electric and magnetic Boundary Conditions (Dielectric – Dielectric, Conductor– Dielectric, Conductor– Free Space interfaces).

UNIT IV - EM Wave Characteristics **12Hrs**

Wave Equations for Conducting and Perfect Dielectric Media, Uniform Plane Waves– Definitions, Relation between E&H, Wave Propagation in Lossless and Conducting Media, Conductors & Dielectrics — Characterization, Wave Propagation in Good Conductors and Good Dielectrics, Skin Depth, Surface Impedance, Wave Polarization. Poynting Vector and Poynting Theorem.

UNIT V Transmission Lines **13Hrs**

Types, Parameters, Equivalent Circuit, Transmission Line Equations, Primary & Secondary Constants, Expressions for Characteristic Impedance, Propagation Constant, Phase and Group Velocities, Infinite Line Concepts, Lossless Lines, Types of Distortions, condition for Distortion less transmission lines, Loading — Types of Loading, Input Impedance, SC and OC Lines, Reflection Coefficient, VSWR, Impedance Transformations - λ/4, λ/2, λ/8 Lines, Smith Chart- Configuration and Applications, Single Stub Matching.

Self-study content

UNIT-I	UNIT- II	UNIT- III	UNIT-IV	UNIT-V
Electrostatic Shielding and Its Applications	*Magnetic Shielding and Its Applications*	*Electromagnetic Compatibility (EMC) and Interference (EMI)*	*Electromagnetic Waveguides and Their Applications*	*Optical Fiber as a Transmission Medium and Their Applications*

TEXT BOOKS:

1. Engineering Electromagnetics — William H. Hayt Jr. and John A. Buck,8thEd., McGrawHill,2014
2. Principles of Electromagnetics –Matthew N.O. Sadiku and S.V. Kulkarni, 6thEd., Oxford University Press, Asian Edition, 2015.

REFERENCES:

1. Electromagnetic Waves and Radiating Systems–E.C. Jordan and K.G. Balmain,2ndEd., PHI,2000.
2. Engineering Electromagnetics — NathanIda, 2ndEd., Springer (India) Pvt. Ltd., New Delhi,2005.
3. Electromagnetic Field Theory Fundamentals –Bhag Singh Guru and Husey in R. Hiziroglu, Cambridge University Press, 2nd Ed., 2006.

COURSE OUTCOMES:

CO1: Acquire the knowledge of Basic Laws, Concepts and proofs related to Electrostatic Fields
CO2: Acquire the knowledge of Basic Laws, Concepts and proofs related to Magneto static Fields.
CO3: Characterize the static and time-varying fields; establish the corresponding sets of Maxwell's Equations and Boundary Conditions
CO4: Analyze the Wave Equations and classify conductors, dielectrics
CO5: Analyze the Design aspect of transmission line parameters and configurations.

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	3	2	1		1				1		2	2
CO-2	3	3	2	1		1				1		1	1
CO-3	3	3	2	1		1				1		1	1
CO-4	3	3	2	1		1				1		2	1
CO-5	3				3						3	1	1

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



II B.Tech. II Semester

Course Name	:	**ANALOG AND DIGITAL COMMUNICATIONS**				
Course Code	:	**2504404**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Pre-requisite: Signals and Systems

COURSE OBJECTIVES:

1. To Understand the generation, detection of various Continuous wave Modulation techniques.
2. To acquire theoretical knowledge of each block in AM, FM transmitters and Receivers
3. To understand and analyze the Concepts of Baseband transmission and optimum Receiver.
4. To Understand the generation, detection of various Digital Modulation techniques
5. To understand the various Coding techniques

UNIT - I

Amplitude Modulation **14Hrs**

Need for modulation, Amplitude Modulation: Time and frequency domain description, Generation — Switching modulator, Detection - Envelope detector, DSB-SC Modulation: Generation — Balanced Modulator, Detection- Synchronous detector, COSTAS Loop, SSB Modulation: Generation — Phase discrimination Method and Demodulation - coherent detection, Vestigial side band modulation and demodulation. **Angle Modulation:** Basic concepts of Phase Modulation, Frequency Modulation: Single tone frequency modulation, Spectrum Analysis, Carson's Rule, Generation of FM Waves- Armstrong Method, Detection of FM Waves - Phase locked loop, Comparison of FM and AM.

UNIT - II

Transmitters & Receivers **12Hrs**

Classification of Transmitters, AM Transmitters, FM Transmitters**,** AM Receiver - Super heterodyne receiver, FM Receivers, Stereo FM multiplex reception, Comparison of AM and FM Receiver. Pre- emphasis, and de-emphasis **Pulse Modulation**: Types of Pulse modulation- PAM, PWM and PPM, Comparison of FDM and TDM.

UNIT - III **12Hrs**

Detection and Estimation: Model of Digital Communication Systems, Detection of Known Signals in Noise, Probability of error, Optimum Receivers Using Coherent Detection: Matched filter Receiver and its Properties, Correlation receiver. **Base Band Shaping for Data Transmission:** Requirements of a line encoding format, various line encoding formats- Unipolar, Polar, Bipolar, Discrete PAM signals, inter symbol interference, Nyquist's criterion, Eye pattern.

UNIT - IV

Digital Modulation Techniques: **12Hrs**

PCM Generation and Reconstruction, Quantization Noise, Non-Uniform Quantization and Companding, DPCM, DM and Adaptive DM, Noise in PCM and DM. Digital Modulation formats, Coherent binary modulation techniques (BPSK, BFSK), Coherent quadrature modulation techniques (QPSK), Non-Coherent binary modulation techniques (BFSK, DPSK), QAM, M-ary Digital modulation techniques (PSK, FSK, QAM), bandwidth efficiency, constellation diagrams.

UNIT - V **10Hrs**

Information theory: Entropy, Information rate, Mutual information, Channel capacity of discrete channel, Shannon-Hartley law; Trade-off between bandwidth and SNR. Source coding - Huffman coding, Shannon Fano coding, Channel coding - Linear block codes and cyclic codes.

Self-study content

UNIT-I	UNIT- II	UNIT- III	UNIT-IV	UNIT-V
Noise performance comparison of AM and DSB-SC systems. Detailed study of Bessel functions and sideband distribution in FM.	*Modern digital radio receivers and SDR (Software Defined Radio) basics*	*Delta modulation stability analysis.*	*Constellation diagrams of QPSK and QAM*	*Error Detection and correction codes*

TEXT BOOKS:

1. Electronics Communication Systems-Fundamentals through Advanced-Wayne Tomasi, 5th Edition, PHI, 2009.
2. Digital and Analog Communication System – K. Sam Shanmugam, Wiley, 2019.

REFERENCES:

1. Electronic Communications – Dennis Roddy and John Coolean, 4th Edition, PEA, 2004.
2. Electronics & Communication System – George Kennedy and Bernard Davis, TMH, 2004.
3. Communication System - Simon Haykin and Michael Moher, Wiley, 5th edition, 2022.
4. Principles of Communication Systems - Herbert Taub, Donald L Schiling, Goutam Saha, 3rd Edition, McGraw-Hill, 2008.

COURSE OUTCOMES:

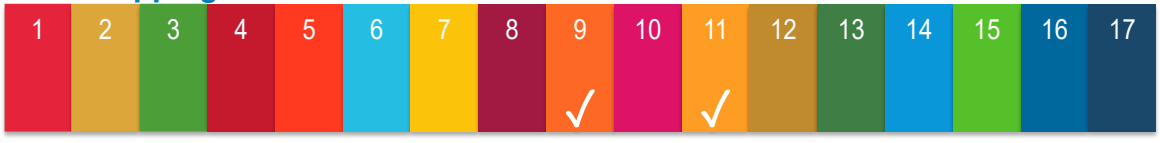
CO1: Design and analyze various Analog Modulation and Demodulation Techniques
CO2: Understand the Concept of Superheterodyne Receiver and Pulse Modulation Techniques
CO3: Emphasize on modern digital data transmission concepts and optimization of receivers.
CO4: Design and analyze various Digital Modulation and Demodulation Techniques
CO5: Analyze and design the various coding techniques

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	3	3	1	-	3	2	-	-	-	1	2	2
CO-2	3	3	3	1	-	2	2	-	-	–	1	2	2
CO-3	3	3	3	1	-	3	2	-	-	-	1	2	2
CO-4	3	3	3	1	-	3	2	-	-	-	1	2	2
CO-5	3	3	3	1	-	3	2	-	-	-	1	2	2

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



II B.Tech. II Semester

Course Name	:	**LINEAR AND DIGITAL IC APPLICATIONS**				
Course Code	:	**2504405**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Pre-requisite: Switching Theory and Logic Design.

COURSE OBJECTIVES:

1. To introduce the basic building blocks of linear integrated circuits.
2. To theory and applications of Analog Multipliers and PLLs.
3. To introduce the concepts of waveform generation and introduce some special function ICs.
4. To understand and implement the working of basic digital circuits.

UNIT - I

Operational Amplifier **12Hrs**

Ideal and Practical Op-Amp Characteristics, Features of 741 Op- Amp, Modes of Operation - Inverting, Non-Inverting, Differential, Instrumentation Amplifier, Differentiators and Integrators, Comparators, Schmitt Trigger, Introduction to Voltage Regulators, Features of 723 Regulator.

UNIT - II

Op-Amp, IC-555 & IC565 Applications **12Hrs**

Introduction to Active Filters, Characteristics of Band pass, Band reject and All Pass Filters, Analysis of 1st order LPF & HPF Butterworth Filters, Waveform Generators — Triangular, Sawtooth, IC555 Timer - Functional Diagram, Monostable and Astable Operations, Applications, IC565 PLL - Block Schematic, principle and Applications.

UNIT - III

Data Converters **12Hrs**

Introduction, Basic DAC techniques, Different types of DACs-Weighted resistor DAC, R-2Rladder DAC, Inverted R-2R DAC, Different Types of ADCs - Parallel Comparator Type ADC, Counter Type ADC, Successive Approximation ADC and Dual Slope ADC, DAC and ADC Specifications.

UNIT - IV

Combinational Logic ICs **12Hrs**

Specifications and Applications of TTL-74XX & CMOS 40XX Series ICs - Code Converters, Decoders, LED & LCD Decoders with Drivers, Encoders, Priority Encoders, Multiplexers, Demultiplexers, Priority Generators/Checkers, Parallel Binary Adder/Subtractor, Magnitude Comparators.

UNIT - V

Sequential Logic IC's and Memories **12Hrs**

Familiarity with commonly available 74XX & CMOS40XX Series ICs– All Types of Flip-flops, Synchronous Counters, Decade Counters, Shift Registers. Memories - ROM Architecture, Types of ROMS & Applications, RAM Architecture, Static & Dynamic RAMs.

Self-study content

UNIT-I	*UNIT- II*	*UNIT- III*	*UNIT-IV*	*UNIT-V*
Why does IC 741 fail in single-supply, High Frequency and Low Power applications	*How does limited bandwidth of IC 741 affect filter accuracy*	*Limitations of IC 741 in Data Converters and Why modern ADC/DAC ICs outperform IC 741 based designs*	*Why are combinational logic ICs preferred over Gate-Level Design*	*List three applications where sequential logic circuits are essential*

TEXT BOOKS

1. Op-Amps & Linear ICs– Ramakanth A. Gayakwad, PHI, 2003.
2. Digital Fundamentals –Floydand Jain, Pearson Education,8th Ed., 2005.

REFERENCE BOOKS

1. Linear Integrated Circuits –D. Roy Chowdhury, New Age International (p) Ltd, 2nd Ed., 2003.
2. Digital Design Principles and Practices–John. F. Wakerly, Pearson 3rd Ed., 2009.
3. Linear Integrated Circuits and Applications – Salivahana, TMH, 2008.
4. OperationalAmplifierswithLinearIntegratedCircuits,4th Ed., William D. Stanley, Pearson Education India, 2009.

COURSE OUTCOMES:

CO1. A thorough understanding of operational amplifiers with linear integrated circuits.
CO2. Attain the knowledge of functional diagrams and design applications of IC555 and IC565.
CO3. Acquire the knowledge and design the Data converters.
CO4. Choose the proper digital integrated circuits by knowing their characteristics.
CO5. Acquire the knowledge of sequential logic ICs and memories

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	3	2	1	-	-	-	-	-	-	-	3	2
CO-2	3	3	3	1	-	-	-	-	-	-	-	-	3
CO-3	3	3	3	1	-	-	-	-	-	-	-	-	3
CO-4	3	3	2	1	-	-	-	-	-	-	-	-	-
CO-5	3	3	3	2	-	-	-	-	-	-	-	-	-

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



II B.Tech. II Semester

Course Name	:	**CONTROL SYSTEMS**				
Course Code	:	**2504406**	L	T	P	C
Course Category	:	**Theory**	3	0	0	3

CIE: 40 SEE: 60 Total: 100

Pre-Requisites: Linear Algebra and Calculus, Ordinary Differential Equations and Multivariable Calculus Laplace Transforms, Numerical Methods and Complex variables

COURSE OBJECTIVES:

1 To understand the different ways of system representations such as Transfer function representation and state space representations and to assess the system dynamic response
2 To assess the system performance using time domain analysis and methods for improving it
3 To assess the system performance using frequency domain analysis and techniques for improving the performance
4 To design various controllers and compensators to improve system performance
5 To understand the State-space models of linear systems.

UNIT - I **14Hrs**

Control System fundamentals: Classification of control systems, Open and Closed loop systems. Mathematical modelling of mechanical systems and their conversion into electrical systems. Block diagram reduction and Signal flow graphs.

UNIT - II **14Hrs**

Time response Analysis: types of input. Transient response of second order system for step input. Time domain specifications. static error coefficients, Routh - Hurwitz criterion for stability.
Root locus techniques: Analysis of typical systems using root locus techniques. Effect of location of roots on system response.

UNIT - III **10Hrs**

Frequency response Analysis: Frequency domain specifications, bode plots, Gain margin and Phase Margin. Polar plot.

UNIT - IV **12Hrs**

Compensators and controllers: Introduction to compensators, Lag compensator, Lead compensator, Lag- Lead compensator, Design of compensators using bode plot. Introduction to controllers, P, I, D, PI, PD, PID controllers.

UNIT - V **10Hrs**

State space representation: Concept of state and state variables. State models of linear time invariant systems, State transition matrix, Solution of state equations. Controllability and observability.

Self-study content

UNIT-I	*UNIT- II*	*UNIT- III*	*UNIT-IV*	*UNIT-V*
Industry examples of control systems	*steady state errors*	*Nyquist plot*	*Applications of controllers and compensators*	*state space analysis using mechanical systems*

TEXT BOOKS:

1. I.J. Nagrath and M. Gopal, Control System Engineering, 5ed., New Age Publishers, 2009.
2. Benjamin C. Kuo, Automatic Control Systems, 7ed., PHI, 2010.

REFERENCE BOOKS:

1. K. Ogata, Modern Control Engineering, 2ed., Prentice Hall, 2010.
2. M. Gopal, Control Systems: Principles and Design, Tata McGraw-Hill, 1997.
3. Norman S. Nise, Control Systems Engineering, 5ed., John Wiley & Sons, 2007.
4. A.K. Jairath, Solutions and Problems of Control Systems, CBS Publishers, 2013.
5. A. Nagoor Kani, Control Systems, 2ed., RBA Publications, 2007.

COURSE OUTCOMES:

CO1: Classify control systems and represents in various models.
CO2: Apply standard test signals to a system to determine their characteristics.
CO3: Make use of stability concepts to obtain the desired characteristics.
CO4: Determine the characteristics of a Linear control system using various time and frequency domain tools.
CO5: Examine the system behavior using various stability analysis techniques.

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO-1	3	3	2	3	2							2	1
CO-2	2	3	2	2	3							2	1
CO-3	3	2	3	3	2							3	2
CO-4	2	3	2	2	3							2	2
CO-5	3	3	3	2	3							2	1

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



II B.Tech. II Semester

Course Name	:	**INNOVATION AND ENTREPRENEURSHIP**				
Course Code	:	**2588401**	L	T	P	C
Course Category	:	**Theory**	2	0	0	2

CIE: 40 SEE: 60 Total: 100

Course Objectives:

1. To familiarize on the basic concepts of innovation, entrepreneurship and its importance.
2. To Identify and analyze the process of problem-opportunity identification, market segmentation, and idea generation techniques.
3. To initiate prototype development and understand minimum viable product.
4. To develop initial Business and financial planning and Go-to-Market strategies
5. To impart knowledge on establishing startups, venture pitching and IPR

Unit I: Fundamentals of Innovation and Entrepreneurship

Innovation: Introduction, need for innovation, Features, Types of innovations, innovations in manufacturing and service sectors, fostering a culture of innovation, planning for innovation. Entrepreneurship: Introduction, types of entrepreneurship attributes, mindset of entrepreneurial and intrapreneurial leadership, Role of entrepreneurs in economic development. Woman Entrepreneurship, Importance of on-campus startups. Understanding to build entrepreneurial mindset attributes and networks individuals while on campus. Core Teaching Tool: Simulation, Game, Industry Case Studies (Personalized for students – 16 industries to choose from), Venture Activity.

Unit II: Problem and Customer Identification

Identification of gap, problem, analyzing the problem from an industry perspective, real-world problems**,** market and customer segmentation, validation of customer problem fit, Iterating problem-customer fit, Competition and Industry trends mapping and assessing initial opportunity, Porter's Five Force Model. Idea generation, Ideation techniques: Brainstorming, Brain writing, round robin, and SCAMPER, Design thinking principles, Mapping of solution to problem. Core Teaching Tool: Several types of activities including: Class, game, Gen AI, 'Get out of the building' and Venture Activity.

Unit III: Opportunity assessment and Prototype development

Identify and map global competitors, review industry trends, and understand market sizing: TAM, SAM, and SOM. Assessing scope and potential scale for the opportunity. Understanding prototyping and Minimum Viable Product (MVP). Developing a prototype: Testing, and validation. Core Teaching Tool: Venture Activity, no-code Innovation tools, Class activity

Unit IV: Business & Financial Models

Introduction to Business Model and types, Lean Canvas Approach: 9-block lean canvas model, building lean canvas for your startup. Business planning: components of Business plan- Sales plan, People plan and financial plan, Financial Planning: Types of costs, preparing a financial plan for profitability using a financial template, understanding the basics of Unit economics, Economies of Scale and analyzing financial performance. Go-To-Market (GTM) approach – Selecting the Right Channel, creating digital presence, and building customer acquisition strategy. Core Teaching Tool: Founder Case Studies – Sama and Securely Share; Class activity and discussions; Venture Activities.

Unit V: Startups and IPR

Startup requirements, building founding team members and mentors, pitch preparation, start-up registration process, funding opportunities and schemes, institutional support to entrepreneurs, startup lifecycle, documentation, legal aspects in startup, venture pitching readiness, National Innovation Startup Policy (NISP) and its features.

Patents, Designs, Patentability, Procedure for grants of patents. Indian Scenario of Patenting, International Scenario: International cooperation on Intellectual Property. Patent Rights: Scope of Patent Rights. Copyright, trademark, and GI. Licensing and transfer of technology.

Core Teaching Tool: Expert talks; Cases; Class activity and discussions; Venture Activities.

Self-study content

UNIT-I	UNIT- II	UNIT- III	UNIT-IV	UNIT-V
SWAYAM/NPTEL MOOCs	*Startup Case Studies*	*Business Model Canvas Preparation*	*. Market survey*	

Suggested Readings:

1. John R Bessant, Joe Tidd, Innovation and Entrepreneurship, 4E, Wiley, Latest Edition.
2. Ajay Batra, The Stratup Launch Book- A Practical Guide for Launching Customer Centric Ventures, Wiley, 2020. (For Core Teaching Tool).
3. Entrepreneurship Development and Small Business Enterprises, Poornima M Charantimath, 3E, Pearson, 2018.
4. D.F. Kuratko and T.V. Rao, Entrepreneurship: A South-Asian Perspective, Cengage Learning, 2013.
5. Robert D. Hisrich, Michael P. Peters, Dean A. Shepherd, Sabyasachi Sinha (2020). Entrepreneurship, McGrawHill, 11th Edition.
6. NISP -Brochure inside pages - startup_policy_2019.pdf

Course Outcomes:

1. Understand the entrepreneurship and the entrepreneurial process and its significance in economic development.
2. Assess the problem from an industry perspective and generate solutions using the design thinking principles.
3. Assess market competition, estimate market size, and develop a prototype.
4. Analyze Business and financial planning models and Go-to-Market strategies.
5. Able to build a start-up, register IP and identify funding opportunities.

Course Articulation Matrix

CO's	Program Outcomes											Program Specific Outcomes	
	1	2	3	4	5	6	7	8	9	10	11	1	2
CO1	1	2	1			2	2	1	1	1	2	0	2
CO2		3	2	1	1	2		1	1	1	2	1	2
CO3		2	3	2	3	1		2	1	2	2	2	3
CO4		2	3	1	2	2	1	2	2	3	2	1	3
CO5		1	2			2	3	3	3	2	2		3

SDG'S Mapping:



II B.Tech. II Semester

		L	T	P	C
Course Name	**: COMPUTATIONAL MATHEMATICS LAB**				
Course Code	**: 2599403**	0	0	2	1
Course Category	: **Practical**				

CIE: 40 SEE: 60 Total: 100

Course Objectives: To learn

- Solve problems of Eigen values and Eigen Vectors using Python/MATLAB.
- Solution of Algebraic and Transcendental Equations using Python/MATLAB
- Solve problems of Linear system of equations
- Solve problems of First-Order ODEs Higher order linear differential equations with constant coefficients

UNIT-I: Eigen values and Eigenvectors: **6P**

Programs:

- Finding real and complex Eigen values.
- Finding Eigen vectors

UNIT-II: Solution of Algebraic and Transcendental Equations : **6P**

Bisection method, Newton Raphson Method

Programs:

- Root of a given equation using Bisection method.
- Root of a given equation Newton Raphson Method.

UNIT-III: Linear system of equations **6P**

Jacobi's iteration method and Gauss-Seidal iteration method

Programs:

- Solution of given system of linear equations using Jacobi's method
- Solution of given system of linear equations using Gauss-Seidal method

UNIT-IV: First-Order ODEs **8P**

Exact and non-exact equations, Applications: exponential growth/decay, Newton's law of cooling.

Programs:

- Solving exact and non-exact equations
- Solving exponential growth/decay and Newton's law of cooling problems

UNIT-V: Higher order linear differential equations with constant coefficients: **6P**

Programs:

- Solving homogeneous ODEs
- Solving non-homogeneous ODEs

Self-study content

UNIT-I	*UNIT- II*	*UNIT- III*	*UNIT-IV*	*UNIT-V*
Iterative eigenvalue methods	*Fixed point convergence*	*Matrix Splitting*	*. Discrete Approximation of Continuous Systems*	*System Reduction*

TEXT BOOKS:

1. MATLAB and its Applications in Engineering, Rajkumar Basal, Ashok Kumar Geo, Manoj Kumar Sharma, Pearson publication.
2. Kenneth A. Lambert, The fundamentals of Python: First Programs, 2011, Cengage Learnings.
3. Think Python First Edition, by Allen B. Downey, Orielly publishing.
4. Introduction to Python Programming, William Mitchell, Povel Solin, Martin Novak et al., NC Lab

Public Computing, 2012.
5. Introduction to Python Programming, ©Jacob Fredslund, 2007.

REFERENCE BOOKS:

1. An Introduction to Python, John C. Lusth, The University of Alabama, 2011.
2. Introduction to Python, ©Dave Kuhlman, 2008.

Course Outcomes: After learning the contents of this paper, the student must be able to

CO-1. Develop the code to find the Eigen values and Eigen Vectors using Python/MATLAB.
CO-2. Develop the code find solution of Algebraic and Transcendental Equations and Linear system of equations using Python/MATLAB
CO-3. Write the code find the system of linear equations using Jacobi's method, Gauss-Seidal method.
CO-4. Write the code to solve problems of First-Order ODEs equations with constant coefficients
CO-5. Write the code to solve problems of homogeneous & non-homogeneous ODEs

- Visualize all solutions Graphically through programmes

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**1**	**2**
CO-1	3	2	1	2	3	-	-	1	1	-	1	2	1
CO-2	3	3	1	2	3	-	-	1	1	-	1	2	2
CO-3	3	3	1	2	3	-	-	1	1	-	1	2	1
CO-4	3	2	1	2	3	-	-	1	1	-	1	1	1
CO-5	3	2	1	2	3	-	-	1	1	-	1	1	1

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



II B.Tech. II Semester

Course Name	:	**ANALOG AND DIGITAL COMMUNICATIONS LABORATORY**				
Course Code	:	**2504407**	L	T	P	C
Course Category	:	**Practical**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Note:

- Minimum 12 experiments should be conducted.
- All these experiments are to be simulated first either using MATLAB, Commsim or any other simulation package and then to be realized in hardware.

COURSE OBJECTIVES:

1.Design and implement various Analog Modulation and demodulation Techniques and observe the time and frequency domain characteristics

2.Design and implement various multiplexing and de multiplexing techniques in various applications

3. Apply different types of sampling with various sampling rates and duty cycles

4.Design and implement various pulse modulation and demodulation techniques and observe the time and frequency domain characteristics

5. Design and implement various Digital Modulation and demonstration techniques and observe the waveforms of these modulated signals Practically

List of Experiments:

1. Generate Amplitude modulated Signal and perform demodulation for different modulation indices. Plot the corresponding waveforms Compare the modulation index theoretically and practically.
2. Generate Frequency modulated Signal and perform demodulation for different modulation indices. Plot the corresponding waveforms, Compare the modulation index theoretically and practically.
3. Generate modulated and demodulate DSB-SC Signal for different modulation indices and plot the corresponding waveforms, Compare the modulation index theoretically and practically
4. Generate and demodulate SSB-SC modulated Signal (Phase Shift Method) for different modulation indices and plot the corresponding waveforms. Also calculate theoretically and practically the modulation index in each case
5. Demonstrate the Frequency Division Multiplexing & De multiplexing practically by transmitting at least 4 different signals simultaneously with respect to time and recovering without distortion.
6. Verify Sampling theorem for different sampling rates, Sampling types and Duty Cycles and Plot the sampled and reconstructed Signals. Write the conclusions, based on practical observations
7. Design and implement a Pulse Amplitude Modulator & Demodulator Circuit using 555 timer and plot the corresponding waveforms from the practical observations
8. Design and implement a Pulse Width Modulator & Demodulator Circuit using 555 timer and plot the corresponding waveforms from the practical observations
9. Design and implement a Pulse Position Modulator & Demodulator Circuit using 555 timer and plot the corresponding waveforms from the practical observations
10. Generate PCM Modulated Signal and demodulate it by designing and implementing the corresponding Demodulator. Plot the corresponding waveforms from practical observations
11. Generate Delta Modulated Signal and demodulate it by designing and implementing the

corresponding Demodulator. Plot the corresponding waveforms from practical observations.

12. Generate FSK modulated Signal and demodulate it by designing and implementing the corresponding Demodulator. Plot the corresponding waveforms from practical observations.
13. Generate practically Binary PSK modulated Signal and demodulate it by designing and implementing the corresponding Demodulator. Plot the corresponding waveforms from practical observations.
14. Generate practically DPSK modulated Signal and demodulate it by designing and implementing the corresponding Demodulator. Plot the corresponding waveforms from practical observations.
15. Generate practically QPSK modulated Signal and demodulate it by designing and implementing the corresponding Demodulator. Plot the corresponding waveforms from practical observations.

Self-study content

- *Digital Modulation Techniques: ASK, FSK, PSK and QPSK*
- *Pulse Modulation Techniques: PAM, PWM, PPM and PCM*

COURSE OUTCOMES:

CO1: Will be able to design and implement various Analog Modulation and Demodulation Techniques and observe the time and frequency domain characteristics of these modulated Signals

CO2: Will be able to Design and implement various multiplexing and de multiplexing techniques in various applications

CO3: Will be able to understand the concept of aliasing and different types of sampling with various sampling rates and duty cycles by implementing practically

CO4: Will be able to design and implement various Pulse Modulation and Demodulation Techniques and observe the time and frequency domain characteristics of these modulated Signals

CO5: Will be able to Design and implement various Digital Modulation and demonstration techniques and observe the waveforms of these modulated signals Practically

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**1**	**2**
CO-1	1	-	3	1	2	2	-	2	3	2	1	3	2
CO-2	1	-	3	1	2	2	-	2	3	2	1	3	2
CO-3	1	-	3	1	2	2	-	2	3	2	1	2	2
CO-4	1	-	3	1	2	2	-	2	3	2	1	3	2
CO-5	1	-	3	1	2	-	-	2	3	2	1	1	-

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



II B.Tech. II Semester

Course Name	:	**LINEAR AND DIGITAL IC APPLICATIONS LABORATORY**				
Course Code	:	**2504408**	L	T	P	C
Course Category	:	**Practical**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Note:

- Minimum 12experiments should be conducted.
- Verify the functionality of the IC in the given application.

List of Experiments:

1. Design an Inverting and Non-inverting Amplifier using Op Amp and calculate gain.
2. Design Adder and Subtractor using Op Amp and verify addition and subtraction process.
3. Design a Comparator using Op-Amp and draw the comparison results of A=B, A<B, A>B.
4. Design a Integrator and Differentiator Circuits using IC741 and derive the required condition practically.
5. Design a Active LPF, HPF cutoff frequency of 2 KHz and find the roll off of it.
6. Design a Circuit using IC741 to generate sine / square / triangular wave with period of 1 KHz and draw the output waveform.
7. Construct Mono-stable Multivibrator using IC555 and draw its output waveform.
8. Construct Astable Multivibrator using IC 555 and draw its output waveform and also find its duty cycle.
9. Design a Schmitt Trigger Circuit and find its LTP and UTP.
10. Design Frequency modulator and demodulator circuit and draw the respective waveforms.
11. Design Voltage Regulator using IC723, IC 7805 / 7809 / 7912 and find its load regulation factor.
12. Design R-2R ladder DAC and find its resolution and write a truth table with respective voltages.
13. Design Parallel comparator type / counter type / successive approximation ADC and find its efficiency.
14. Design a Gray code converter and verify its truth table.
15. Design an even priority encoder using IC74xx and verify its truth table.
16. Design a 8x1multiplexer using digital ICs.
17. Design a 4-bit Adder / Subtractor using digital ICs and Add / Sub the following bits.
    (i) 1010 (ii) 0101 (iii) 1011
    0100 0010 1001.
18. Design a Decade counter and verify its truth table and draw respective waveforms.
19. Design a Up/downcounterusingIC74163anddrawread/write waveforms.
20. Design a Universal shift register using IC74194 / 195and verify its shifting operation.
21. Design a 16x4 RAM using 74189 and draw its read /write operation.
22. Design a 8x3 encoder / 3x8 decoder and verify its truth table.

Self-study content

- *To design an instrumentation amplifier using three op-amps and study its high CMRR.*
- **To design a precision rectifier using op-amp and diode and verify rectification of low-level signals.*

Course Outcomes: Upon completing this course, the student able to

1. Design and implementation of various analog circuits using 741 ICs.
2. Design and implementation of various Multivibrators using 555 timers.
3. Design and implement various circuits using digital ICs.
4. Design and implement ADC, DAC and voltage regulators.
5. Design and implement the MUX, counters, and encoders.

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	2		4	5	6	7	8	9	10	11	1	2
CO-1	1		3	3	3	-	-	-	3	3	1	3	3
CO-2	1		3	3	3	-	-	-	3	3	1	2	3
CO-3	1		3	3	3	-	-	-	3	3	1	2	2
CO-4	1		3	3	3	-	-	-	3	3	1	3	3
CO-5	1		3	3	3				3	3	1	2	1

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:



II B.Tech. II Semester

Course Name	:	**Web and Mobile Applications**				
Course Code	:	**2504409**	L	T	P	C
Course Category	:	**Skill Development Course**	0	0	2	1

CIE: 40 SEE: 60 Total: 100

Course Objectives:

1. Learn to install and configure Android Studio for mobile application development.
2. Create simple user interfaces using various layouts, input boxes, and buttons, and handle user interactions such as displaying messages on button clicks.
3. Design multi-activity applications using Intents to pass data between different activities.
4. Integrate device capabilities such as sending SMS messages programmatically using Intents.
5. Implement advanced UI components like fragments, radio buttons, spinners, date pickers, menus, and persistent data storage using a database for user registration.

List of Experiments:

1. Installation of Android studio
2. Development Of Hello World Application
3. Create an application that takes the name from a text box and shows hello message along with the name entered in text box, when the user clicks the OK button
4. Create a screen that has input boxes for User Name, Password, Address, Gender (radio buttons for male and female), Age (numeric), Date of Birth (Date Picker), State (Spinner) and a Submit button. On clicking the submit button, print all the data below the Submit Button (use any layout)
5. Design an android application to create page using Intent and one Button and pass the Values from one Activity to second Activity
6. Design an android application Send SMS using Intent
7. Create an android application using Fragments
8. Design an android application Using Radio buttons
9. Design an android application for menu
10. Create a user registration application that stores the user details in a database table

Self-Study Content:

- ❖ Create a Multi-Screen App with Shared Preferences for User Login.
- ❖ Recycler View with Room Database – Display Registered Users.

Major Equipment required for Laboratories:

1. PC
2. ANDROID STUDIO

Course Outcomes:

Upon completing this course, the students will be able to

1. Successfully install and configure Android Studio to create and run Android projects.
2. Design interactive user interfaces using various layouts, input controls (Edit Text, Spinner, Date Picker), and radio buttons with proper event handling.
3. Develop applications that can navigate between multiple activities and pass data using Intents.
4. Send SMS messages from an Android application using Intents.
5. Create a complete user registration application that stores and manages user details in a database table

Course Articulation Matrix

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**1**	**2**
CO-1	3	2	2	1	3	1	1	2	1	1	2	3	2
CO-2	3	3	3	2	3	2	1	2	2	1	2	3	3
CO-3	3	2	3	1	3	1	1	2	2	1	2	3	2
CO-4	3	3	3	2	3	2	2	2	2	2	2	3	3
CO-5	3	3	3	3	3	3	2	3	3	2	3	3	3

*(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using **mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial**)*

SDG'S Mapping:




SUSTAINABLE DEVELOPMENT GOALS
1 NO POVERTY
2 ZERO HUNGER
3 GOOD HEALTH AND WELL-BEING
4 QUALITY EDUCATION
5 GENDER EQUALITY
6 CLEAN WATER AND SANITATION
7 AFFORDABLE AND CLEAN ENERGY
8 DECENT WORK AND ECONOMIC GROWTH
9 INDUSTRY, INNOVATION AND INFRASTRUCTURE
10 REDUCED INEQUALITIES
11 SUSTAINABLE CITIES AND COMMUNITIES
12 RESPONSIBLE CONSUMPTION AND PRODUCTION
13 CLIMATE ACTION
14 LIFE BELOW WATER
15 LIFE ON LAND
16 PEACE, JUSTICE AND STRONG INSTITUTIONS
17 PARTNERSHIPS FOR THE GOALS

R25 SYLLABUS MAPPPED WITH SDGs

Sem	Course	SDG Mapping	Main connection	How the Course Supports the Particular SDG(s)
I-I	Matrices and Calculus	SDG 3,4,8,9,10,16	Mathematical foundation for engineering analysis, modelling and innovation	Builds mathematical knowledge and analytical skills for quality engineering education, mathematical modelling, engineering analysis and innovation.
I-I	Engineering Physics	SDG 4, 7, 9,10,11,13	Semiconductors, solar cells, advanced materials, lasers and optical technologies	Supports health-related technologies, clean/solar energy and technological innovation through semiconductors, advanced materials, lasers and optical technologies.
I-I	C Programming & Data Structures	SDG 4, 8, 9	Computational problem-solving and digital engineering skills	Develops computational problem-solving, programming and digital skills supporting quality education, employability and technological innovation.
I-I	Computer Aided Engineering Graphics	SDG 4,8, 10, 11	CAD, engineering design, infrastructure planning and efficient design	Develops CAD and engineering-design skills supporting innovative infrastructure, sustainable development and responsible resource use.
I-I	Basic Electrical Engineering	SDG 7, 9, 12	Electrical energy, machines, installations and energy efficiency	Develops knowledge of electrical energy, machines and efficiency, supporting clean energy, resilient systems and resource efficiency.
I-I	Engineering Physics Lab	SDG 3,4, 7,8, 9,11	Solar-cell, semiconductor, optical-fiber and material experiments	Provides practical work on solar cells, semiconductors, optical fiber and materials, supporting clean energy and technological innovation.
I-I	C Programming & Data Structures Lab	SDG 4, 8, 9	Practical programming and algorithm development	Provides hands-on programming and algorithm practice supporting digital learning, employability and innovation.
I-I	Basic Electrical Engineering Lab	SDG 7, 9, 12	Electrical machines, transformers, efficiency and electrical systems	Provides practical experience with electrical machines, transformers, efficiency and systems, supporting energy efficiency and sustainable engineering.
I-I	IT Workshop	SDG 4, 8, 9,17	Digital literacy, computing, Internet and cyber security	Builds digital literacy, computing, Internet and cyber-security skills supporting education, employment and digital innovation.
I-I	Induction Program	SDG 4	Orientation, higher education and lifelong learning	Supports orientation, higher education and lifelong learning, strengthening quality education.
I-II	Ordinary Differential Equations & Vector Calculus	SDG 4, 9,17	Mathematical modelling of engineering and physical systems	Develops mathematical modelling skills for engineering and physical systems, supporting quality education and innovation.

I-II	Engineering Chemistry	SDG 3, 6, 7,9,11, 12, 13,14	Water treatment, energy sources, corrosion, biodegradable polymers, pollution analysis and green hydrogen	Addresses water treatment, energy sources, corrosion, biodegradable polymers, pollution and green hydrogen, supporting health, water, clean energy, responsible production and climate action.
I-II	English for Skill Enhancement	SDG 3,4, 8,9,10,16	Communication, technical writing, employability and entrepreneurship	Develops communication, technical writing, employability and entrepreneurship skills supporting quality education and productive employment.
I-II	Network Analysis and Synthesis	SDG 7, 9, 12	Efficient circuit/network analysis and engineering-system design	Develops efficient circuit/network analysis and engineering-system design skills supporting energy systems, innovation and resource efficiency.
I-II	Electronic Devices & Circuits	SDG 3, 7, 9		
I-II	Engineering Chemistry Lab	SDG 3, 6, 7,9,11, 12,13	Water analysis, corrosion, polymers, solar-cell applications and chemical measurements	Provides practical work in water analysis, corrosion, polymers, solar-cell applications and chemical measurements, supporting water quality, clean energy and responsible production.
I-II	Engineering Workshop	SDG 4,8, 9,11, 12	Practical engineering, fabrication and responsible production skills	Develops practical engineering and fabrication skills while promoting responsible production and innovation.
I-II	English Language & Communication Skills Lab	SDG 4, 8,16	Professional communication, teamwork and employability	Strengthens professional communication, teamwork and employability, supporting quality education and productive employment.
I-II	Electronic Devices & Circuits Lab	SDG 3, 7, 9	Practical semiconductor and electronic circuit applications	Provides practical semiconductor and electronic-circuit experience applicable to health technologies, energy systems and innovation.
II-I	Probability Theory & Stochastic Processes	SDG 4, 9	Statistical modelling and engineering decision-making	Develops statistical modelling and engineering decision-making skills supporting quality technical education and innovation.
II-I	Signals & Systems	SDG 9, 11	Signal processing, communication, sensing and smart systems	Develops signal-processing, communication and sensing concepts supporting technological innovation and smart systems.
II-I	Digital Logic Design	SDG 9	Digital-system and electronic-infrastructure development	Builds digital-system and electronic-infrastructure design skills contributing to technological innovation.
II-I	Electronic Circuit Analysis	SDG 7, 9	Electronic-system analysis and efficient technology	Develops skills for analysing electronic systems and improving technology efficiency, supporting energy-related systems and innovation.
II-I	Python Programming	SDG 4, 8, 9	Software development, computational skills and employability	Develops software and computational skills supporting

				quality education, employability and technological innovation.
II-I	Modelling & Simulation Lab	SDG 9, 11	Engineering simulation, modelling and system analysis	Provides practical engineering simulation, modelling and system-analysis experience supporting innovative and smart infrastructure.
II-I	Digital Logic Design Lab	SDG 9	Practical digital hardware design	Provides hands-on digital hardware design practice supporting innovative electronic and digital systems.
II-I	Electronic Circuit Analysis Lab	SDG 9	Practical circuit analysis, simulation and design	Provides practical circuit analysis, simulation and design skills supporting technological innovation.
II-I	Python Programming Lab	SDG 4, 8, 9	Practical programming and software-development skills	Provides hands-on programming and software-development practice supporting digital learning, employability and innovation.
II-I	Linux & Shell Scripting	SDG 8, 9	Computing infrastructure, automation and technical employment	Develops computing-infrastructure, automation and technical employment skills supporting productive work and technological innovation.
II-I	Environmental Science / Indian Knowledge System	SDG 6, 11, 12, 13, 14, 15, 16	Natural resources, ecosystems, pollution, climate change, biodiversity, sustainability and environmental governance	Builds understanding of natural resources, ecosystems, pollution, climate change, biodiversity, sustainability and environmental governance.
II-II	Numerical Methods & Complex Variables	SDG 4, 9	Numerical/computational methods for engineering problem-solving	Develops numerical and computational methods for engineering problem-solving, supporting quality technical education and innovation.
II-II	Electromagnetic Fields & Transmission Lines	SDG 7, 9	Communication and electrical/electromagnetic infrastructure	Develops knowledge for communication and electrical/electromagnetic infrastructure, supporting energy systems and technological innovation.
II-II	Analog & Digital Communications	SDG 9, 11	Communication networks and information infrastructure	Develops communication-network and information-infrastructure concepts supporting technological innovation and smart connected systems.
II-II	Linear & Digital IC Applications	SDG 7, 9, 12	ICs, regulators, ADC/DAC and efficient electronic systems	Covers ICs, regulators and ADC/DAC applications for efficient electronic systems, supporting energy efficiency, innovation and responsible resource use.
II-II	Control Systems	SDG 9, 11	Automation, industrial control and smart infrastructure	Develops automation and industrial-control skills supporting technological innovation and smart infrastructure.
II-II	Innovation & Entrepreneurship	SDG 1, 8, 9	Entrepreneurship, economic opportunity, innovation and technology startups	Promotes entrepreneurship, economic opportunity, innovation and technology-startup development, supporting productive employment and innovation.
II-II	Computational	SDG 4, 9	Computational engineering analysis and modelling	Provides practical computational engineering analysis and modelling

	Mathematics Lab			skills supporting quality education and technological innovation.
II-II	Analog & Digital Communications Laboratory	SDG 9, 11	Practical communication, multiplexing and modulation systems	Provides practical communication, multiplexing and modulation experience supporting innovative communication networks and information infrastructure.
II-II	Linear & Digital IC Applications Laboratory	SDG 7, 9, 12	Regulators, ADC/DAC, ICs and digital-system implementation	Provides practical work with regulators, ADC/DAC, ICs and digital systems, supporting efficient energy use, innovation and responsible engineering.
II-II	Web & Mobile Applications	SDG 8, 9	Digital applications, software development and digital services	Develops digital applications, software and digital-service skills supporting employability and technological innovation.

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